

Particulate Matter

Week in Review

Week ending 19 September 2026 | 13 September 2026 to 19 September 2026 | 148 records pooled from 7 daily issues

148

RECORDS POOLED
(148 UNIQUE DOIs)

44%

MEASUREMENT SIDE
(65 RECORDS)

10

SUBTOPICS
ACTIVE

47

POOLED EFFECT
ESTIMATES (42 WITH A
CI)

15

ABSTRACT-FREE
(10% OF WEEK)

The week in one paragraph

Last week's two open questions both resolved, and both in the same direction: the collapse in exposure modelling was an artefact, and the abstract-free problem was fixable. *Exposure* went 20 → 38 and *METADATA ONLY* went 16% → 10%. The pipeline change that did the second is the more consequential of the two.

Scale. 148 records over seven issues on daily totals of 21, 16, 22, 21, 18, 30 and 20 — a 1.9× range, the flattest week this watch has recorded and flatter again than last week's 2.6×. Against recent rollups: 148 after 126, 150, 109 and 112, so the level is high-normal. The flatness matters more than the level, because every cluster movement below is being read against a stable denominator rather than through a single deposit spike. All **ten** subtopics are populated.

Exposure assessment & modelling rebounded 20 → 38 and overshot. The 12 September rollup recorded a halving from 47 to 20, declined to call it a turn in the field, and set an explicit test: whether *Exposure* returns to the mid-20s. It did not return to the mid-20s — it went past the pre-collapse level to 38, the second-highest reading in the series. **The test is answered and the answer is unambiguous:** the 20 was a one-week sampling floor, not a contraction. *Sensing* moved only 31 → 27, so the two measurement clusters again moved in opposite directions, and the measurement-side share reads 47, 54, 40, 44% across four weeks — still oscillating, still not trending, and the 29 August warning against building a claim on it still stands.

The abstract-free problem was halved by a pipeline change, not by the publishers. 15 of 148 (10%) shipped as *METADATA ONLY*, against 20 of 126 (16%) last week, and inside *Sensing* the count fell 10 → 4. Last week's rollup named an abstract-retry pass as the highest-value outstanding pipeline work. What actually delivered the improvement was something else: the PMC and AGRICOLA retro-index gate moved into the screening script on 18 September, so records that used to enter abstract-less and be carried are now rejected at source as out-of-window artefacts. **That is a different mechanism with the same visible effect**, and the distinction matters — the retry pass is still not built, and the five same-day Elsevier, ACS and Springer deposits in the 19 September issue confirm the underlying publisher behaviour is unchanged.

Estimate density returned to the series norm and precision improved sharply. 47 pooled estimates against 148 records, one per 3.1, from a four-week series of 2.9, 4.4, 2.7, 3.1. More useful than the density: 42 of 47 carry a confidence interval (89%), against 40 of 47 last week and far above the series floor. **Four of the five bare points come from a single paper** — the wildfire versus non-wildfire PM_{2.5} hospitalisation analysis of 18 September, whose abstract reports eight headline risks and not one interval.

Subtopic trend and inflection

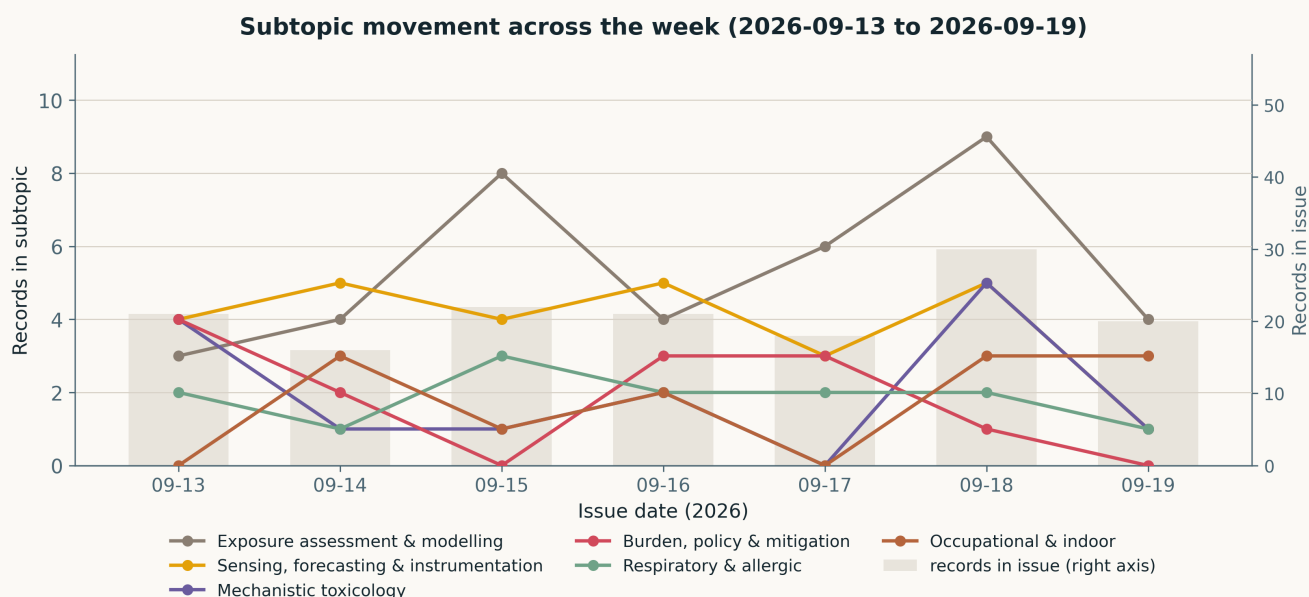


Fig. Per-issue subtopic counts across the week from *state/metrics.csv*, seven issues. Daily totals 21, 16, 22, 21, 18, 30 and 20 — a $1.9\times$ range, the flattest week in the series, against $2.6\times$ last week and $5.4\times$ the week before. The weekend suppression that dominated the last two rollups is largely absent here: Sunday 13 September returned 21 records, not the collapse seen on 6 and 12 September, because the Crossref-by-ISSN and Europe PMC legs now carry the weekend entirely. The 30-record Friday is the 18 September issue, which was built twice — a partial morning pass and a full-window re-harvest at 23:0x — and is a complete day rather than a deposit anomaly.

Expanding. *Exposure assessment & modelling* 20 → 38 is the week's dominant movement and is discussed above. *Other clinical endpoints* 4 → 8 doubles, driven almost entirely by the 19 September issue, in which six of twenty records landed in that residual band — frailty, acute-care utilisation, renal disease, paediatric respiratory infection, ocular surface and psoriasis. *Mechanistic toxicology* 9 → 14 and *Cardiovascular & metabolic* 8 → 11 both continue the recovery begun last week. *Respiratory & allergic* 12 → 13 holds.

Contracting. *Occupational & indoor* 15 → 12 eases for a second consecutive week off the 19-record surge of 5 September, but remains above the pre-surge regular-week maximum of 9 for a third straight week. The 5 September rollup declined to call the indoor turn a trend and the 12 September rollup set a second week as the test; it has now held a third. **On this watch's own stated criteria the indoor turn is now a trend and not a fluctuation.** *Sensing* 31 → 27 and *Burden, policy & mitigation* 14 → 13 are mild. *Reproductive & developmental* 6 → 5 stays at the bottom of the table, where it has sat every week of this series.

The inflection worth naming. It is not a subtopic movement at all — it is that **two consecutive weekly rollups set explicit falsifiable tests and both were answered this week.** *Exposure* was tested against a mid-20s return and overshot to 38; the indoor turn was tested against a third week and held. A watch that only reported counts could not have done either. The concrete thing to watch next week is *Occupational & indoor*: having now qualified as a trend, the interesting question is no longer whether it persists but whether any of it moves from chambers and models into occupied buildings with health outcomes, which so far **none** of it has.

Study architecture

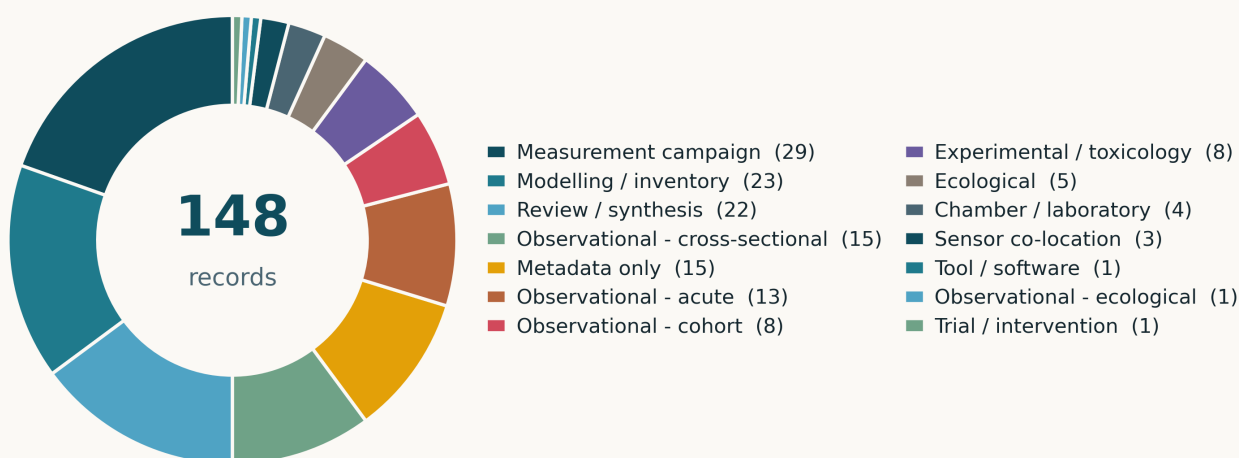


Fig. Study architecture over the pooled week. *Measurement campaign* (29) leads *Modelling / inventory* (23) and *Review / synthesis* (22), extending the measurement-ahead-of-modelling ordering restored last week. *Metadata only* falls to 15 and to fourth place, having been second at 20 a week ago — the clearest single-number statement of the screening change. Primary observational epidemiology (cross-sectional 15, acute 13, cohort 8, ecological 6) totals 42, the largest grouping in the week if read as one. **One** interventional record appears in the 148, against two last week.

Pooled analytics

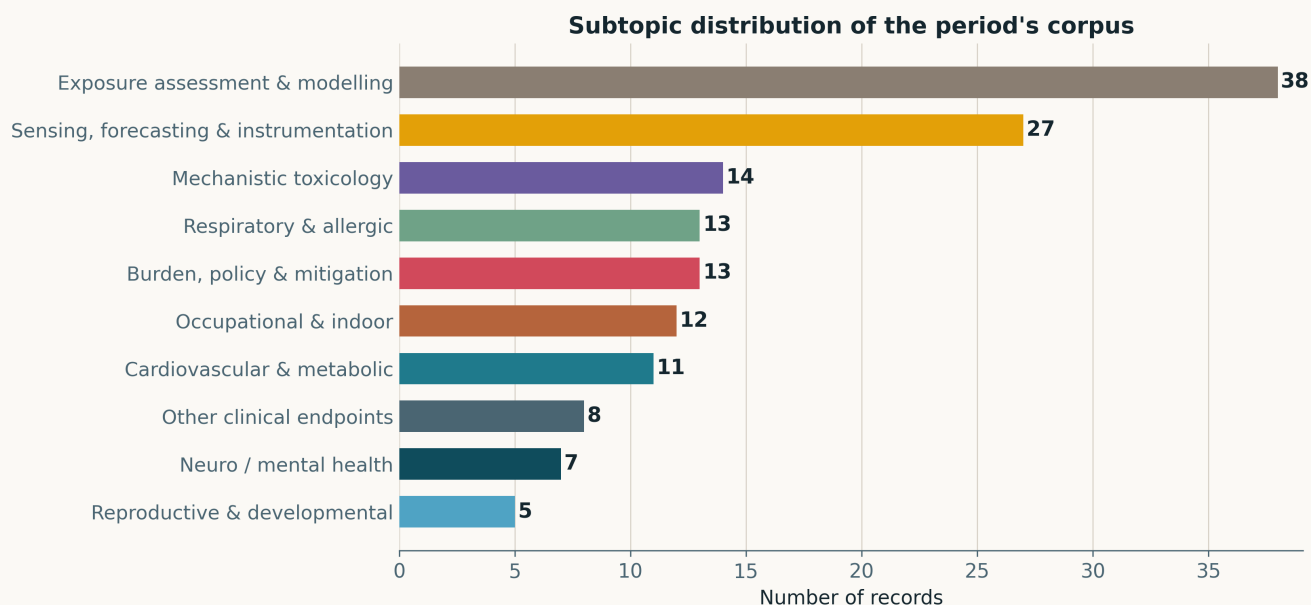


Fig. Pooled subtopic distribution, **all ten bands populated**. *Exposure assessment & modelling* at 38 and *Sensing* at 27 put the measurement side at **65 of 148 (44%)**. The health side is unusually evenly spread beneath them — *Mechanistic toxicology* 14, *Respiratory* 13, *Burden* 13, *Occupational & indoor* 12, *Cardiovascular & metabolic* 11 — five clusters inside a four-record band, which is the tightest health-side distribution in the series and means no single clinical literature dominated the week.

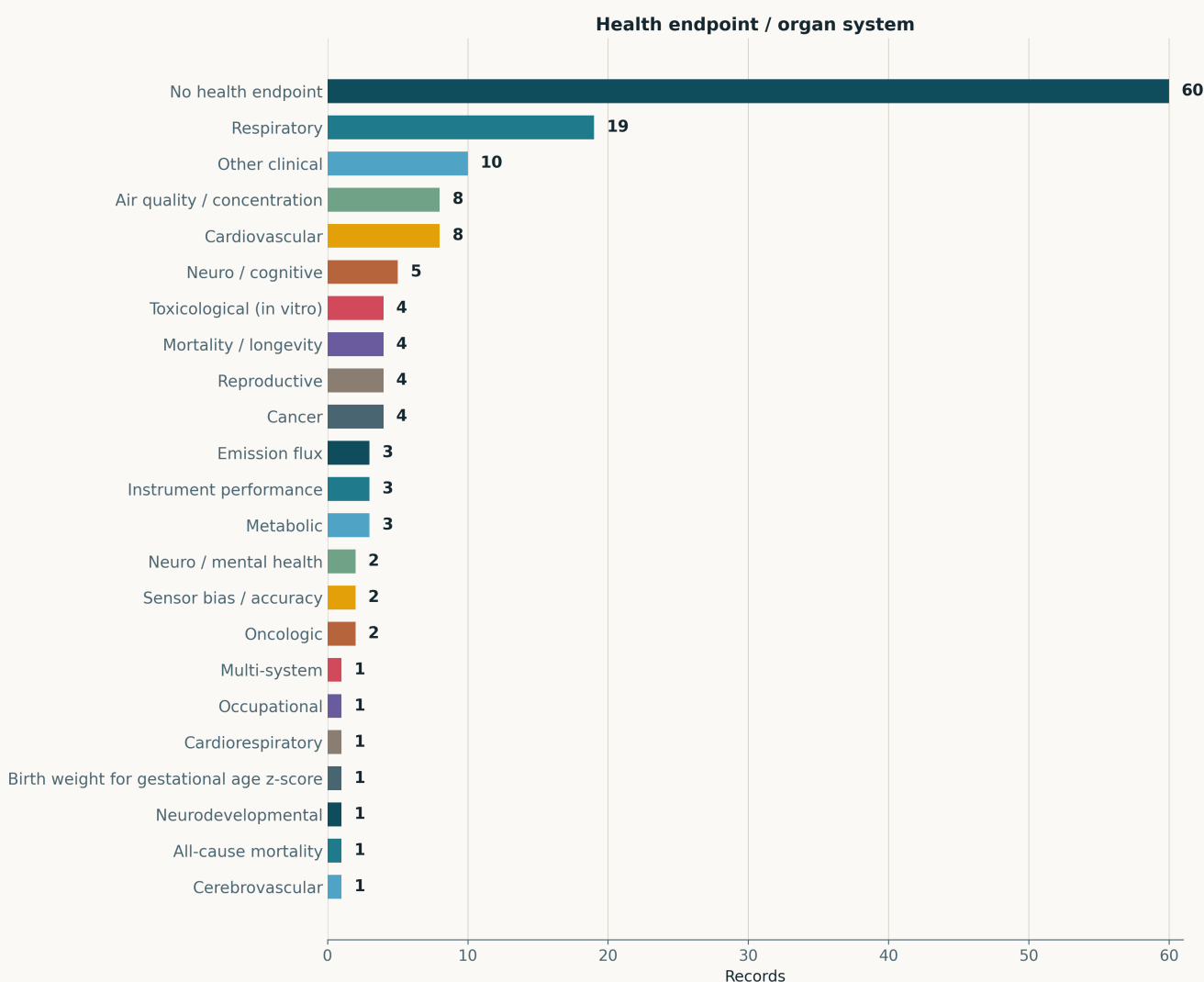


Fig. Health endpoint over the pooled week, canonicalised. “No health endpoint” is 60 of 148 (41%), the aerosol-chemistry, emissions, instrument-performance and not-applicable records pooled. Among the records that do report a human outcome, **respiratory leads at 19**, then other clinical at 10 and cardiovascular at 8. Note that several labels in this chart are *not* human outcomes despite reading like metrics — *air quality / concentration*, *instrument performance*, *sensor bias /*

accuracy, emission flux — and the gap between those and a clinical endpoint is the subject of the heatmap below.

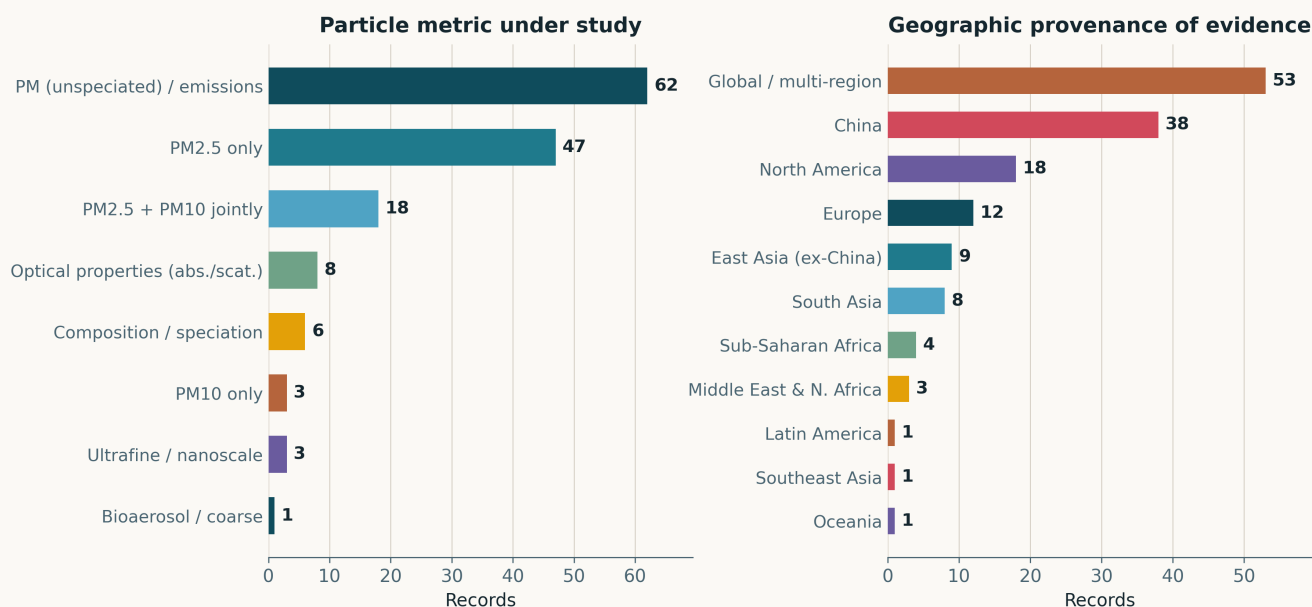


Fig. Left — particle metric. **PM_{2.5} alone is 47 of 148** and unspciated PM or emissions terms 62; joint PM_{2.5} + PM₁₀ is 18. The tail is the finding: optical properties 8, **composition or speciation 6**, **ultrafine or size-resolved number 3**, PM₁₀ alone 3, bioaerosol 1. Right — geographic provenance. *Global / multi-region* at 53 is the largest bar and, as every week, overstates reach — it pools genuine multi-country work with reviews that have no field site and with records whose location is unrecoverable. **China at 38 is the largest genuine single-country bar**, ahead of North America at 18 and Europe at 12.

Life-course windows addressed this period (records may span several stages)

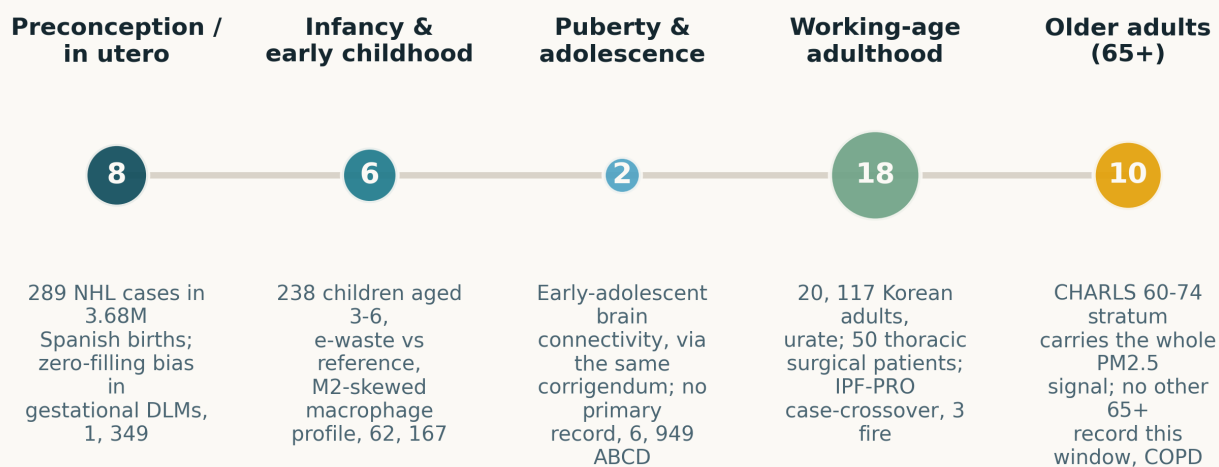
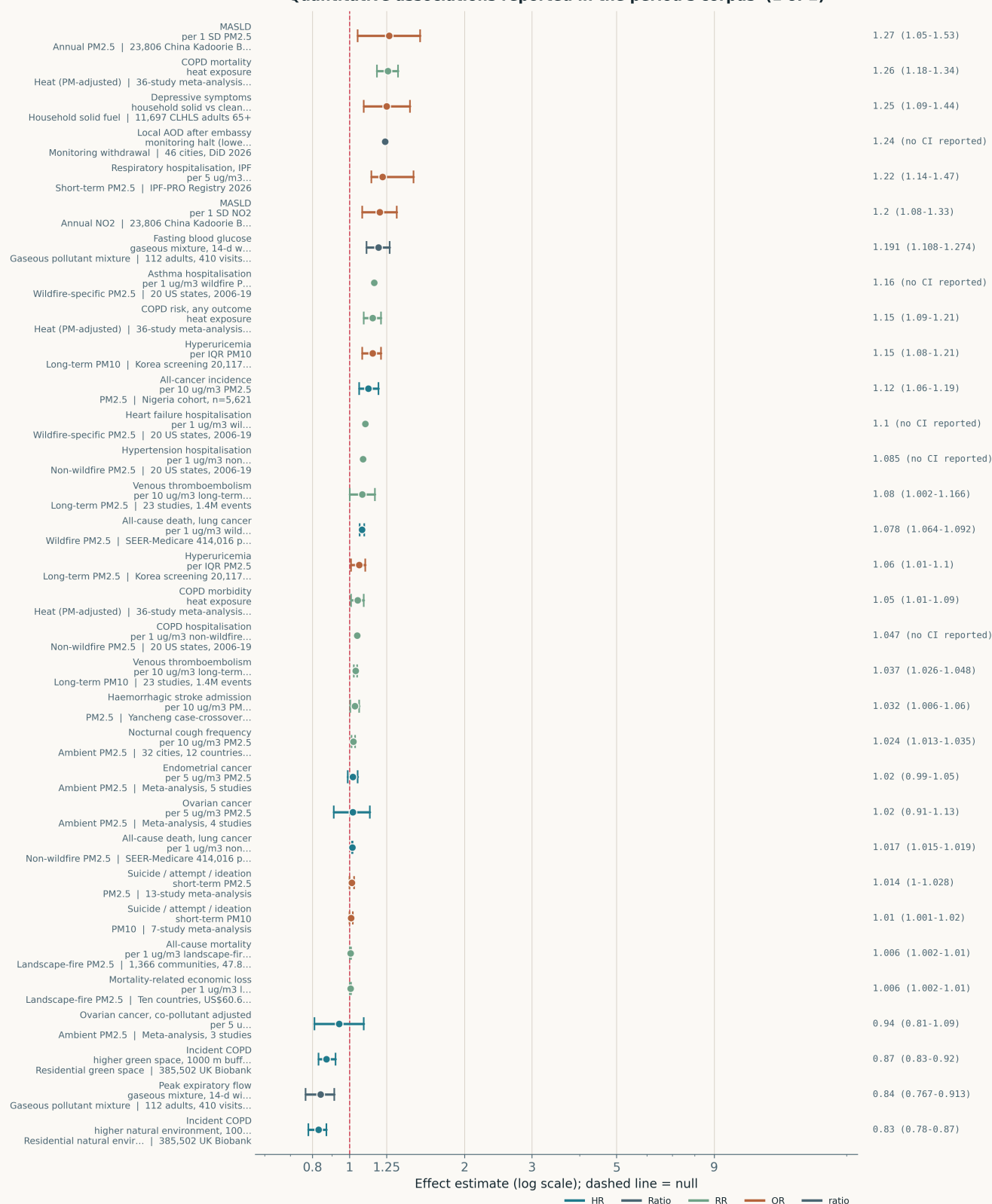


Fig. Life-course windows across the pooled week; counts are non-exclusive. **Working-age adulthood dominates at 18**, as it has every week. The notable reading is at the young end: **preconception and in utero is 8** and infancy and early childhood 6, and unlike last week these are carried by completed epidemiology rather than by measurement-side records — 289 non-Hodgkin lymphoma cases in a 3.68-million Spanish birth series, a zero-inflated gestational PM_{2.5} bias analysis, and a 238-child e-waste macrophage study. **Puberty and adolescence remains the weakest window at 2**, as in every rollup in this series.

Quantitative associations reported in the period's corpus (1 of 2)



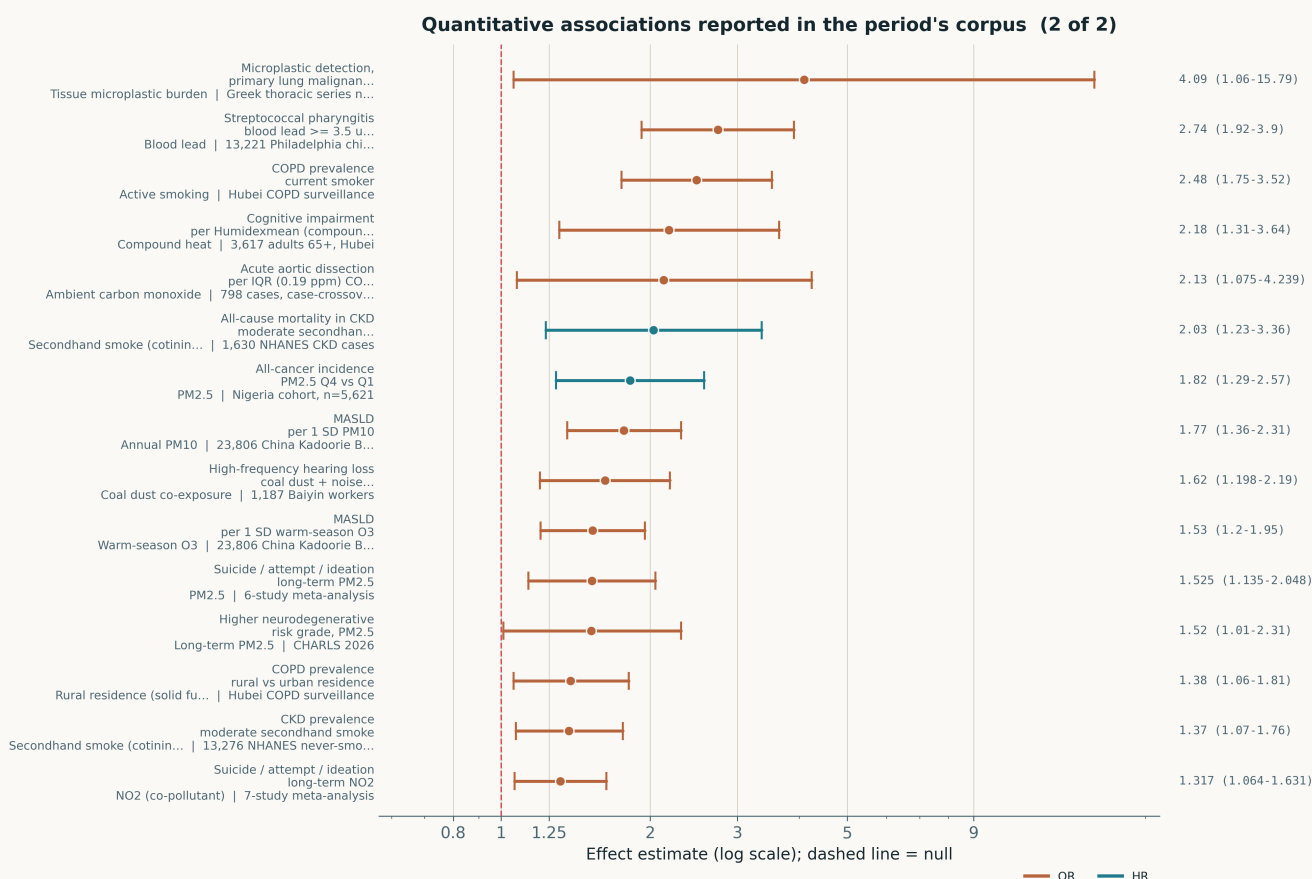


Fig. All 47 pooled effect estimates on a common logarithmic axis, split across two panels. **These are not mutually comparable and are not meta-analysed:** contrasts include per-IQR, per- $10 \mu\text{g m}^{-3}$, per- $1 \mu\text{g m}^{-3}$, per-SD, per-exposure-category and mixture-index units, on OR, HR, RR, IRR and ratio scales. **Heterogeneity here is structural, not statistical**, so no I^2 is reported and pooling these would be a category error. Four things are readable. First, precision improved: **42 of 47 report an interval (89%)**, and **four of the five bare points come from one paper** — the wildfire versus non-wildfire $\text{PM}_{2.5}$ hospitalisation analysis, which publishes eight headline risks without a single interval. Second, **the largest effects are again the least precise**: the widest interval on the axis belongs to the largest point estimate, **OR 4.09 (1.06–15.79)** for microplastic detection in primary lung malignancy, an interval spanning fifteenfold and resting on a handful of cases. Third, **the tightest interval belongs to one of the smallest effects** — **HR 1.0171 (1.0149–1.0192)** for all-cause death in lung cancer per $1 \mu\text{g m}^{-3}$ of non-wildfire $\text{PM}_{2.5}$ — which is the characteristic shape of this literature: population-scale precision on effects of a few percent. Fourth, several estimates are *co-pollutant* rather than PM (O_3 , NO_2 , SO_2 , CO) and are plotted because the papers report them alongside PM. Read the labels before the positions.

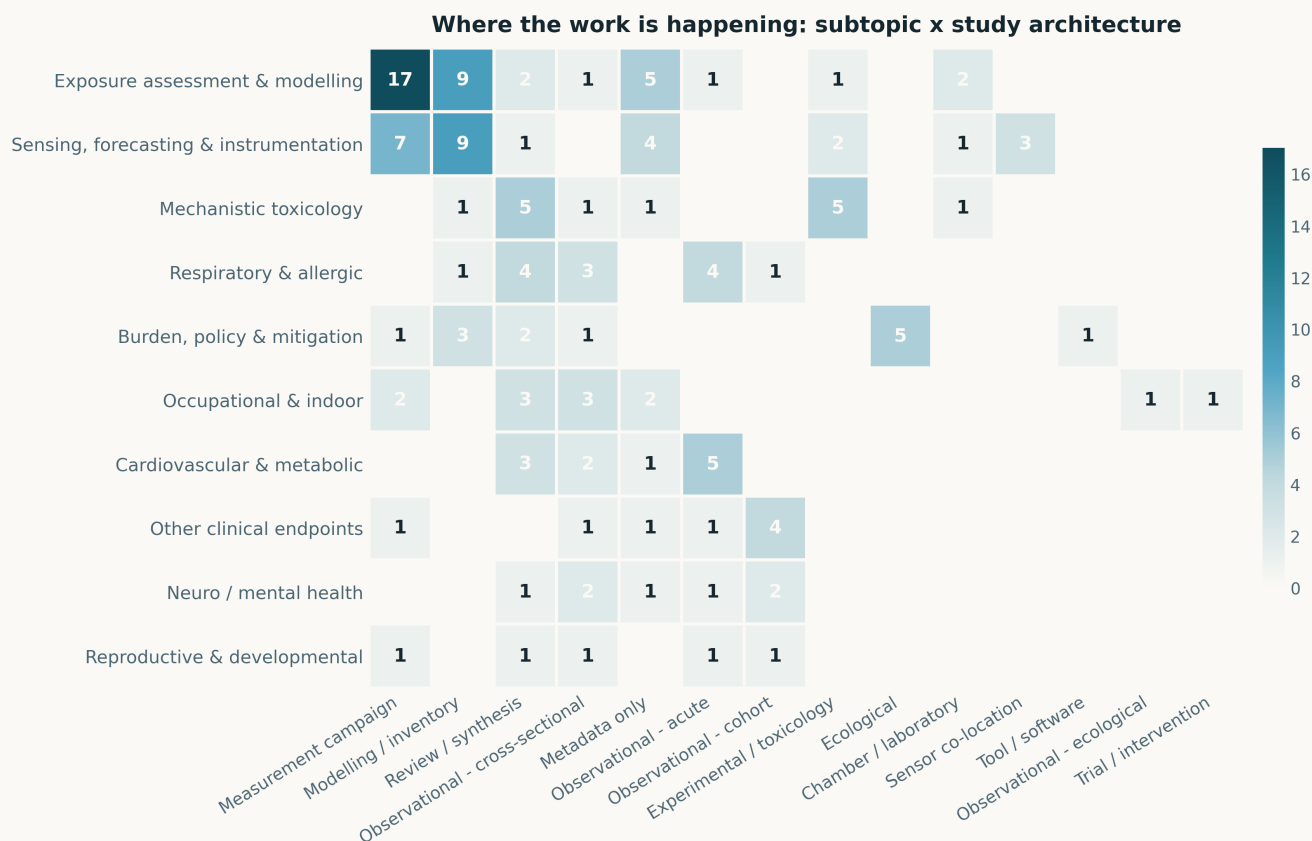


Fig. Subtopic × architecture, pooled. The densest cell in the week is *Exposure assessment & modelling* × *Measurement campaign* at 17, displacing last week's *Sensing* × *Metadata only* (which has fallen from 10 to 4). That single swap is the clearest structural evidence of the screening change. The standing finding survives an eighth week and is now measured on a larger base: of the 65 records in the *Sensing* and *Exposure* rows, 4 carry any human health endpoint, and only 3 of those observe one — the fourth is a modelled inhalation cancer risk, not a measured outcome. Instrumentation and epidemiology continue to be written by different people in different journals.

Venues, trial watch, and what the pipeline did

Venues. *Environ Sci Technol* 11 of 148 leads, followed by *Environ Pollut* (9), *Front Public Health* (8), *Atmos Chem Phys* (6) and *Atmosphere* (5), then a four-way tie at 4 between *Atmos Meas Tech*, *Aerosol Air Qual Res*, *Environ Monit Assess* and *Zhonghua Liu Xing Bing Xue Za Zhi*. Concentration is low — the leading venue holds 7% of the week — and the distribution is a harvesting artefact as much as an editorial one: titles on the Crossref by-ISSN list enter at their own publication rate, those reachable only through PubMed enter at PubMed's. The presence of *AMT* and *ACP* at all is entirely due to the by-ISSN leg, since PubMed does not index them. **Preprints supplied 2 records**, down from 8, both labelled in the register.

Recurring authors. Unchanged and still unresolvable. Surname frequency is led by Zhang, Li, Huang and Chen at two or three each, which is a naming artefact of a corpus that is 26% Chinese-affiliated and not a research-group signal. **No repeat-group cluster is claimed for this week.** Adding a first-author affiliation field remains the open schema item that would make this section mean anything; it has been on the list since 3 September and is now the oldest unaddressed gap in the series.

Trial watch — five in-window records and, for the first time, nothing new to state. ClinicalTrials.gov returned 5 PM or air-pollution registrations or updates across 13–19 September, and all five were already in *state/trials.json*, captured by the daily sweeps during the week. That is a first: on every previous weekly reconciliation the weekly sweep found records the daily leg had missed. **The daily trial leg is now ahead of the weekly sweep rather than behind it**, which is the result the daily sweeps were added to produce. The four topical records are NCT07540000 (INTERLUNG, Johns Hopkins, Bhaktapur, Nepal, $n = 800$, a PHASE4 community-health-worker respiratory intervention with a spirometric primary endpoint — still the strongest trial in the accumulated set), NCT07793123 (Robertet SA, oral antioxidant versus placebo on skin radiance, $n = 60$), NCT07713264 (VISIONS, Stanford, outdoor-worker education, $n = 50$) and NCT07817836 (Pavia, completed observational, DNA-methylation biological age, $n = 151$). The fifth, NCT02820402 (Project Viva), is a 1999 birth cohort whose air-pollution work is secondary and is deliberately not stored. **Three of the four topical records measure no air-pollution variable at all** — exposure is inferred from urban residence or occupation — which is the fourth consecutive week in which the registry's air-pollution trials are mostly trials about something else. *trials.json* now holds 12 trials across 14 windows.

What the harvesting legs did. Crossref-by-ISSN was again the workhorse, highest-yield leg on every day it ran and the only route to the instrumentation journals. Europe PMC was the second leg and the one that made the weekend survivable — Sunday 13 September returned 21 records where the two previous Sundays collapsed. PubMed ran as two separate queries on the health and instrumentation axes all week; the instrumentation query returned zero on every single day, which after seven consecutive days is no longer a sampling observation but a confirmation that PubMed structurally does not index this literature. OpenAlex returned HTTP 429 every day, unchanged since 28 July. arXiv returned HTTP 406 every day and contributed no record to any issue this week — third consecutive week of that, and the query is overdue for inspection. Consensus returned a nil every day for the fourth consecutive week, its hits consistently the right literature with Crossref created dates years outside the window. One new abstract-recovery route was tested and failed: Semantic Scholar, queried per DOI on 19 September, returned 404 on all five same-day deposits, having not yet indexed them. Together with the OpenAlex per-work failure logged on 18 September, that closes the two cheapest remaining recovery options, and leaves the deferred-retry pass as the only mechanism that can work.

Open gaps

- ▶ **The abstract-retry pass is still not built, and is now the only thing that can work.** The 10% metadata-only rate is an improvement produced by *rejecting* retro-indexed records, not by *recovering* withheld ones. Two recovery routes were tested and failed this week and last — OpenAlex per-work, and Semantic Scholar per DOI — both for the same reason: they cannot return an abstract the publisher never deposited or that no index has yet ingested. **The only route left is time:** re-query Crossref, Europe PMC and PubMed for the standing held list a week after first sight. That list now holds **18 records**.
- ▶ **Four of 65 measurement-side records carry a human health endpoint, and only three observe one.** Eighth consecutive week, now on the largest single-week base yet recorded. This remains the most durable structural finding in the series.
- ▶ **Ultrafine and size-resolved number: 3 of 148.** Unchanged as a share from last week. The one instrumentation record in the 19 September issue that would speak directly to this — a sub-10 nm unipolar aerosol charger in *J Aerosol Sci* — shipped without an abstract, which is the gap and the pipeline failure landing on the same record.
- ▶ **One interventional record in 148,** and it does not test a monitoring technology. The *Occupational & indoor* cluster now qualifies as a trend on this watch's own criteria, and **none** of it has yet been evaluated against health outcomes in occupied buildings rather than against chambers and models.
- ▶ **Pipeline defects carried forward:** no first-author affiliation field (oldest open item); the *Cancer/Oncologic* split in `ENDPOINT_CANON`; a 6-element `LIFECOURSE` array in the 2026-08-03 corpus where the schema requires 5 — outside this window but it **will** corrupt the September monthly; and `.git` lock and temp debris that the delete-restricted mount cannot remove, wanting manual cleanup on a local checkout. **Resolved this week:** `metrics.csv` has been verified unquoted-safe for thirteen consecutive issues (428 rows, 0 malformed), so the “known defect” note in the task specification is stale and should be struck.

Paper-level digest — the week by subtopic

Every record that appeared in a daily issue keeps that issue's critical summary verbatim; nothing here is re-written or re-assessed. Records surfaced without an abstract are shown as such. Ordered by subtopic, then by the date of the issue that first carried the record.

Sensing, forecasting & instrumentation

27 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (27 OF 27 RECORDS IN THIS SUBTOPIC)

AAQR 2026 Aerosol Air Qual Res

A Sensor-Driven State-Space Framework for PM_{2.5}-Informed Natural Ventilation Opportunities Under Coupled Pollution-Thermal-Humidity Constraints. Four residential communities - Kuala Lumpur, Bogor, Singapore and Palmerston (AU) - over 1 Jan to 1 Feb 2026, 724-744 hourly bins each. Crossing a sustained-PM_{2.5} screen with a T-RH envelope (24-30 °C, RH ≤ 65%) shows the two constraints are close to anti-correlated: clean-hour fractions were 14.5 / 34.3 / 30.8 / 97.4%, while standalone thermal feasibility ran 97.0 / 71.5 / 98.5 / 20.6%. The site with clean air has almost no comfortable hours and vice versa. Limits are real - one month, one season, four sites, uncalibrated AirGradient data taken as-is with no co-location against reference monitors, and fixed rather than locally derived thresholds. Still the most directly usable result of the issue for LCS network design: it turns a raw sensor feed into an actionable control signal.

doi: 10.1007/s44408-026-00167-6

Sun et al. 2026 Atmos Chem Phys

Aerosol properties during a transatlantic smoke transport event from Aeolus observations

ALADIN, a spaceborne high-spectral-resolution lidar, retrieves extinction and backscatter directly rather than through an assumed lidar ratio, and is used to follow a September 2020 western-US smoke plume to Europe. Mean extinction near 200 Mm⁻¹ across selected cross-sections; AOD fell 0.54 → 0.33 along the path; the layer rose from ~4.4 km over the US to above 6 km over Europe and arrived above cloud, with lidar ratios first falling then rising. One event constrains generality. For satellite-to-surface PM_{2.5} fusion the relevant point is blunt: this plume was invisible at ground level throughout.

doi: 10.5194/acp-26-12953-2026

Bai et al. 2026 Atmos Meas Tech

Low-level jet evolution and vertical aerosol redistribution: a Doppler wind lidar study

Two dust episodes over Hefei, East China. April 2021: the northwesterly jet core descended from 2.5 km to 0.3 km then re-ascended to 1.0 km while surface PM₁₀ peaked at 410 µg/m³, with shear above 0.04 s⁻¹ and increased spectral width at the lower jet interface and a downward shift of enhanced backscatter. March 2022: a stable jet core and persistent lower-tropospheric stability held aerosol aloft, the PM₁₀ peak arriving at 16:00 LT, ~8 h after inferred frontal onset. Two regimes — rapid downward mixing during jet descent versus delayed surface impact from a decoupled layer. Two cases is mechanism illustration, not climatology, and the inertial-oscillation attribution rests on hodograph rotation alone. The clearest statement yet of why a surface-only network can read clean for hours.

doi: 10.5194/amt-19-5951-2026

Wang et al. 2026 Environ Sci Technol

14-day PM_{2.5} and O₃ forecasting over eastern China with a CNN-Transformer

Hybrid architecture combining multilevel regional meteorological fields with emission inventories at a 14-day lead. Relative R² gains over baseline are 40% for O₃ (0.58 to 0.81) and 18% for PM_{2.5} (0.44 to 0.52); 850 hPa fields add 5% for O₃ and emissions add 9.8% for PM_{2.5}; 66–83% of O₃ episodes are captured in the north. Read honestly, PM_{2.5} skill remains poor — R² 0.52 at 14 days is not decision-grade, and the paper's own framing concedes O₃ is the stronger case. No episode-capture rate is reported for PM_{2.5}, which is the number a forecast user would act on, and gridded reanalysis truth hides station-scale error.

doi: 10.1021/acs.est.6c05923

Clean air advocacy data approaches 2026 Curr Environ Health Rep

Examining Data-Driven Approaches for Clean Air Advocacy in a Changing Climate: a Scoping Review.

Fifty studies from 41 countries and over 100 cities, sorted into five approaches: sensor-based monitoring, modelling, digital platforms and visualisation, community-centred approaches, and lived-experience assessment. 86% combined at least two. The review's operational finding for network builders is that low-cost monitoring has changed data availability far faster than it has changed governance: projects generate evidence but rarely outlive their funding cycle, and open-data formats are not standardised enough for community data to enter policy processes reliably. Scoping reviews do not grade evidence quality, so a single-site pilot and a multi-city network count equally here; no quantitative synthesis of advocacy outcomes is attempted, and publication bias toward successful campaigns is unaddressed.

doi: 10.1007/s40572-026-00563-5

Zhou et al. 2026 Proc Natl Acad Sci U S A

Air pollution after the shutdown of US embassy monitoring

Matched difference-in-differences across 46 cities with same-country controls, using real-time satellite AOD. No differential pre-trend; a prompt 24–33% rise in local pollution after US reporting stopped, largest where substitutes were poorest. Calibrating AOD to concurrent pre-shutdown embassy PM_{2.5} gives +1.9 µg/m³ on the authors' conservative translation, with alternatives spanning 1.5–16.3 — an order-of-magnitude spread that sits in the calibration step, not the DiD, and no CI accompanies the headline. The design is the contribution: monitoring treated as an intervention with a measurable atmospheric effect.

doi: 10.1073/pnas.2605467123

Gökozan 2026 Chemosensors

Machine-learning calibration of a low-cost PM_{2.5} sensor with location-specific covariates and feature engineering

Ablation design separating environmental covariates, auxiliary gases and feature engineering for one outdoor residential sensor, validated on a leakage-safe time-based split. Feature engineering helped in every configuration: raw PM_{2.5} plus engineered features raised Random Forest test R² from 0.724 to 0.847 and cut RMSE from 6.96 to 5.19 µg/m³; humidity interactions and cyclic time encodings carried the importance. One sensor at one site is the limit, and 5.19 µg/m³ is large against typical ambient means — but the operational claim, that useful correction is reachable without continuous co-location, is the one network operators need tested. Connector-sourced; outside the entry window.

doi: 10.3390/chemosensors14070154

Front Bioeng 2026 Front Bioeng Biotechnol*Bionic breathing-type airway exposure system for inhalation exposure studies.*

A 3D-printed trachea-to-G3 airway replica with millisecond breathing control (five waveforms, oral/nasal/combined routes), CO₂ exchange and multi-zone exposure interfaces. Validation recovers the expected right-dominant airflow split (right main bronchus 51.6+/-0.1% vs left 48.4+/-0.1%) and separates deposition by breathing pattern (large-loop 78.0+/-1.6% vs small-loop 61.7+/-1.3%). BEAS-2B cells showed dose-dependent cytotoxicity; hiPSC-derived lung models gave selective inflammatory responses. The weakness is that airflow-split fidelity is a low bar - it follows from the geometry, not the dynamics - and no comparison against in vivo human deposition data is offered, so anatomical realism is asserted rather than demonstrated. For sensing work it matters as a dosimetry bench: it converts a measured mass concentration into a branch-resolved deposited dose.

doi: 10.3389/fbioe.2026.1871869

Hu et al. 2026 Atmos Chem Phys*Biomass burning aerosols during the 2023 Canadian wildfires from dual-site Raman and fluorescence lidar*

Dual-site observation of transported smoke, May–September 2023, at ATOLL (France, multi-wavelength Raman with a 466 nm fluorescence channel) and GPI (Russia, five-channel broadband fluorescence at 355 nm excitation). Upper-troposphere/lower-stratosphere layers showed higher depolarisation, slightly lower lidar ratios, lower Ångström exponents and a redshifted fluorescence peak against free-troposphere layers; fluorescence magnitude and spectral shape agreed across two sites and two instrument designs, the strongest evidence yet that these signatures are transferable rather than instrument-specific. The consequential finding is negative: **near-absence of hygroscopic growth at 90–100% RH**, contradicting the standard assumption that aged biomass-burning aerosol is hygroscopic — with direct consequences for every κ -Köhler humidity correction applied to optical PM sensors during smoke episodes. One fire season, two sites; altitude correlations at r^2 0.61–0.68 that the authors concede proxy for injection height and plume origin.

doi: 10.5194/acp-26-13027-2026

Huang et al. 2026 Environ Sci Technol*Interpretable spatiotemporal coupling network for emission prediction in sludge gasification*

PISTIC-Net couples a directed process topology with per-feature optimal time lags, reaching R^2 of **91.18% for SO₂, 80.37% for NO_x and 78.09% for PM** on industrial-scale data. Under simulated abrupt sensor failure it continued tracking emission trajectories via learned process inertia, and hierarchical attention shifted from upstream thermochemical zones for SO₂ to whole-process attention for PM, flagging disturbance propagation **5–8 minutes** before emission peaks. That PM is the worst-predicted species is the informative result: particle formation runs through nucleation, condensation and agglomeration paths a topology-plus-lag representation captures least well, whereas gas-phase sulfur is nearly a mass balance. Single plant, no external validation site, and interpretability is inferred from attention maps, which are a saliency artefact rather than a causal readout.

doi: 10.1021/acs.est.6c09135

J Aerosol Sci 2026 J Aerosol Sci*High-efficiency plasma-near-passage unipolar aerosol charger for sub-10 nm particles.*

METADATA ONLY. The only instrumentation record in the window and the one whose missing abstract costs most. Crossref deposit of 19 September with no abstract; not indexed by Semantic Scholar, and the OpenAlex per-work fallback is dead for same-day deposits. No charging efficiency, particle loss fraction or flow rate can be quoted, so none is. Context, not content: sub-10 nm extrinsic charging efficiency is the binding constraint on any mobility-sizing instrument at the ultrafine end, and the published state of the art for this class sits well below unity at 5 nm, so a genuine 'high-efficiency' claim here would matter to anyone trying to put size distribution rather than mass on a distributed node. Priority pickup on the next run.

doi: 10.1016/j.jaerosci.2026.106898

Kalluri et al. 2026 — metadata only Atmos Environ*Long-term (2008–2024) black carbon over a semi-arid rain-shadow region in southern peninsular India*

The expected same-day Elsevier deposit gap: no abstract in Crossref, Europe PMC or PubMed on per-DOI retry. Carried on title because a 17-year single-site black-carbon record in a semi-arid rain-shadow setting is the kind of long baseline trend detection needs and South Asian networks mostly lack, and because absorption-coefficient records are what constrain the optical-to-mass conversion every filter- and scattering-based instrument depends on. No trend, magnitude or source attribution is asserted — the title names methods, not results. On the retry list.

doi: 10.1016/j.atmosenv.2026.122366

Krajnc Pogačnik et al. 2026 — metadata only Build Environ*Selective real-time monitoring of respiratory droplets for indoor air safety in educational environments*

Elsevier same-day deposit, abstract absent from Crossref, Europe PMC and PubMed on per-DOI retry. Retained because selective real-time discrimination of respiratory droplets against a background of ambient particles is the hard problem for any optical counter deployed indoors: a scattering sensor reports total number in a size bin and cannot by itself separate exhaled aerosol from resuspended dust, so any claimed selectivity mechanism is worth reading. Whether that selectivity is optical, chemical or inferential is not recoverable from the title, and no performance claim is asserted. On the retry list.

doi: 10.1016/j.buildev.2026.115281

LLM-compiled adaptive sampling policies 2026 Sensors*Agentic LLM Compilation of Adaptive Sampling Policies for Low-Power IoT Sensing*

Directly relevant to battery-limited PM nodes. A generator LLM proposes adaptive-sampling policies as a handful of thresholds and integer periods under 100 bytes, a deterministic evaluator scores each exactly on the recorded trace, and an optimiser LLM refines over eight rounds; no weights are fitted and the model sees only eight summary statistics, never the raw series. Benchmarked on the Beijing PM_{2.5} series against classical baselines, budget-matched random search, NSGA-II and ParEGO on the same grammar, the loop beat zero-shot generation and random search (pooled Wilcoxon $p < 1e-5$): at a 0.20 duty cycle mean RMSE 21.79 vs 25.09, at 0.30 15.98 vs 20.57. Gains narrow at the tightest budgets, bounded by signal curvature. Single dataset, offline compilation only.

doi: 10.3390/s26175519

Li et al. 2026 Atmos Meas Tech*A multi-angle and polarization-based retrieval algorithm for aerosol layer height of smoke and dust*

Optimal-estimation retrieval adding multi-angle polarimetry to conventional spectral aerosol-layer-height methods, exploiting polarisation sensitivity to the Rayleigh-versus-aerosol scattering contrast. Information-content analysis quantifies the gain as increased degrees of freedom for signal over radiance-only observation — measured, not asserted. Applied to HARP2 on PACE and validated against ATLID on EarthCARE, **RMSE 1.03 km** across all collocated samples. That is the number to hold: about a kilometre of vertical uncertainty against a 1–2 km boundary layer constrains plume altitude but does not resolve surface coupling. Validated against a single spaceborne lidar with limited collocation statistics and no ground profile network. **Why it matters here:** vertical distribution is the dominant error term in AOD-to-surface-PM_{2.5} conversion, and smoke and dust are exactly where it bites.

doi: 10.5194/amt-19-5843-2026

Ni et al. 2026 Atmos Meas Tech*An AI-based algorithm for retrieving aerosol optical depth and single scattering albedo from All-Sky Imager observations*

Machine-learning retrieval trained on co-located sun-photometer AOD and SSA at two sites, Beijing_PKU in East China and SGP in the central US. Independent validation gave $r > 0.86$ **for AOD at all wavelengths with RMSE below 0.25**, and for SSA $r = 0.74$ (**Beijing_PKU**) and **0.84 (SGP) with RMSE below 0.09**. Retrieving SSA from a consumer-grade sky camera is the striking part — absorption is normally the hardest aerosol property to retrieve and demands a calibrated multi-channel photometer. Two sites means two aerosol regimes and two instrument configurations, and an AOD RMSE of 0.25 is large wherever ambient AOD is low. As a densification strategy this is the same economics that produced PurpleAir — cheap hardware plus a trained model substituting for a reference instrument — with the same unanswered transferability question.

doi: 10.5194/amt-19-5871-2026

Organic aerosol self-diffusion MD 2026 ACS Omega*Predicting the Self-Diffusion Coefficient of Atmospheric Organic Aerosol Components via Molecular Dynamics Simulations.*

Molecular dynamics estimates of self-diffusion coefficients for atmospherically relevant organics in the particle phase, with three tiers of method: direct simulation at 298 K for mobile species, high-temperature simulation with WLF or VFT extrapolation for less mobile ones (oxomalonic, tartronic and cis-pinonic acids, dihydroxyacetone), and 420 K only for species too immobile to converge. Mobility falls in the order $-\text{COOH} > -\text{OH} > \text{C=O}$ for monofunctional compounds, and multiple or adjacent polar groups suppress it further through dense hydrogen-bond networks. Molecular weight and elemental ratios correlate with D only within chemical families. Relevant because particle viscosity governs gas-particle partitioning and water uptake, and so the hygroscopic growth that biases optical sensors. Pure simulation, no measured diffusivities for validation.

doi: 10.1021/acsomega.6c04326

Komiljon et al. 2026 Atmosphere*Non-cumulative, size-specific calibration of low-cost PM sensors under simulated drilling*

Controlled laboratory evaluation of PMS5003 and Sniffer4D under simulated concrete-slab drilling, with six correction models — linear, polynomial, Random Forest, XGBoost, ANN, Kalman — fitted independently per size fraction. Random Forest and ANN perform best, $R^2 > 0.89$ **for PM₁** and **> 0.87 for PM_{2.5}** at 50 s averaging. The premise is the useful part: reporting non-cumulative rather than cumulative bins, since cumulative output conflates fractions with different deposition behaviour and different regulatory meaning. Against that, a single drilling source in a chamber produces a mineral-dust size distribution that will not transfer to combustion aerosol, and PM₁₀ performance is not reported alongside the finer fractions. *Connector-sourced; Crossref resolves only the preprint of record.*

doi: 10.20944/preprints202604.1260.v1

Vance et al. 2026 J Occup Environ Hyg*Evaluation of the Partector 2 Pro for ultrafine particles from 3D printing*

Wind-tunnel comparison of a portable diffusion-charging monitor against SMPS and CPC during controlled printing of PLA, PETG, TPU and nylon. Number-concentration agreement is strong across all four filaments — regression slopes **0.88–1.05**, Pearson $r > 0.96$, mean deviations **−9.6% to +6.2%**, inside the $\pm 30\%$ manufacturer specification. Sizing is systematically biased low (slopes 0.82–0.90) and worst at the fine end: **−27.0% to +7.5% at D₁₀** and **−28.3% to −7.3% across D₂₅–D₇₅**. That pattern is the operational point for any distributed ultrafine network — the instrument is fit for concentration surveillance but its size distribution must not be used to infer deposition fraction. Wind tunnel only, with no humidity or co-pollutant interference testing.

doi: 10.1080/15459624.2026.2724020

Peng et al. 2026 Environ Pollut*Inhibition of condensable particulate matter by co-firing biomass and high-sulfur coal*

50 kW one-dimensional self-sustaining furnace, high-sulfur coal co-fired with corn stover or pine wood at 10% and 20% blend. Baseline condensable PM was **6.72 mg/MJ**, **83.3% inorganic**, falling to 5.09 and 3.85 mg/MJ for corn stover at 10% and 20% and to 5.36 and 4.55 for pine wood — **a 43% reduction at 20% corn stover**, traced to suppression of sulfate-associated inorganics and organic precursors. PAHs and sulfur-containing organics fell below detection at 20% blending. **Why it matters here:** condensable PM is invisible to hot-filter methods and to every optical low-cost sensor, yet can rival filterable PM in mass, so inventories built on filterable-only sampling are biased low. One bench furnace, one coal, two feedstocks, no atmospheric ageing — and condensable PM measurement is method-dependent enough that absolute values should not travel across sampling protocols.

doi: 10.1016/j.envpol.2026.129163

Gupta et al. 2026 Aerosol Air Qual Res*Evaluating and calibrating low-cost air quality sensors in contrasting aerosol regimes*

PurpleAir PA-II co-located with FEM monitors for 2–4 weeks in Raleigh, North Carolina and Delhi, India, deliberately spanning a clean and a heavily loaded regime. A model on PA PM_{2.5}, temperature and RH holds mean absolute bias within **12% hourly and 6% daily**. The headline is negative: pre-deployment calibration specific to location or region is a prerequisite for scientific use, i.e. no single transferable correction exists. Two to four weeks is short against the seasonal cycle in either city, so drift and seasonal composition change lie outside what this design can see. *Connector-sourced; outside the 14 September entry window.*

doi: 10.1007/s44408-026-00146-x

Raju et al. 2026 J Environ Sci Health A*Investigations of black carbon aerosol sources and strengths over a rural-valley location in Haryana, India*

Three years of seven-wavelength **AE33 aethalometry** (January 2020 – December 2022) at Panchgaon, a rural Aravali valley site at 285 m AMSL in northwest India. Annual means: eBC 4.06 ± 2.15 , fossil-fuel eBCff 3.10 ± 2.40 , biomass-burning eBCbb $1.42 \pm 1.19 \mu\text{g}/\text{m}^3$, with eBCff carrying 68% of absorption at 880 nm. Winter maxima (eBCff 4.19, eBCbb $2.51 \mu\text{g}/\text{m}^3$) and monsoon minima (1.90, 0.52) track boundary-layer height and wet scavenging. The aethalometer-model split rests on **assumed absorption Ångström exponents that are never reported**, and the eBCff/eBCbb ratio of 3.02 ± 3.80 has a standard deviation larger than its mean — the apportionment is unstable even where the record is not. A three-year reference-grade absorption series at a rural South Asian site is exactly the co-location anchor low-cost networks there lack.

doi: 10.1080/10934529.2026.2734478

Rural wintertime SIA metal catalysis 2026 Atmos Res*Important role of acidity-regulated metal catalysis in rural wintertime secondary inorganic aerosol formation in the North China Plain*

METADATA ONLY. Crossref deposit of 13 September with no abstract released, so only the claim in the title is verifiable here: that transition-metal-catalysed oxidation, modulated by aerosol acidity, is a significant route to secondary inorganic aerosol in the rural North China Plain in winter. If sustained, this bears on low-cost optical sensing, because the sulfate and nitrate produced by this route are strongly hygroscopic and drive the humidity-dependent bias that every nephelometric PM sensor inherits, and because rural residential coal burning supplies both the metals and the acidity. Placed on the abstract retry list; no design, sample or effect size can be assessed until the record deposits, and the record should be revisited rather than cited.

doi: 10.1016/j.atmosres.2026.109322

Wathore et al. 2026 Aerosol Air Qual Res*Environmental and temporal performance of ML calibration for a low-cost $\text{PM}_{2.5}$ sensor*

Long-term co-location in Vishakhapatnam comparing Random Forest and XGBoost against linear regression, adding temperature, relative humidity, hour-of-day and month-of-year stepwise, plus a hybrid ensemble. The best single model cuts RMSE 58% against the linear baseline and the hybrid reaches 63%, meeting US EPA low-cost $\text{PM}_{2.5}$ performance criteria. The contribution is the evaluation design rather than the models: performance is reported binned across environmental and temporal strata and across AQI categories, which exposes where a calibration degrades instead of burying it in a pooled R^2 . No absolute RMSE in $\mu\text{g m}^{-3}$ is given, and single-site training leaves transferability untested — the same limitation already flagged for the Sofia calibration-transfer work in this corpus. *Connector-sourced; outside the 15 September entry window.*

doi: 10.1007/s44408-026-00094-6

Zhang et al. 2026 J Hazard Mater*Adaptive robotic sensing for exposure-oriented mapping of hazardous airborne particle dispersion in indoor environments.*

CFD virtual sampling plus a full-scale mechanically ventilated aerosol release. The claim worth taking is the region-of-interest one: adaptive sampling cut normalised reconstruction error in the high-concentration ROI to 9.5% at 65 points against 18.3% for uniform scanning, and cut mean error at four fixed reference locations from 59.7% to 34.3% in the physical experiment. Note the honest counter-result the authors report — optimised uniform scanning beat adaptive sampling on final global NRMSE (2.5% vs 3.0%), so the gain is purely in plume localisation, not in field-wide accuracy. Single ventilation condition, single room, no repeat releases, so the 34.3% figure has no dispersion around it. Directly relevant to mobile-node deployment: it quantifies what a moving sensor buys over a fixed sparse array for exactly the transient indoor plume case fixed networks miss.

doi: 10.1016/j.jhazmat.2026.143612

Zhao et al. 2026 PLoS One*Deep learning $\text{PM}_{2.5}$ prediction integrating improved MSTL decomposition and the FATA optimizer*

FATA-GSPMSTL-CNN-LSTM: RFECV plus Pearson feature screening, a grid-searched power-spectral variant of Multiple Seasonal-Trend decomposition using Loess, the Fata Morgana metaheuristic for hyperparameter search, and a CNN-LSTM head. Evaluated separately for heating and non-heating seasons in Guangzhou and Xianyang, and the seasonal split is the sensible part given how differently those two cities are sourced. Beyond that it follows a familiar pattern: four stacked components, **no ablation isolating which one carries the gain**, and improvement measured only against author-selected baselines. No prediction intervals, so the output is a point forecast with no stated uncertainty and cannot drive threshold alerting. Two cities, no external-city test.

doi: 10.1371/journal.pone.0358271

Zhou et al. 2026 Atmos Meas Tech*Performance modeling of a small mixing-type condensation particle counter*

Validated CFD model of the sMCPC particle channel resolving turbulent flow, temperature and species transport, with Kelvin, non-continuum and latent-heat terms in the growth calculation. Raising the saturator-cooler temperature difference ΔT raises supersaturation and lowers the critical activation diameter $D_{p,50}$; raising total flow Q_g suppresses supersaturation and shifts its peak downstream; both ΔT and vapour fraction steepen the $D_{p,50}$ slope. Everything is numerical — counting efficiency is asserted through one validation case, not measured across the swept space. This is design-space guidance of the kind a portable ultrafine network node needs before it goes to the field.

doi: 10.5194/amt-19-5815-2026

Exposure assessment & modelling

38 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (38 OF 38 RECORDS IN THIS SUBTOPIC)

Kim et al. 2026 PLoS One*LASSO-based reduced-form CMAQ for ozone and $\text{PM}_{2.5}$ response to emission change*

Surrogate for CMAQ fitted by LASSO over **118 emission scenarios** spanning seven sectors and 17 provincial regions, with an adaptive logit transform to respect concentration bounds. Trained on 100 simulations and tested on 18: mean RMSE **0.1 ppb** for ozone and **0.1 $\mu\text{g m}^{-3}$** for $\text{PM}_{2.5}$ against the parent model. The sparse set of retained sector-region terms is the practical output, making the control-design problem interpretable rather than opaque. Error is quoted against CMAQ and not against observations, so every CMAQ bias passes through undiagnosed; 118 scenarios is thin for a 119-dimensional predictor space and no out-of-envelope extrapolation test is reported.

doi: 10.1371/journal.pone.0347073

Aji et al. 2026 Environ Monit Assess*Fine-mode combustion aerosol events near Indonesian coal plants from AE-filtered Himawari-8*

41,119 daily Himawari-8 observations around 13 Java coal plants, 2015–2025, with events defined as Ångström exponent > 1.2 and AOD > 0.4 (8.52% of days). Suralaya has the highest mean AOD at 1.01 with 51.4% very-high-pollution days; Pacitan the lowest at 0.46. Random forest and XGBoost return accuracy, precision, recall, F1 and AUC-ROC all **exactly 1.000** — on a target that is a deterministic threshold function of the two inputs, so the models relearn the labelling rule rather than predict anything, and the 2026–2060 Monte Carlo projection built on them is uninterpretable. Column AOD without surface validation cannot separate plant plume from boundary-layer aerosol.

doi: 10.1007/s10661-026-15804-1

Zhang et al. 2026 Environ Pollut*Surface-based temperature inversion climatology across China and its relation to PM_{2.5}*

Hourly microwave-radiometer profiles from **33 stations** over 2024–2025, validated against radiosondes at 25 sites, used to build a surface-based inversion climatology and relate it to PM_{2.5} at hotspot stations. Winter SBI occurrence **39.2%** against **12.1%** in summer, with a northwest-to-southeast gradient peaking in the Tarim Basin and Hexi Corridor. SHAP and Spearman analysis place boundary-layer variables in the top predictor tier monthly and daily, with effective mixing-layer height third among all hourly predictors in every season. Two years is short for a climatology, the network is sparse relative to the terrain it spans, and SHAP importance over correlated meteorological predictors does not establish the causal pathway it is read as establishing.

doi: 10.1016/j.envpol.2026.129172

Kim & Park 2026 Atmosphere*Loading-rate-based analysis of PM_{2.5} and precursor gases in Korean agricultural areas*

547 days of hourly data from eight agricultural sites, comparing concentration-based with loading-rate-based interpretation, where loading rate is concentration scaled by a wind-speed-derived air exchange rate. Concentration correlations among PM_{2.5}, NH₃, NO₂ and SO₂ are weak and inconsistent; loading-rate correlations are strong and uniform at $r \geq 0.816$, and the diurnal shape flips from bimodal to unimodal. The authors concede the improvement may reflect the shared wind-speed term — it almost certainly does, since dividing every series by one common quantity induces correlation by construction. Recorded as a method to treat with suspicion: any dense open-field network normalising by a common meteorological variable manufactures the same result.

doi: 10.3390/atmos17090898

Atmos Pollut Res 2026 Atmos Pollut Res*Canyons and Canopies: Effects of building configuration and green infrastructure on urban air quality.*

METADATA ONLY. Crossref deposit of 19 September with no abstract; Semantic Scholar has not indexed the DOI and the OpenAlex per-work route returns no inverted index, the same dead end logged on 18 September. Carried on title and venue alone. Nothing is asserted about the method, the domain or the direction of the green-infrastructure effect, because the record does not support it. Flagged for pickup: street-canyon geometry crossed with canopy cover is directly a siting question for a distributed network - it determines whether two nodes a block apart are measuring the same air - so this is worth re-fetching once the abstract appears.

doi: 10.1016/j.apr.2026.103205

China NOx isotopes 2026 Environ Pollut*Driving factors of atmospheric nitrogen oxides emissions and pollution in China.*

Twenty-two years of nitrate isotope data put fossil-fuel combustion at ~53% of national NO_x and non-fossil sources (biomass burning, microbial N cycling) at 47%. The paper is unusually candid about what it cannot resolve: regional source-fraction differences 'lack statistical significance', and the isotope-derived NO_x trend is a non-significant 7% decline ($p=0.19$), which the authors say precludes robust inference of emission-deposition decoupling - even though a 35% post-2011 drop in surface NO₂ against only 19% less nitrate deposition is the headline. A NO_y deposition saturation threshold of 15.0 kg N/ha/yr is proposed. Relevant to PM as nitrate is a major secondary component; the saturation claim rests on a fitted inflection with no uncertainty stated.

doi: 10.1016/j.envpol.2026.129167

Delhi N2O5 2026 Atmos Chem Phys*Enhanced nocturnal N2O5-CINO₂ chemistry in Delhi.*

Early-spring 2023 measurements in Delhi against a winter 2019 baseline. Nocturnal N₂O₅ and ClNO₂ averaged 13 and 80 ppt, roughly double 2019, and the authors attribute this to nighttime NO falling from 124+/-25 to 44+/-9 ppb, which removes the main N₂O₅ sink. Particulate chloride at 4.7 $\mu\text{g m}^{-3}$ is among the highest reported anywhere and drives efficient N₂O₅-to-ClNO₂ conversion, raising NO₃ and Cl radical production and organic oxidation. The comparison is the weak joint - different seasons (spring vs winter) and different years are confounded, so the NO-decline explanation competes with a straightforward seasonal one. Matters for PM because this chemistry feeds nitrate and secondary organic mass, the fraction low-cost optical sensors systematically mishandle under high humidity.

doi: 10.5194/acp-26-13121-2026

Dou et al. 2026 — metadata only Environ Pollut*Volatile organic compounds in Chengdu: functional zone disparities and four-year dynamics in a basin city of southwest China*

Same-day Elsevier deposit; no abstract in Crossref, Europe PMC core, Europe PMC fullTextXML or the OpenAlex per-work record. The title indicates a **four-year** ambient VOC record for Chengdu resolved by urban functional zone — the same Sichuan Basin whose humidity-driven nitrate chemistry Sun et al. model in this issue. In scope as precursor monitoring: functional-zone contrasts are the spatial scale distributed networks are built to resolve, and a four-year baseline is long enough to separate trend from meteorological variability. Carried forward for pickup.

doi: 10.1016/j.envpol.2026.129164

Environ Pollut 2026 Environ Pollut*Underlying Surface Heterogeneity Modulates Meteorological Drivers of PM_{2.5} and O₃ Pollution in the Sichuan Basin.*

METADATA ONLY. Crossref deposit of 19 September, abstract absent from the deposit and not yet indexed anywhere queried. Carried on title and venue. The claim in the title - that land-surface heterogeneity modulates how meteorology drives PM_{2.5} and O₃ - is the kind of effect that breaks a spatially stationary land-use regression, and the Sichuan Basin is the standard stagnation test case, but no magnitude, period or model class can be stated from what was deposited. Held for pickup.

doi: 10.1016/j.envpol.2026.129201

Fireman et al. 2026 Int J Public Health*Biomonitoring of urban ultrafine particle exposure and associated health effects*

99 hospital staff — 50 in the industrial Haifa Bay area, 49 Tel Aviv controls — with spirometry, FeNO, exhaled breath condensate and saliva at baseline and one year. Latent class analysis found a cluster with higher BMI, raised EBC ultrafine particle concentration and elevated salivary LDH-calcium, predominantly Haifa residents, reporting more cough, sputum and atopic complaints. The concept is worth tracking — EBC particle burden as an internal-dose biomarker would close the gap between a fixed monitor and what a person actually inhaled — but the execution does not yet support it. $n = 99$ **with LCA on a handful of variables will find clusters whether or not they exist**; no ambient UFP measurements accompany the biomarker, so the external-to-internal dose relation is never tested; and BMI loading on the same cluster makes an exposure reading hard to separate from a metabolic one.

doi: 10.3389/ijph.2026.1609855

Firozabad size-segregated lung deposition 2026 Environ Monit Assess*Seasonal variation of particulate matter deposition in human lungs through different exposure pathways in an industrial city.*

Two-season sampling in the Firozabad glass-industry belt: mean $PM_{1.0}$ 77.14, $PM_{2.5}$ 105.40 and PM_{10} 277.64 $\mu g\ m^{-3}$, winter about 1.2x summer. Modelled pulmonary-region deposition fractions were highest for infants and children (78-79% for $PM_{1.0}$ and $PM_{2.5}$), and total regional deposition was near-identical for nasal and oral breathing (16.67-16.68%) - a suspiciously exact agreement that suggests the two pathways were run with the same default ventilation parameters rather than route-specific ones. Cr exceeded the $1e-4$ carcinogenic benchmark in both seasons and both age groups; Ni and Co dominated non-carcinogenic risk. Deposition is modelled, not measured, and no personal or activity data were collected.

doi: 10.1007/s10661-026-15885-y

Frueh et al. 2026 J Expo Sci Environ Epidemiol*Source apportionment of volatile organic compounds in a Philadelphia fenceline community: results from THRIVEair*

Year-long community-designed VOC campaign in South Philadelphia run with Philly Thrive, collecting one-week integrated passive thermal-desorption samples of **38 VOCs at 20 sites**, July 2023 – June 2024, around a former oil refinery under redevelopment. Concentrations were higher in industrial than non-industrial and high- than low-traffic areas. **Benzene, the community's declared concern, was low overall** (median 1.21 $\mu g/m^3$, range 0.31–9.05), while (*m,p*)-xylenes emerged as the industrial-area pollutant of concern; toluene:benzene and xylene:ethylbenzene ratios were higher at industrial and high-traffic sites, indicating fresher emissions, and PMF apportioned sources. **One-week integration resolves none of the short plumes fenceline residents actually report**. The design is the template distributed PM networks should copy: community-set siting and a declared pollutant of concern rather than compliance checking.

doi: 10.1038/s41370-026-00976-2

Godri Pollitt 2026 — metadata only Environ Sci Technol*Beyond inhalation: air as an integrative matrix for exposome research*

ACS deposited this DOI on 18 September without an abstract and no text was recoverable from Crossref, Europe PMC or PubMed on per-DOI retry. Carried because it is the **second air-as-exposome-matrix Perspective in this same issue**, alongside Sunyer-Caldú, which makes the pair an editorial signal rather than a coincidence: air sampling is being argued into the exposome programme as an integrative matrix rather than merely an inhalation route. No argument or claim is asserted beyond the title. On the retry list.

doi: 10.1021/acs.est.6c05073

Han et al. 2026 China CDC Wkly*Environmental Mixture-Health Associations: Current Analytical Practice and Reporting Recommendations.*

97 original mixture-exposure studies in four high-impact journals, 2021-25. Median sample size 729 (IQR 396-1,992) and median 9 exposures (IQR 6-15); BKMR in 80.4%, WQS 36.1%, quantile g-computation 30.9%, with 42.3% running at least two methods. 88.7% used internal (biomarker) exposures and only 67.4% of those stated a detection-rate threshold; feature reduction was reported in 25.8%. Post-hoc variable-importance, interaction and nonlinearity testing appeared in 80-90% of papers, which with median $n=729$ and 9 correlated exposures is a multiple-comparison problem the field is not reporting. The review is descriptive and the four-journal frame is not the literature, so prevalence figures are upper bounds on best practice. The checklist matters here because PM component mixtures are the archetypal case and, as the frailty and MASLD records in this same issue show, component-level claims are being made routinely without the preprocessing documentation this paper finds missing.

doi: 10.46234/ccdcw2026.179

Kim et al. 2026 Environ Health*Residential indoor air monitoring for long-term exposure assessment in susceptible populations*

130 older adults with idiopathic pulmonary fibrosis, Seoul Metropolitan Area, 2021–23. Indoor hourly $PM_{2.5}$ averaged **14.90 $\mu g/m^3$ (SD 36.12)** against **20.44 (61.63)** outdoors, higher in winter (20.17) than summer (11.19) and by day than night, with hourly patterns that contrast rather than track outdoors. Infiltration factors from median indoor/outdoor ratios of sulfur, nickel and iron were **51–70%**. Source contrasts were large: pets 24.09 vs 13.35, humidifiers 19.19 vs 13.16, smokers 16.76 vs 11.79, cooking 15.14 vs 10.96 $\mu g/m^3$. Every standard deviation exceeds its mean, so the distribution is episodic and a seasonal mean is a poor summary; five days per season is thin support for a long-term claim and no health estimate is reported yet. The transferable piece is the non-contact protocol — instruments redeployed by postal mail, which is how an unattended network is actually serviced.

doi: 10.1186/s12940-026-01334-3

Konon et al. 2026 — metadata only Atmos Environ*Seasonal variation in volatile organic compound emissions and secondary organic aerosol formation from unconventional oil and gas development*

Same-day Elsevier deposit; nothing in Crossref, Europe PMC core, Europe PMC fullTextXML or the OpenAlex per-work record. The title indicates a seasonally resolved measurement campaign of VOC emissions from unconventional oil and gas development and the secondary organic aerosol those emissions form. In scope as a precursor study: **SOA is the fraction of basin $PM_{2.5}$ that no primary-particle emission inventory captures**, and seasonality in the precursor flux propagates straight into any exposure model built on annual-average emission factors. Carried forward for pickup.

doi: 10.1016/j.atmosenv.2026.122368

Ahmad et al. 2026 Atmosphere*Carbonaceous composition and multi-wavelength optical properties of PM in Lahore*

Six-week urban campaign in Lahore, 30 quartz filters collected, 25 usable for gravimetry and optics, **only the first 15 analysed for OC/EC**. PM_{2.5} ranged **40–417 $\mu\text{g m}^{-3}$** with a mean of **137 $\pm$ 84**. Organic carbon was 76.6% of total carbon against 23.4% elemental. Mean absorption Ångström exponent **1.17 $\pm$ 0.30**, with 56% of samples in 1.0–1.5, 32% below 1.0 and 12% above 1.5. The authors are appropriately careful that AAE intervals describe spectral behaviour rather than attribute sources. Fifteen OC/EC filters in one pre-monsoon season cannot characterise Lahore's annual cycle, and the gravimetric series has no co-located reference for traceability. Directly relevant to optical low-cost sensors, whose response depends on exactly this absorption behaviour.

doi: 10.3390/atmos17090899

Li et al. 2026 Atmosphere*A theoretical model for predicting visibility during dust events*

Physically based chain from surface wind erosion through dust emission and near-surface concentration to horizontal visibility, with an analytical concentration–visibility relation closed heuristically and a moment-balance entrainment model carrying median particle size and soil cohesion into the threshold friction velocity. Against three published datasets the relation explained $R^2 = 0.62$ and had the highest R^2 of the formulations compared under identical calibration. The appeal is economic: visibility is observed almost everywhere dust matters and PM₁₀ usually is not, so a defensible physical inversion turns an existing network into a concentration proxy. $R^2 = 0.62$ supports event detection and screening, not exposure assignment. The authors are explicit that **only the concentration–visibility link was evaluated** — the emission and entrainment stages are untested, and closing the chain needs synchronised observations that do not currently exist.

doi: 10.3390/atmos17090902

Han et al. 2026 Environ Pollut*Source apportionment of PM_{2.5} oxidative potential in metropolitan Melbourne*

Published PMF source contributions linked by multiple linear regression to oxidative potential measured on **three complementary acellular assays** (DTT, AA, DCFH); models explain **73–86%** of OP variance. Combustion dominates OP_v-DTT at 52–53% with biomass burning above traffic; OP_v-AA is driven by metal-rich sources including diesel at 20–22%; secondary nitrate contributes 28–33% and secondary sulphate 10% at the second site. The assay-dependent ranking is the result — a single-assay OP metric would nominate a different mitigation target. Reusing an earlier PMF solution propagates its rotational ambiguity, and no uncertainty is quoted on the source-resolved shares. First Southern Hemisphere estimate of its kind.

doi: 10.1016/j.envpol.2026.129159

Melssen Pfahl et al. 2026 Environ Sci Technol*Renoxification via Reduction of Inorganic Nitrate by Photoacids in Aerosols.*

Chamber study, no human subjects. Catechol and related phenols directly reduce particulate nitrate to HONO and NO under UV-visible light; nebulised gallic acid with nitrate produced 23 times more NO_y than nitrate photolysis alone. HONO formation persisted at pH 8, above the pK_a of HONO, implicating the phenols as photoacids rather than as bulk acidifiers. The limitation is the usual one for chamber aerosol chemistry - gallic acid at nebuliser concentrations is a proxy for a biomass-burning phenol pool that is far more heterogeneous, and no ambient measurement anchors the 23-fold enhancement. Relevance to distributed sensing is indirect but real: a smoke plume that renoxifies this efficiently generates ozone downwind, so a PM-only network mis-reports the health burden of the same plume hours later and tens of kilometres away.

doi: 10.1021/acs.est.6c04742

Lai et al. 2026 Atmosphere*Monte Carlo model for wind-driven PM emissions from industrial granular materials*

First quantitative evaluation of a saltation-and-sandblasting emission model for wind erosion of granular stockpiles, with the Monte Carlo impact count fixed rather than run to convergence. Applied to lead and zinc sulphide concentrates at a Sardinian plant, cutting simulated impacts from **10,000 to 500** changed the median emission by a small relative difference at a twentyfold saving, and the revised model reproduced emission magnitude, sandblasting efficiency and their friction-velocity dependence. Validation is internal and physical-consistency based — no measured emission flux is compared against, which is the central weakness. Fugitive stockpile emission is exactly the source class regulatory networks miss and fenceline arrays are asked to capture.

doi: 10.3390/atmos17090891

Montreal smoke advisory mobility 2026 Health Place*Urban mobility responses to wildfire smoke-related air quality advisories in Montreal, Canada.*

Time-series forecasting against four Montreal mobility streams - bike-share, metro, bus and recreational running - with counterfactuals controlling season, weather, holidays and events. Advisory days shifted travel from active to enclosed modes: bike-share fell, bus and metro rose, running did not move, and the effect varied by neighbourhood. This matters for exposure assessment because mode substitution changes both ventilation rate and microenvironment concentration in opposite directions, so an ambient-monitor exposure metric mis-specifies dose on exactly the days it matters most. The abstract reports no effect magnitudes or intervals, only significance, and advisory issuance is a coarse proxy for smoke concentration; running data are self-selected app users.

doi: 10.1016/j.healthplace.2026.103736

Multi-Tree 2026 J R Stat Soc Ser C Appl Stat*Multi-Tree Model for Precision Environmental Health with Longitudinally Assessed Mixture Exposure.*

A Bayesian ensemble of tree triplets - two interacting binary tree types - that estimates individualised and subgroup distributed-lag effects for mixture exposures assessed repeatedly, with built-in modifier selection over a high-dimensional candidate set. Applied to Colorado birth registry data it recovers a PM_{2.5}-birth weight association concentrated in mothers under 25 with low income. The applied finding is the weaker half: effect-modification by maternal age and income is already well established, so it functions as a sanity check rather than a discovery, and no interval is reported for the subgroup estimate in the abstract. The methodological contribution is the part worth using - most distributed-lag mixture methods return one population-average curve, and this returns a per-individual one.

doi: 10.1093/jrsssc/qlag047

ORACLES 4STAR 2026 Atmos Chem Phys*Subseasonal and spatial variability of biomass burning aerosol radiative properties observed over the Southeast Atlantic during ORACLES.*

All three ORACLES deployment months (2016–2018) of 4STAR sky-scans over the Southeast Atlantic smoke-over-stratocumulus regime. Single scattering albedo rises through the burning season - the plume brightens - and the absorption Ångström exponent analysis gives no evidence of a brown carbon contribution, so the authors attribute it to composition change rather than a change of aerosol type. Comparison against a 30-year climatology from 31 southern African AERONET sites shows the same seasonal direction over land and sea. Sampling is the caveat the paper handles well: the September SSA dip is shown to be an artefact of 2016's more southerly flight tracks. Three campaign years cannot separate a seasonal cycle from interannual variability, which they concede.

doi: 10.5194/acp-26-13083-2026

Ilies et al. 2026 Front Public Health

Urban PM₁₀ and population exposure from satellite, meteorological and socio-spatial data

MODIS-MAIAC AOD combined with four ground stations and meteorology in a regression for PM₁₀ over Oradea, Romania, then folded with population density, built form, NDVI and elevation into a composite exposure index. Cold-season maxima reach **41.8 $\mu\text{g m}^{-3}$** against an annual mean near 20.5 — above WHO, inside EU limits — and **20.1%** of the metropolitan population falls in high-predisposition areas. The AOD-PM₁₀ relation alone is weak (**R = 0.28**) and meteorology carries the model, which is the authors' own admission and the standing reason satellite-only surfaces underperform at city scale. Four stations cannot validate an intra-urban field, and the composite index is unweighted and never checked against measured personal exposure.

doi: [10.3389/fpubh.2026.1940025](https://doi.org/10.3389/fpubh.2026.1940025)

PRD limona ketone 2026 Environ Pollut

Ground-gridded observations of high concentrations of ambient limona ketone suggest overlooked limonene oxidation in subtropical megacities.

Large-scale gridded plus intensive diurnal measurements of limona ketone across the Pearl River Delta found unusually high concentrations during spring and autumn ozone episodes. A parameterised inversion using the product as a constraint implies more limonene precursor than inventories carry, with a pronounced nighttime enhancement; modelled production is OH-dominated while the loss pathway shifts from OH by day to the nitrate radical at night. The inference is indirect by construction - precursor abundance is inverted from a product, so it inherits every error in the assumed yield and loss rates, and the authors only claim the nocturnal enhancement is robust 'across plausible parameter ranges' rather than quantifying it. Episode-specific sampling limits it further. Matters as biogenic SOA is an under-constrained part of urban PM mass.

doi: [10.1016/j.envpol.2026.129177](https://doi.org/10.1016/j.envpol.2026.129177)

Paygude et al. 2026 Environ Monit Assess

Heavy metal exposure and source profiling from leaf-deposited particulate matter in Pune

27 roadside, industrial, peri-urban, highway and rural sites. Elemental order **Ca > K > Fe > Na > Mg > Mn > Zn > Cu > Cr > Pb**, consistent with crustal material, construction and resuspended road dust; enrichment factors flagged Cu, Zn and Cr at traffic- and industry-influenced sites; PCA separated construction/road-dust, vehicular, biomass-or-food and localised industrial factors. Hazard quotients ordered **Cr > Pb > Cu > Zn** with hazard indices below 1 for adults and children. The method is the contribution: leaf-deposited dust is a zero-power, zero-maintenance passive sampler that resolves metal-enriched microenvironments a sparse reference network cannot. Its limits are equally clear — deposition velocity varies with leaf morphology, species and washoff history so concentrations are not convertible to airborne mass, integration time is uncontrolled, and an **inhalation hazard quotient computed from deposited dust requires an airborne concentration that is never observed here**.

doi: [10.1007/s10661-026-15906-w](https://doi.org/10.1007/s10661-026-15906-w)

Ajibade et al. 2026 Proc Natl Acad Sci U S A

Gentrification, displacement, disadvantage and exclusivity predict urban environmental and health inequities

PCA on 2010 and 2020 ACS and Census data yields four contemporary segregation typologies for 342 Portland tracts, compared against historical HOLC zones across 104 environmental and health variables. HOLC grade predicted persistent exclusivity in the urban core but not injustice elsewhere; **persistently disadvantaged tracts, often outside HOLC boundaries entirely**, carried the highest pollution and hazard exposure and the worst health and disability outcomes. Exclusive neighbourhoods had more canopy and better preventive care *and* more air pollution — an awkward result the paper does not resolve. Wholly ecological and cross-sectional. For monitor siting the message is that a single historical redlining layer is an inadequate stratifier.

doi: [10.1073/pnas.2422760122](https://doi.org/10.1073/pnas.2422760122)

Shukla et al. 2026 Atmos Chem Phys

Organic vapors from savannah and boreal fire emissions under photochemical and dark aging

Controlled burns of understudied African savannah and European boreal fuels sampled by high-resolution PTR-MS, then aged photochemically and in the dark in a 29 m³ Teflon chamber. Total primary organic-vapour emission factors spanned **69–161 g/kg** by fuel type. Photochemical ageing depleted furanics, phenolics and oxygenated aromatics and enriched carbonyls and small O-containing species; dark low-NO_x ageing produced almost no compositional change, and clustering attributed **73% of group-level compositional variance to oxidation regime**. Chamber burns of prepared fuels do not reproduce field moisture, combustion phase or dilution, so these are upper-bound laboratory factors. Any smoke PM_{2.5} forecast assuming a fixed downwind secondary yield is contradicted here.

doi: [10.5194/acp-26-12977-2026](https://doi.org/10.5194/acp-26-12977-2026)

Sen et al. 2026 Environ Microbiol Rep

Genomic and phenotypic classification of Staphylococcus species isolated from Delhi air

Outdoor culturable bioaerosol sampling at six Delhi sites in November 2019 — the peak pollution month, pre-pandemic — using a Merck MAS Eco-100. *S. gallinarum*, *S. haemolyticus* and *S. arlettae* were recovered and characterised by whole-genome sequencing, phenotypic microarray, antibiotic-resistance profiling and biofilm formation in the presence of black carbon. Comparative genomics found the urease cluster *ureABCDEFG* in *S. arlettae* Naraina and *S. haemolyticus* Yojna Vihar 1, unusual because urease is normally reported negative in both species and may confer a niche fitness advantage. The black-carbon biofilm assay is the link to PM and the part worth following, but **no quantitative result for it is reported**. Culture-based sampling recovers a small biased fraction of the airborne microbiome, six sites in one month give no seasonal or spatial contrast, and no particle mass was measured alongside the biology, so the association with the pollution season is contextual.

doi: [10.1111/1758-2229.70411](https://doi.org/10.1111/1758-2229.70411)

Liu et al. 2026 J Environ Manage

Decadal changes in PM_{2.5} sources and secondary aerosol drivers, Shenzhen 2014–2024

Eleven years of speciated filter sampling with PMF apportionment and machine-learning attribution of secondary formation drivers. Annual mean fell **37.7 to 17.3 $\mu\text{g m}^{-3}$** . Secondary sulphate was historically dominant and declined fastest; vehicles became the largest single contributor after 2019; **secondary nitrate barely moved and is now the binding constraint**. NO_x and VOC cuts drove nitrate and SOA declines but rising atmospheric oxidation capacity offset them, while sulphate tracked oxidation and regional transport over local SO₂. Attributing a decadal trend to oxidation capacity from a correlational model at one site cannot exclude meteorological confounding, and the PMF factor series carry no uncertainty bounds.

doi: [10.1016/j.jenvman.2026.130913](https://doi.org/10.1016/j.jenvman.2026.130913)

Sichuan Basin 2026 Environ Pollut*Topographic Modulation of High-Temperature Ozone Suppression: Core-Periphery Threshold Structure in the Sichuan Basin.*

High-temperature suppression - ozone flattening or falling under extreme heat rather than rising with the usual climate penalty - appeared in 16 of 18 analysed Sichuan Basin cities. The critical suppression temperature fell from 36.8 C in the basin interior to 33.8 C at the periphery, and the decline rate above it reached $-9.65 \mu\text{g m}^{-3}/\text{C}$ in the inner zone, about 2.1x the outer. SHAP attributions put boundary-layer ventilation ahead of cloud and circulation terms. Ozone, not PM, so it sits at the edge of scope; included because the same basin ventilation structure governs PM accumulation and because the threshold-by-topography result is a caution against fitting a single regional temperature response. Five years of data and a closed basin limit transfer elsewhere.

doi: 10.1016/j.envpol.2026.129161

Groenen et al. 2026 Sci Total Environ*Characteristics and trend of wildfire activity in the Southern Hemisphere*

Two decades of MODIS active-fire detections across Southern Africa, South America and Oceania. Seasonality is strongly heterogeneous: Southern Africa shows a stable 20-week May–October season with low inter-annual variability, South America a 14-week July–October season with high variability, and **Oceania no clear season at all**, which the authors read as co-existing fire regimes. Active-fire counts are a detection proxy, not emitted mass — insensitive to smouldering, biased by overpass timing and cloud — so no PM emission inventory follows directly. Note that the abstract's stated period (2004–2024) disagrees with the title's (2003–2022); check the paper before citing either.

doi: 10.1016/j.scitotenv.2026.182240

Sun et al. 2026 Environ Sci Technol*Nonlinear nitrate formation governed by aerosol water and VOC-sensitive oxidation in a humid urban basin*

Wintertime field observations in a humid, topographically confined Chinese urban basin, combined with a box model and thermodynamic calculations. Nitrate became the dominant $\text{PM}_{2.5}$ component during pollution episodes, co-varying with NO_2 , O_x and aerosol liquid water content. Pathway-resolved modelling split formation almost evenly between daytime $\text{OH}+\text{NO}_2$ (52%) and nighttime heterogeneous N_2O_5 hydrolysis (46%), the latter rising sharply during episodes. Binning regressions located thresholds at RH 87% and ALWC $76.5 \mu\text{g}/\text{m}^3$ above which formation efficiency rose abruptly; VOCs exerted slightly stronger control than NO_x , with oxygenated VOCs and alkenes dominant. One site, one winter, one box model — the thresholds are not validated out of sample. The RH threshold is directly load-bearing for sensor correction: it is the same humidity range in which optical low-cost sensors diverge from reference monitors.

doi: 10.1021/acs.est.6c08621

Sunyer-Caldú 2026 Environ Sci Technol*Beyond blood and urine: decoding the human chemical exposome across matrices and time*

A Perspective arguing that high-resolution mass spectrometry can profile thousands of chemical features per sample while study design has not kept pace — no investigation has combined external source monitoring, multi-matrix internal biomonitoring and toxicokinetic normalisation in one longitudinal framework. Three elements are proposed: multi-matrix sampling extending beyond blood and urine to accumulation archives, developmental matrices and external media; a Unified Exposure Scale placing heterogeneous matrices on a common toxicokinetically normalised dosimetric axis; and a two-track hazard-prioritisation pipeline. No data, no validation, and the Unified Exposure Scale is circular where it matters — it needs toxicokinetic parameters that do not exist for the molecular dark matter it is meant to rank. Logged for how air monitoring is being positioned inside the Global Human Exposome Project.

doi: 10.1021/acs.est.6c06509

Yu et al. 2026 Atmos Chem Phys*Continental pollutants modulate organic nitrogen and light absorption of marine organic aerosols*

Shipboard sampling over the Yellow and Bohai Seas across three seasons. Aerosol organic nitrogen averaged $0.35 \pm 0.25 \mu\text{g N m}^{-3}$, 4–60% of total nitrogen, highest in autumn. Anthropogenic secondary pollutants — aged biomass burning and secondary nitrate — supplied 36–76% of ON, 46–83% of the water-soluble and 39–89% of the water-insoluble fraction; in spring dust accounted for 55% of ON and above 80% during a dust episode, while summer carried a marine biogenic component. Organic-aerosol light absorption rose with both ON concentration and ON/OC ratio, so absorption capability per unit carbon increased with nitrogen enrichment rather than tracking mass. Three cruises give no annual coverage and apportionment is receptor modelling on a small campaign. The sensing consequence is direct: an optical instrument calibrated on one absorption regime is misapplied when composition shifts.

doi: 10.5194/acp-26-13175-2026

Zero-filling DLM bias 2026 Int J Epidemiol*Gestational misalignment with fixed exposure windows: the potential dangers of zero-filling in distributed lag models*

Distributed lag models require equal-length exposure histories, and perinatal analyses routinely pad short gestations with zeroes to the maximum length. The authors show analytically that zero-filling manufactures an association between gestational age and late-pregnancy exposure, introducing confounding absent from the observed data. In simulation, zero-filling produced the largest bias, worst coverage and highest RMSE of all approaches tested; applied to a Colorado birth cohort it produced implausibly strong late-pregnancy effects that adjustment for gestational age attenuated. Carrying the last pre-birth value forward, using observed post-birth exposures, or truncating at 37 weeks eliminated the bias. This invalidates a common analysis choice across the critical-window literature, and the fix is trivial. Simulations are the authors' own generative model.

doi: 10.1093/ije/dyag162

Zhao et al. 2026 Environ Sci Technol*Unveiling the importance of organic hydroxy acids-related autoxidation pathway in isoprene secondary organic aerosol formation*

Year-long $\text{PM}_{2.5}$ measurements at **nine Pearl River Delta sites** quantifying six isoprene-derived highly oxygenated molecules (iHOMs). Annual mean $2.69 \pm 3.52 \text{ ng}/\text{m}^3$, higher in summer and autumn, dominated by methyltartaric acids (71%). Particle-phase tracer shares for the HOM, HMML and IEPOX pathways were 10%, 11% and 79%; box-modelled gas-phase intermediates gave 25%, 22% and 53%, so IEPOX dominates isoprene SOA in the polluted PRD with the other two non-negligible in winter. Pathway partitioning tracked the ISOPO₂ isomer split. The standard deviation exceeds the mean, so the annual average describes episodes rather than a typical day, and tracer-to-SOA scaling relies on chamber yields measured well above ambient. These are **nanogram-level organics** — the composition layer no optical network sees.

doi: 10.1021/acs.est.6c06525

Cardiovascular & metabolic**11 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (11 OF 11 RECORDS IN THIS SUBTOPIC)

COPDB Beijing 2026 J Hazard Mater

Associations between air pollution and cardiorespiratory function and metabolic indicators: New insights into mixture exposures.

112 older adults in the Beijing COPDB study across 410 follow-up visits. Single-pollutant BKMR flagged SO₂ and UFPs for fasting blood glucose, PM_{2.5} Ca²⁺ for heart rate, and O₃ and UFPs for reduced peak expiratory flow. The novel step is removing those pollutants and re-fitting: the residual mixture associations got stronger, with the gaseous group over a 14-day window tied to a 19.1% FBG increase (10.8-27.4%) and a 16.0% PEF decrease (8.7-23.3%). That is a real warning about single-pollutant screening, but the design invites it - dropping the strongest predictors and finding the rest matter more is partly a collinearity artefact, and n=112 with 410 visits is thin for pairwise interactions across a dozen-plus exposures. Component-resolved measurement is the strength.

doi: 10.1016/j.jhazmat.2026.143550

Chiang et al. 2026 J Am Heart Assoc

Ambient carbon monoxide as a delayed trigger of acute aortic dissection: a time-stratified case-crossover study in a tropical industrial metropolis

Time-stratified case-crossover of **798 emergency-department-confirmed acute aortic dissections**, 2018–2021, exposures assigned from **24 monitoring stations**, conditional logistic regression across lags 0–3 adjusted for meteorology. **No pollutant was associated on the event day.** At 2-day lag each 1 ppm CO increment carried a 2.67-fold risk (95% CI 1.207–5.900) and each IQR increment of 0.19 ppm raised risk **113% (95% CI 1.075–4.239)**; PM_{2.5}, PM₁₀, NO₂, O₃ and SO₂ showed nothing at any lag. Two problems: a 1 ppm contrast is roughly five IQRs, so the headline number extrapolates past the observed range; and with six pollutants across four lags the one surviving association is not multiplicity-corrected. The PM nulls are the relevant part here — station-assigned exposure at this network density returns nulls that a denser deployment might resolve.

doi: 10.1161/jaha.126.049936

Fang et al. 2026 J Thromb Haemost

Venous Thrombosis and Air Pollution: A Systematic Review and Meta-Analysis of 1.4 Million Adult Patients.

23 studies, >1.4 million venous thrombotic events, searched to 1 July 2025. Long-term particle exposure was associated with VTE - RR 1.0804 (95% CI 1.0015-1.1656) per 10 µg m⁻³ PM_{2.5} and 1.0369 (1.0260-1.0479) per 10 µg m⁻³ PM₁₀ - while short-term particle exposure and every gaseous metric were null. The PM_{2.5} lower bound sits at 1.0015, so the headline association is barely separated from the null and rests on a handful of long-term cohorts; PM₁₀, with the far tighter interval, is the more defensible estimate despite being the cruder metric. The projected 98,233 events averted by meeting the 5 µg m⁻³ WHO guideline is a linear no-threshold extrapolation well below the exposure range of the contributing studies and should be read as illustrative. The long-short contrast is the useful signal: chronic annual mean, not the daily excursion a dense sensor network resolves best, is what carries the thrombotic risk here.

doi: 10.1016/j.jtha.2026.09.022

Fleming & Fleming 2026 — metadata only Am J Med

Tire wear pollution and cardiovascular risk

Elsevier deposited title and authorship on 18 September with no abstract in Crossref, Europe PMC core, Europe PMC fullTextXML or the OpenAlex per-work record. The title places it as a commentary linking tire-wear particle pollution to cardiovascular risk; it asserts no finding here. Logged because the particle class matters for measurement: **non-exhaust emissions now dominate traffic PM in fleets with modern aftertreatment**, and tire wear differs from combustion PM in both size distribution and composition — a combination that mass-only optical sensors and exhaust-derived emission factors each mis-handle. Carried forward for pickup when an abstract appears.

doi: 10.1016/j.amjmed.2026.09.003

Huang et al. 2026 Zhonghua Liu Xing Bing Xue Za Zhi

Association between air pollution exposure and metabolic dysfunction-associated steatotic liver disease in adults in China.

23,806 adults, China Kadoorie Biobank third resurvey, mean age 65.3 (SD 9.1); 13,428 with MASLD of whom 2,457 had significant fibrosis. Per 1 SD: PM_{2.5} OR 1.27 (95% CI 1.05-1.53), PM₁₀ 1.77 (1.36-2.31), NO₂ 1.20 (1.08-1.33), warm-season O₃ 1.53 (1.20-1.95). One unit of the mixture score gave 40% higher MASLD odds (18-65%) and 42% higher fibrosis odds (1-100%) - the fibrosis interval touching unity makes that second estimate uninformative. PM_{2.5} was non-linear (P=0.007), flat at low concentration and steepening above it. The design is cross-sectional, so reverse causation from residential sorting by health status is unaddressed, and the reported PAFs of 25.4-66.4% are causal-language attached to a single-timepoint contrast and should not be quoted as avoidable burden.

doi: 10.3760/cma.j.cn112338-20251103-00783

Hyperuricemia low-concentration Korea 2026 Clin Rheumatol

Long-term exposure to ambient air pollution and an increased risk of hyperuricemia.

Health-screening series of 20,117 Korean adults (2015-2019), 15.3% hyperuricaemic. All five pollutants were associated with hyperuricaemia per IQR: PM_{2.5} OR 1.06 (95% CI 1.01-1.10), PM₁₀ 1.15 (1.08-1.21), NO₂ 1.07 (1.00-1.14), SO₂ 1.13 (1.07-1.19), CO 1.06 (1.01-1.11), with adjustment for eGFR so the signal is not simply renal clearance. PM₁₀ outperforming PM_{2.5} is unusual and points at a coarse-fraction or resuspended-dust component rather than combustion aerosol. Cross-sectional design with a single serum urate measurement cannot establish temporality, and CMAQ assignment at address gives no indoor or commuting contribution - the exposure error that street-scale sensing exists to reduce.

doi: 10.1007/s10067-026-08398-z

Li et al. 2026 Zhonghua Liu Xing Bing Xue Za Zhi

Joint effects of short term exposure to heat and ozone on circulatory disease mortality in Beijing-Tianjin-Hebei region and surrounding areas.

59 counties, warm seasons 2013-18, median 4 circulatory deaths per county-day. Independently, each 10 µg m⁻³ of O₃ raised circulatory mortality 0.16% (95% CI 0.02-0.31%) and a P75-to-P99 temperature rise raised it 21.43% (16.32-26.76%). The interaction is the finding: heat-associated mortality rose from 9.93% (2.52-17.89%) at 65 µg m⁻³ O₃ to 27.17% (20.48-34.23%) at 214 µg m⁻³, and the O₃ coefficient only became significant above roughly the 89th temperature percentile, peaking at 0.92% circulatory and 1.26% cerebrovascular at P99. Ozone is not PM, but the structure transfers directly: a pollutant coefficient estimated as a main effect is an average over a temperature distribution and will be wrong at both tails. Four deaths per county-day is thin, and the counties are not independent - regional synoptic episodes drive heat and O₃ together, so the product term partly encodes shared meteorology rather than biological synergy.

doi: 10.3760/cma.j.cn112338-20260118-00051

Pollutant mixtures atherosclerosis 2026 Int J Mol Sci*A Perfect Storm of Pollutants: Environmental Mixtures in the Pathogenesis of Atherosclerosis*

Structured narrative review of 37 studies on how co-exposure to particulate matter, metals, pesticides, PAHs and endocrine-disrupting chemicals converges on atherogenesis. The mechanistic through-line is oxidative stress, endothelial dysfunction, chronic inflammation and lipid dysregulation feeding monocyte recruitment, foam-cell formation, smooth-muscle remodelling and plaque instability, with epigenetic changes in DNA methylation and non-coding RNA proposed as the long-term susceptibility carrier. The authors note that additive, synergistic and antagonistic interactions cannot be predicted from single-agent data and argue for mixture-based risk assessment. As a narrative review with no protocol, search strategy or quality appraisal, the 37 studies are a selected rather than systematic sample, and the human evidence cited is predominantly associative.

doi: 10.3390/ijms27177624

Shahriary et al. 2026 Toxicol Lett*Cardiovascular and pregnancy-related health effects of ultrafine particulate matter*

Mechanistic synthesis of experimental, controlled-human-exposure and epidemiological evidence linking ultrafine PM to cardiovascular injury, with emphasis on hypertensive disorders of pregnancy. Pathways covered: pulmonary and systemic inflammation, oxidative stress, mitochondrial dysfunction, autonomic imbalance, endothelial injury and disturbed angiogenic balance, extended to maternal cardiovascular adaptation and placental vascular development. The review's honest core is its attribution problem — it states plainly that **UFPM-specific findings are hard to separate from PM_{2.5}, diesel exhaust particles, engineered nanoparticles and traffic-related pollution generally**, and that direct developmental evidence is largely animal. Uncertainties named: particle translocation, exposure metric, critical gestational windows. No pooled estimates. The exposure-metric point is the operational one — UFP is a number-concentration phenomenon and mass-based networks, including nearly all low-cost optical nodes, do not measure it.

doi: 10.1016/j.toxlet.2026.113203

Xu et al. 2026 Front Public Health*Ambient fine particulate matter and haemorrhagic stroke in Yancheng, China*

Individual-level time-stratified case-crossover on 2,312 haemorrhagic stroke admissions across four Yancheng hospitals, 2022–2024, conditional logistic regression. Strongest at lag01: each 10 $\mu\text{g m}^{-3}$ of PM_{2.5} associated with a **3.24% increase (95% CI 0.55–6.00)**, with lag0 2.50% (0.11–4.95) and lag1 2.76% (0.32–5.27). The association survives adjustment for O₃ but **loses significance once SO₂ or NO₂ enters** — which is the finding that matters, because it means the PM_{2.5} coefficient is not separable from co-emitted combustion products at this station density. City-average exposure assigned to residence leaves within-city gradients unmeasured, and a four-hospital catchment is not the full case population.

doi: 10.3389/fpubh.2026.1882030

Zhang et al. 2026 Nat Commun*Cardiopulmonary hospitalisation risks from wildfire-specific and non-wildfire PM_{2.5} in 20 US states*

Self-controlled design over 20 states, 2006–19, contrasting two-year average wildfire-specific against non-wildfire PM_{2.5}. Per 1 $\mu\text{g/m}^3$, wildfire PM_{2.5} carried **10% (heart failure) to 16% (asthma)** higher hospitalisation risk; non-wildfire PM_{2.5} **4.7% (COPD) to 8.5% (hypertension)**. Associations were stronger among minorities, metropolitan residents, the less educated and more deprived communities. The toxicity contrast at equal mass is the contribution, and it undercuts the mass-only metric every regulatory network and low-cost sensor reports. Two limits: no confidence interval is given for any of the eight headline risks, so precision cannot be judged; and wildfire-specific PM_{2.5} is itself modelled, so the toxicity difference is conditional on a smoke-attribution step whose error is not propagated.

doi: 10.1038/s41467-026-76974-7

Neuro / mental health**7 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (7 OF 7 RECORDS IN THIS SUBTOPIC)

Pearson et al. 2026 J Epidemiol Community Health*Environmental exposure and risk of suicide, attempted suicide and ideation*

Forty-five studies screened across air, noise, light and water pollution, 29 pooled. Short-term PM_{2.5} **OR 1.0139 (95% CI 1.0001–1.0279)** over 13 studies and PM₁₀ **1.0100 (1.0007–1.0195)** over 7; long-term PM_{2.5} **1.5254 (1.1351–2.0481)** over 6 and NO₂ **1.3173 (1.0641–1.6308)**. The short-term PM_{2.5} lower bound sits at 1.0001, so significance is nominal. More seriously, the **38-fold gap between short- and long-term PM_{2.5} point estimates is an artefact of pooling per-increment against high-versus-low categorical contrasts under one label**; the two are not on a common scale and must not be read as a duration–response. Newcastle-Ottawa assessment only, with no weighting for exposure misclassification.

doi: 10.1136/jech-2025-225491

CHARLS neurodegenerative composite 2026 Environ Health*The association of long-term air pollution with a composite neurodegenerative risk score among Chinese elderly: evidence from the CHARLS.*

CHARLS middle-aged and older adults, outcome a composite of grip strength, CES-D-10, MMSE and IADL at 2015 follow-up adjusted for the 2011 baseline score. Only PM_{2.5} tracked the composite: adjusted beta 0.010 (95% CI 0.008–0.016) and ordered-logit OR 1.52 (95% CI 1.01–2.31) for a higher risk grade, confined to the 60–74 stratum. PM₁₀, NO₂ and O₃ were null, and neither hs-CRP nor WBC modified the effect, which undercuts the systemic-inflammation pathway the authors set out to test. The composite is not a validated neurodegeneration instrument and mixes motor, affective and cognitive domains, so the estimate is not interpretable as dementia risk; a three-year exposure contrast is also short for a neurodegenerative endpoint.

doi: 10.1186/s12940-026-01335-2

Corrigendum 2026 Brain Behav Immun Health*Corrigendum to 'Prenatal and childhood air pollution exposure, cellular immune biomarkers, and brain connectivity in early adolescence'.*

METADATA ONLY. Corrigendum notice only; the record carries no abstract beyond the pointer to the 2024 article it corrects, and the deposited text does not state which results changed. Logged rather than summarised because the corrected paper sits squarely on a line this digest tracks - modelled early-life PM_{2.5} against immune biomarkers and adolescent brain connectivity - and because a correction to that specific inference chain is worth flagging to anyone citing the 2024 original. No effect estimate can be extracted from the notice, and the direction and magnitude of the correction remain unknown until the corrected version is read directly. Treat any citation of the 2024 estimates as provisional until then.

doi: 10.1016/j.bbimh.2026.101317

Hubei Memory & Aging 2026 Environ Pollut

Associations between compound heat exposure and cognitive impairment and the potential mediating role of plasma sphingolipid metabolism.

3,617 non-dementia adults aged 65+ in the Hubei Memory and Aging Cohort, repeatedly assessed 2016–2025. Excess cumulative temperature in heatwaves was linearly related to cognitive impairment for daytime-only, nighttime-only and compound heatwaves; temperature-humidity indices gave near-U-shaped curves with Humidexmean best fitting (OR 2.18, 95% CI 1.31–3.64). Among 77 plasma sphingolipid measures, only the ceramide C22:0 / sphingomyelin C26:0 ratio mediated, and at -14.3% - a suppression effect, not an explanatory pathway, which the abstract soft-pedals. PM_{2.5} enters only as a covariate, so this is a heat paper that adjusts for particles rather than a PM paper. Testing 77 metabolites for a single mediation signal invites a multiplicity problem that is not addressed.

doi: 10.1016/j.envpol.2026.129182

Morrel et al. 2026 Biol Psychiatry Glob Open Sci

Outdoor air pollution, perivascular space morphology, and cognition in preadolescence

Cross-sectional analysis of 6,949 ABCD participants aged 9–10, with annual-average PM_{2.5}, O₃, NO₂ and 15 PM_{2.5} components assigned to residential addresses by spatiotemporal models, perivascular-space (PVS) count and volume from T1- and T2-weighted MRI, and cognition from the NIH Toolbox. Zinc, ammonium and bromine were positively associated with regional PVS count; higher count in five regions tracked poorer cognition; calcium, Zn and NH₄⁺ were associated with poorer cognition (FDR-corrected $p < 0.01$); frontal-lobe PVS count mediated the Zn–total-cognition association ($p < 0.01$); and weighted-quantile-sum PM_{2.5} mixtures predicted higher temporal and cingulate PVS with poorer working memory and crystallised intelligence. The design cannot order exposure, PVS and cognition, and the abstract reports significance without a single effect size. The component-specific pattern is the measurement claim: speciation, not mass, carries the signal.

doi: 10.1016/j.bpsgos.2026.100790

Temporao et al. 2026 medRxiv (preprint)

Daily associations between outdoor air pollution and sleep quality

Budapest Sleep, Traits and Experiences Study — seven days per participant of daily activity and sleep-quality reports plus home EEG giving sleep efficiency, %REM, onset latency and total sleep time. Linear mixed models with participant random intercepts, adjusted for age, sex, BMI, education, weekday, season, temperature, physical activity, screen exposure and travel. Day-1 ozone predicted lower day-2 sleep efficiency and poorer perceived quality, surviving multiple-comparison correction, multi-pollutant models and sensitivity analyses for depression, insomnia and chronotype; no robust association for PM_{2.5} or NO₂. The null is confounded with exposure-model quality — city-aggregated station data give near-zero within-city contrast, and misclassification is worse for PM_{2.5} than for the more spatially homogeneous ozone. Sample size is not stated in the abstract.

doi: 10.64898/2026.09.15.26363130

Zhang et al. 2026 Front Public Health

Household solid fuel use, kitchen ventilation and depressive symptoms among older adults in China

11,697 CLHLS participants aged 65+, 2018 wave. Depressive symptoms (CES-D-10) in 26.87%, solid fuel use in 29.67%. Solid fuel carried OR 1.25 (95% CI 1.09–1.44) against clean fuel and B = 0.57 (0.33–0.82) on the continuous score — consistent but small, roughly a fifth of a scale point per unit. The charcoal subgroup showed a larger estimate on $n = 39$ exposed and the authors correctly flag it as unreliable. No interaction was found between fuel type and kitchen ventilation, which undercuts the paper's own framing that ventilation modifies the exposure. Exposure is entirely self-reported category with no PM_{2.5} measurement, so dose is unknown and misclassification is certain to be substantial; and the cross-sectional design cannot exclude reverse causation, since depressed older adults may differ in both household energy choice and reporting.

doi: 10.3389/fpubh.2026.1920233

Respiratory & allergic

13 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (13 OF 13 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026 Int J Chron Obstruct Pulmon Dis

Residential green and blue space, genetic susceptibility and incident COPD

385,502 UK Biobank participants, Cox models, greenness at 300 m and 1000 m buffers against a COPD polygenic risk score. At 1000 m: green space HR 0.87 (95% CI 0.83–0.92), natural environment HR 0.83 (0.78–0.87); blue space weak and inconsistent. No multiplicative or additive interaction with genetic risk, and the natural-environment association held in every genetic stratum — so the protective signal is not running through a susceptible subgroup. The inferential problem is what is absent: no air-pollution term is reported. Residential greenness is inversely correlated with traffic PM_{2.5} and NO₂ nearly everywhere, so an unadjusted greenness–COPD association may simply restate a pollution–COPD association.

doi: 10.2147/copd.s627458

Kim et al. 2026 Int J Environ Health Res

Heat exposure and COPD morbidity and mortality: systematic review and meta-analysis

Thirty-six observational studies pooled in a three-level random-effects model that handles non-independent estimates. Heat raised overall COPD risk (RR 1.15, 95% CI 1.09–1.21), with mortality (1.26, 1.18–1.34) well above morbidity (1.05, 1.01–1.09). Heat-wave and threshold definitions both gave RR 1.21 against 1.05 for continuous temperature metrics, and the continuous metrics were also the least heterogeneous — so definitional choice, not biology, drives much of the spread. The finding that earns this record its place here is negative: adjusting for air pollution did not substantially alter the associations, which argues against PM confounding of heat–COPD associations while saying nothing about interaction. No publication-bias or exposure-misclassification assessment is reported in the abstract.

doi: 10.1080/09603123.2026.2734444

IPF acute PM_{2.5} hospitalisation 2026 Ann Am Thorac Soc

Short-term air pollution exposure and hospitalizations in idiopathic pulmonary fibrosis: data from the IPF-PRO Registry.

Case-crossover on respiratory hospitalisations among IPF-PRO Registry patients at 46 US sites. Each 5 µg m⁻³ higher PM_{2.5} on lag day 0 raised the odds of respiratory hospitalisation 22% (95% CI 14% to 47%); lags 1–6 were null, and there was no effect modification by age, sex, Social Vulnerability Index, MUC5B genotype, telomere length or antifibrotic use. The same-day-only signal is biologically plausible for a fibrotic lung with no reserve, but the reported interval is not symmetric about the point estimate on the log scale (0.07 below, 0.19 above), which suggests a transcription or rounding error somewhere in the reporting chain and should be checked against the published table. Exposure is monitor-based, not personal, so Berkson error will attenuate. Nearest-monitor assignment is exactly the resolution a dense low-cost network would replace.

doi: 10.1093/annalsats/aaog285

Korean J Intern Med 2026 Korean J Intern Med*Health effects of air pollution on chronic respiratory diseases.*

Narrative synthesis of the epidemiological evidence tying air pollution to incidence, exacerbation and mortality across asthma, COPD, interstitial lung disease and lung cancer, then a section on non-pharmacological exposure-reduction measures at individual and societal level. No search strategy, no inclusion criteria, no quantitative pooling, so it cannot be read as evidence in its own right - the value is the intervention section, which is the part least covered by the recent primary literature. Included as a reference point for what the field currently treats as settled rather than for any new estimate. The interventions discussed (purifiers, mask use, activity rescheduling) all presuppose exposure information at a spatial resolution regulatory networks do not provide.

doi: 10.3904/kjim.2025.423

Meng et al. 2026 Front Public Health*Exposure-response thresholds for ambient PM_{2.5} and SO₂ vs COPD and small airway dysfunction, Yunnan*

11,095 residents aged 20+, multistage random cluster sampling 2021–22, with spirometry in questionnaire-screened high-risk individuals. COPD prevalence 27.2%, small airway dysfunction 53.6%. Restricted cubic splines gave significant nonlinearity for both pollutants with thresholds at 21.33 µg/m³ PM_{2.5} and 8.67 µg/m³ SO₂; above the PM_{2.5} threshold, COPD and SAD prevalence were 33.4% and 58.4%. Random-forest importance ranked air pollution above smoking and biomass fuel — a strong claim from a cross-sectional design that should be read as variable importance, not effect size. No temporality, exposure assigned without residential history, and the enriched high-risk sampling frame means these prevalences do not transfer to a general population. A 21 µg/m³ threshold sits well above the WHO 5 µg/m³ guideline and close to the regional mean, so it is plausibly a data-support artefact rather than a biological inflection.

doi: 10.3389/fpubh.2026.1874156

Zhu et al. 2026 J Asthma Allergy*Extreme cold events and paediatric asthma visits, modified by ambient CO and SO₂*

62,167 paediatric asthma outpatient visits at one tertiary hospital in Nanjing, 2019–2024, quasi-Poisson DLNM over lag 0–14. Extreme cold events show **no overall association**, but cumulative RR reaches 2.60 (95% CI 1.13–5.95) in the high-CO stratum and 3.41 (1.15–10.11) in the high-SO₂ stratum. The intervals are very wide and the strata are subsets of an already null main effect, so this is hypothesis-generating; notably PM_{2.5} and PM₁₀ did not emerge as modifiers. Single-hospital catchment with city-average exposure. Relevant here mainly as a reminder that pollutant-by-meteorology interaction terms demand far denser exposure data than one municipal station provides.

doi: 10.2147/jaa.s483101

Martinez & Nadeau 2026 Curr Allergy Asthma Rep*Early-life exposure to air pollutants and childhood asthma*

Narrative synthesis of recent cohort and mechanistic work on the prenatal and early postnatal window. Reports epigenetic modification in airway epithelium, a convergence point shared between tobacco smoke exposure and genetic asthma predisposition, new serum biomarkers of exposure, and a murine result in which asthma-predisposing information passes from exposed dam to offspring **via NK cells**. No pooled estimate, no effect size, and no stated search strategy or inclusion criteria, so study selection is unauditable and the piece cannot be weighted against the systematic reviews in this window. Useful as a pointer to the NK-cell transmission mechanism and the biomarker candidates, and for nothing quantitative.

doi: 10.1007/s11882-026-01293-2

San Juan-Reyes et al. 2026 Environ Sci Pollut Res Int*Short-term ambient air pollution and respiratory hospitalisation in children under five*

266 children under five admitted to one regional IMSS hospital in the Toluca Valley Metropolitan Zone during 2023; matched case-crossover with conditional logistic regression, primary window lag 1–7, adjusted for temperature and humidity. PM_{2.5} was associated with admission for viral pneumonia, bacterial pneumonia and asthma, CO with bronchiolitis, with heterogeneous municipality-level associations. **The abstract reports direction only — not one odds ratio, interval or unit appears** — so the effect size is unverifiable from the published summary. 266 events split across four diagnoses, five pollutants, several lag windows and three municipalities is thin, and exposure is entirely fixed-station with no personal or indoor component in a setting where indoor sources plausibly dominate a young child's dose.

doi: 10.1007/s11356-026-38237-2

Kremer et al. 2026 Nat Microbiol*Pneumococcal population structure modifies the effect of air pollution on invasive disease*

Nineteen years of South African national surveillance — 59,017 invasive pneumococcal disease cases with 4,350 sequenced isolates — in Bayesian spatiotemporal models linking environment to pathogen diversity. PM_{2.5} is positively associated with IPD, but **the lag structure varies by serotype**: only serotypes 4, 8 and 23F confer immediate risk, and high prevalence of lineage GPSC21 (serotype 19F) shifts the population-level response to immediate. Cumulative 8-week effects for moderate humidity and cold minimum temperature are +5% and +4%. No confidence interval accompanies the headline pollution association and the serotype-stratified lag models present a large multiple-comparison surface. Still the first demonstration that pathogen genotype modifies an air-quality exposure-response, and a direct argument for pairing dense PM monitoring with pathogen surveillance.

doi: 10.1038/s41564-026-02458-5

Tang et al. 2026 Zhonghua Liu Xing Bing Xue Za Zhi*Analysis on epidemiology trends of chronic obstructive pulmonary disease in residents aged 40 years and above in Hubei Province, 2014–2024.*

China COPD surveillance, Hubei, waves 2014–15, 2019–20 and 2024, complex weighted sampling. Prevalence in adults 40+ fell 12.5% (95% CI 10.1–14.8) to 10.7% (7.8–13.6) to 8.1% (5.8–10.5), but the trend test was null (t=–2.06, P=0.079) - the point estimates drop by a third and the study still cannot reject a flat line, which is what three waves and intervals this wide buy you. Occupational dust exposure, household coal heating and respiratory symptom rates all fell significantly over the same decade while protective-equipment use rose, so the particulate exposure story is consistent even where the outcome trend is not. Adjusted risks: rural 1.38 (1.06–1.81), male 3.45 (1.80–6.61), age 70+ 8.43 (4.51–15.73), current smoker 2.48 (1.75–3.52). The protective BMI gradient (obese OR 0.07) is almost certainly reverse causation from advanced disease, not a benefit of adiposity.

doi: 10.3760/cma.j.cn112338-20251218-00906

Thoracic tissue microplastics malignancy 2026 J Clin Med*Microplastics in Human Thoracic Tissues: Increased Burden in Patients with Lung Malignancy*

Fifty adults undergoing thoracic surgery for suspected malignancy; 62 particles were recovered from 50 specimens, with microplastics detected in 32 patients (64%). Fragments made up 97%, median burden one particle per specimen. Detection was more frequent in primary lung malignancy than in metastatic or benign disease (81.8% vs 50.0%, p = 0.020), and the association held after adjusting for age and smoking, OR 4.09 (95% CI 1.06–15.79). Particles in mediastinal nodes were smaller, consistent with size-dependent translocation. The interval spans an order of magnitude on 50 patients and the lower bound is barely above unity; reverse causation is unaddressed, since tumour tissue architecture may retain particles rather than particles causing tumours. Contamination control is described but blank counts are not given.

doi: 10.3390/jcm15176503

Vance et al. 2026 Scand J Public Health

Greenspace, air pollution and respiratory health outcomes: a systematic review of cohort studies

14 cohort studies to May 2026, all rated good on the Newcastle-Ottawa Scale, predominantly European. Greenspace was generally protective against air-pollution-associated respiratory mortality, and the methodological finding is the transferable one: **effects appeared more consistently through mediation than moderation**, implying greenspace acts by improving local air quality rather than by raising individual resilience. That has a direct modelling consequence — if the pathway is mediation, greenspace and PM_{2.5} should not be entered as independent covariates without care. No pooled estimate is reported, so the synthesis is narrative and magnitude is unavailable; greenspace and pollution are both strongly confounded by socioeconomic position; and near-complete European coverage limits transfer.

doi: 10.1177/14034948261481328

Xia et al. 2026 Commun Health

Short-term associations between air pollution and nocturnal cough frequency.

Nightly cough counts inferred by an AI cough detector from Sleep Cycle audio, aggregated to city-day across 32 cities in 12 countries over ~500 days. Each 10 µg m⁻³ PM_{2.5} was associated with a 2.4% rise in nocturnal cough frequency (RR 1.024, 95% CI 1.013–1.035), with a non-linear exposure-response showing increases at low concentrations and a visible excursion during a wildfire episode. Two threats dominate: the cough detector's accuracy is never validated against a clinical reference in this paper, so outcome misclassification is unquantified, and Sleep Cycle users are a self-selected smartphone-owning population whose composition varies by city. Relevant as proof that a consumer sensor stream can act as a population respiratory endpoint alongside a PM network.

doi: 10.1038/s44528-026-00030-5

Reproductive & developmental

5 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (5 OF 5 RECORDS IN THIS SUBTOPIC)

Ammons et al. 2026 Environ Res

Environmental chemical exposures and ovarian and endometrial cancers: a systematic review

145 studies screened — 54% ovarian-cancer only, 53% North American, 40% cohort — with outdoor air pollution the single most studied exposure class (27 studies, 18%). The pooled PM_{2.5} result is a clean null for ovarian cancer: **HR 1.02 (95% CI 0.91–1.13)** per 5 µg/m³ across four studies, and **0.94 (0.81–1.09)** restricted to the three with co-pollutant adjustment. Endometrial cancer was weakly positive and implausibly precise — 1.02 (0.99–1.05) overall but **1.01 (1.00–1.01)** with co-pollutant adjustment and 1.01 (1.01–1.02) for incident cases only. **A confidence interval that narrow from five studies signals near-zero between-study heterogeneity or shared cohorts rather than genuine precision**, and no *I*² is reported. Worth recording as a negative: gynaecologic cancer is one endpoint where the PM_{2.5} signal does not appear, against a literature that increasingly reports positive associations wherever it looks.

doi: 10.1016/j.envres.2026.125661

Sun et al. 2026 Environ Res

Air pollutant mixtures and endometrial receptivity in the peri-implantation window

1,349 women across 14 cities in Liaoning, with quantile g-computation and BKMR on a six-pollutant mixture against sonographic endometrial thickness and volume. A simultaneous one-quartile increase in the mixture is associated with **0.5 mm** less thickness and **0.28 ml** less volume, with CO and O₃ carrying the largest weights and PM_{2.5} and PM₁₀ not dominating. **No confidence intervals are reported for the mixture effects**, which for a BKMR result is a substantive omission. A 0.5 mm shift is at or below the reproducibility of transvaginal endometrial measurement, so the clinical meaning is unclear even if the association holds. Monthly gridded exposure with no personal or indoor component, against an implantation window measured in days.

doi: 10.1016/j.envres.2026.125682

Nelin et al. 2026 Ann Epidemiol

Associations of short-term changes in air pollution with spontaneous preterm birth: A time-stratified case-crossover analysis.

A well-powered null: 109,735 spontaneous preterm births in Michigan (2003–2020) and 5,321 in metropolitan Philadelphia (2008–2020), analysed with nested daily lags, averaged lags and DLNMs, all temperature-adjusted, with likelihood ratio tests comparing linear-lag and DLNM fits. No significant association emerged in either cohort under any specification, with point estimates near null and narrow intervals - the narrowness is what makes this informative rather than merely underpowered. The exposure is modelled, not measured, so non-differential error at the residential address would bias exactly toward this result, and the authors do not quantify that error. Case-crossover also removes only time-invariant confounding. Worth pairing with the positive short-term literature rather than reading alone.

doi: 10.1016/j.annepidem.2026.110291

Prenatal PM childhood NHL Spain 2026 Cancers

Exposure to Particulate Matter (PM) During Pregnancy and Non-Hodgkin Lymphomas in Children

Population-based case-control study across Spain (2009–2016) with 289 incident non-Hodgkin lymphoma cases from the Spanish Childhood Tumor Registry against a base population of 3,682,038 children under 15. Continuous prenatal PM_{2.5} was associated with higher NHL odds and PM₁₀ more weakly; tertile analyses put the PM_{2.5} signal in the third trimester, and PM₁₀ in the first and second trimesters among children diagnosed before age five. The abstract reports no odds ratios or intervals at all, only directions, so effect size cannot be judged from it. With 289 cases split across three tertiles and three trimesters the trimester-specific cells are small, and random-forest exposure surfaces at residential level carry error that is itself spatially structured.

doi: 10.3390/cancers18172746

Shriyan et al. 2026 Wellcome Open Res

Ambient and Indoor Air Pollution in Pregnancy and the risk of Low birth weight and Ensuing Effects in Infants (APPLE): A cohort study in Bangalore, South India.

516 pregnant women recruited before 18 weeks in Bangalore urban slums, infants followed to age one. This is a protocol, not a result - no effect estimate is reported, and it is a fourth-version revision of a design first posted years ago, so its appearance in this window is a deposit event rather than new evidence. Carried because the measurement plan is the relevant part: paired indoor and ambient PM₁₀, PM_{2.5} and CO in solid-fuel slum households is exactly the setting where a fixed regulatory monitor is furthest from personal exposure, and where a low-cost indoor node earns its keep. The planned mediation analysis of LBW on infant adiposity is ambitious for n=516 and will be underpowered for anything but a large effect.

doi: 10.12688/wellcomeopenres.14830.4

Mechanistic toxicology

14 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (14 OF 14 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026 Environ Sci Technol*A Quasi-Michaelis-Menten Mechanism Explains the Nonlinear Relationship between Transition Metals and PM_{2.5} Oxidative Potential Measured by the Dithiothreitol (DTT) Assay.*

62 ambient PM_{2.5} samples, Beijing, one year, plus laboratory standards. The authors derive a quasi-Michaelis-Menten scheme for DTT, transition metals and oxygen, and the resulting Cu and Mn equations reproduce both the saturating single-metal response and the antagonism seen in Cu-Mn mixtures. A three-term model on water-soluble Cu, Mn and WSOC - WSOC entered linearly - accounts for 98.6% of measured OP-DTT. The weakness is that 98.6% is an in-sample fit on 62 filters from one city with no held-out set and no cross-site test, so the coefficients are calibration, not validation. Still, this is the most directly useful result of the day for sensing: it says OP-DTT is a saturating, not additive, function of composition, so any attempt to infer toxicity-weighted PM from a low-cost mass or optical channel must carry a nonlinear composition term or it will systematically overstate hazard in metal-rich air.

doi: 10.1021/acs.est.6c09293

Dulong et al. 2026 Biochimie*Effects of fine particulate matter on neuroblastoma cell line: metabolic and lipid deregulations and their consequences on the onset of neurological diseases*

Narrative review built around the **SH-SY5Y** human neuroblastoma line as an in vitro neurodegeneration model. The proposed cascade runs mitochondrial damage → ER stress → oxidative stress → calcium dyshomeostasis → lipid dysregulation, converging on apoptosis and autophagy-dependent **ferroptosis**, with lipid peroxidation advanced as the emerging driver. No pooled estimate, no exposure-response and no dose metric: the review does not harmonise PM_{2.5} mass, source or exposure duration across the studies it assembles, and the in vitro doses it draws on generally sit far above ambient, so the chain is qualitative. It concedes its own in-vitro-to-human translation gap. Every mechanism it privileges is **composition-dependent** — redox-active metals, organic carbon — which a mass-only network cannot resolve.

doi: 10.1016/j.biochi.2026.09.011

Gao et al. 2026 Front Public Health*CCL22-associated immune signatures in air pollution-related atopic dermatitis*

Shared pollutant-atopic dermatitis targets enriched in cytokine and chemokine signalling, pattern-recognition receptor activity, IL-17 and Toll-like receptor pathways. A framework over 15 algorithms and 175 combinations selected plsRglm + AdaBoost, with AUC 0.963 in training and 1.000, 0.909 and 0.966 across three validation cohorts, giving a ten-gene signature (CCL22, CCL5, CSF2, F2RL1, HRH4, ICAM1, IFNG, IL10, IL17A, IL18). CCL22 was upregulated in lesional samples, localised to dendritic cells and macrophages, and confirmed in DNFB-induced murine dermatitis. **The exposure link is the weak joint: no particulate matter was measured or administered anywhere**, so air pollution enters only as a database annotation. An AUC of exactly 1.000 indicates leakage or overfitting rather than performance, and DNFB is a contact hapten, not a particle.

doi: 10.3389/fpubh.2026.1906298

Gulyaev et al. 2026 Biomed Rep*Polyphenols and inhalation-induced oxidative stress in chronic lung diseases: a systematic review*

Systematic review to January 2026 across PubMed, Cochrane and Web of Science, with RoB 2 and SYRCLE risk-of-bias assessment. Preclinical evidence was consistent — attenuated oxidative damage, suppressed inflammatory signalling and enhanced antioxidant defence, largely via Nrf2 activation and NF-κB modulation, with histopathological improvement across several exposure models. **Clinical evidence was not:** some trials improved oxidative and inflammatory markers or symptom scores, but spirometric effects were inconsistent. That preclinical-to-clinical gap is the familiar signature of antioxidant intervention research, and the authors say so. Heterogeneity precluded meta-analysis, so there is no pooled estimate; trials were small and short. Recorded as the mitigation-side counterpart to the exposure literature, not as evidence that dietary intervention substitutes for reducing exposure.

doi: 10.3892/br.2026.2194

Kou et al. 2026 Front Toxicol*Pollutants regulate changes in pathological markers of neurodegenerative diseases: a new perspective in environmental toxicology*

Review organised around core neurodegenerative pathological markers — **amyloid-β, tau, α-synuclein, TDP-43** — rather than around individual pollutants, detailing how air pollutants, metals and persistent organic pollutants modulate the generation, aggregation, propagation and toxicity of each, under a proposed “pollutants-pathogenic links-neural network failure” framework. No quantitative synthesis, no effect estimates, no exposure metric and no study-quality appraisal; it reorganises existing mechanistic claims rather than adding evidence, and **air pollutants are treated as a single undifferentiated class** alongside metals and POPs. The one useful implication for measurement is that marker-level convergence makes pollutant class, not PM mass, the exposure variable worth resolving.

doi: 10.3389/ftox.2026.1814795

Ma et al. 2026 Ecotoxicol Environ Saf*Combined PM_{2.5} and microplastics exposure exacerbates allergic asthma via OPA1/SLC7A11-mediated ferroptosis*

Ovalbumin-induced murine asthma plus OVA-stimulated MRE cells exposed to PM_{2.5} and/or microplastics. Co-exposure amplified inflammatory infiltration, goblet-cell proportion and lung-function impairment beyond either agent alone, with raised ROS and MDA and reduced T-AOC, CAT and SOD. Ferroptosis markers moved coherently — Fe²⁺ accumulation, lipid peroxidation, NCOA4/FTH1/ACSL4 up, SLC7A11/GPX4/GCLC/GSS down — alongside mitochondrial depolarisation, ATP depletion and reduced respiratory complex activity. **OPA1 knockdown and ferrostatin-1 both attenuated the injury**, two independent interventions converging on one pathway, which is the strongest part of the design. What it cannot support: no dose-response against realistic ambient concentrations, and instillation or high-dose culture routinely exceeds inhaled doses by orders of magnitude. PM_{2.5} composition is unspecified, so this is one sample, not a class.

doi: 10.1016/j.ecoenv.2026.120808

PM diabetic kidney MR transcriptomics 2026 Int J Mol Sci*Particulate Matter Exposure and Diabetic Kidney Dysfunction: Insights from Integrated Transcriptomic and Bioinformatics Analyses*

Bidirectional Mendelian randomisation put PM_{2.5}-10 causally upstream of lower eGFR, more strongly in diabetes, with no reverse causality. Transcriptomic integration yielded 168 genes shared between PM response and diabetic kidney disease, enriched for AGE-RAGE, IL-17, TNF and PI3K-Akt signalling; seven feature genes (AVPI1, DUSP1, FOSB, JUNB, PDK2, TPPP3, VIM) supported diagnostic models across training and external cohorts, with VIM localised to interstitial and fibrotic regions and TPPP3 to podocytes. The MR instrument set for coarse PM is weak and geographically confounded in UK Biobank-derived GWAS, which is the standard objection to air-pollution MR. Everything after the MR step is in silico pattern matching with no experimental validation, and the sanguinarine docking result is speculative.

doi: 10.3390/ijms27177615

PM skin AhR-Nrf2 framework 2026 Int J Mol Sci*Particulate Matter-Induced Skin Injury: A Dual-Pathway AhR-Nrf2 Framework for Epidermal Homeostasis and Therapeutic Targeting*

Review proposing that PM-induced skin barrier damage is best read as coordinated AhR and Nrf2 signalling rather than two separate pathways. Transition metals generate ROS through Fenton chemistry while PAHs are metabolised via AhR in keratinocytes, amplifying oxidative stress; Nrf2 then induces HO-1 and other antioxidant enzymes, and sustained PM exposure is argued to exhaust that compensation, degrading barrier protein expression and raising inflammatory cytokines. The dual-pathway framing is a reasonable organising device and identifies AhR-Nrf2 crosstalk as a therapeutic target, but it is a hypothesis assembled from single-pathway studies rather than a test of coordination, and the review does not distinguish which effects are PM-specific from those common to any oxidant or PAH source.

doi: 10.3390/ijms27177573

Vahdatbin et al. 2026 iScience*Is recycling sustainable: the promise and risks of scrap tire valorization*

Roughly **1.5 billion end-of-life tires** accumulate annually, and this review argues that devulcanization, pyrolysis and mechanical grinding each release hazardous species including VOCs, PAHs, heavy metals, microplastics and 6PPD-quinone. It is qualitative: no emission factors, exposure concentrations or dose-response relationships are given, so the risk claim is directional and cannot rank the three routes. Its value here is the explicit call for high-fidelity monitoring of recycling facilities — a fenceline low-cost-network application with essentially no published deployment literature. Tire wear is already a leading non-exhaust PM source; tire *reprocessing* as a second, point-source pathway is comparatively unmeasured.

doi: 10.1016/j.isci.2026.117462

Garner et al. 2026 Reprod Toxicol*Gestational nanoparticle inhalation disrupts placental zone structure in rats*

Pregnant Sprague Dawley rats exposed by whole-body inhalation to nano-TiO₂ at **9.23 ± 0.39 mg m⁻³** from gestational day 5 to 19, placentas harvested GD20 for histology. Placental weight and placental efficiency were **unchanged**, but decidua and labyrinth zone sizes fell, with compensatory increases in blood space number and maternal blood space area. The compensation is the interesting part: structural remodelling without a mass-level phenotype means placental weight is a poor screening endpoint for ultrafine exposure. The exposure is roughly three orders of magnitude above ambient ultrafine mass and engineered TiO₂ is a poor surrogate for combustion ultrafines in surface chemistry, so this bounds mechanism and not risk. No litter-level statistics are reported.

doi: 10.1016/j.reprotox.2026.109361

UPM instillation reproduction null 2026 Cells*Repeated Short-Term Intratracheal Exposure to Urban Particulate Matter Has No Apparent Adverse Effects on Female Reproductive Outcomes in Mice*

A clean null worth recording. Repeated intratracheal urban PM in female mice produced robust dose-dependent pulmonary inflammation - CD45+, F4/80+ and Ly6G+ infiltration, Il-6, Il-1b, Tnfa, Col1a1 and Tgfb1 induction - with no detectable effect on oocyte recovery, fertilisation competence, gestation length, litter size or postnatal pup development, and no measurable inflammatory transcript in ovary or uterus. Direct exposure of two-cell embryos in vitro did not impair blastocyst formation, though benzanthrane alone slightly reduced it. The result argues against a systemic-inflammation route from lung to ovary at these doses. Intratracheal bolus at 40-200 mg/kg is far above any ambient-equivalent deposited dose and bypasses the nasal and upper-airway deposition that governs real inhalation.

doi: 10.3390/cells15171522

Vera-Colón et al. 2026 — metadata only Part Fibre Toxicol*Airborne fine particulate matter and human embryonic stem cell osteogenic differentiation*

BMC deposited this DOI on 18 September with no abstract; Crossref, Europe PMC core, Europe PMC fullTextXML and the publisher landing page all returned nothing on per-DOI retry — **the first time this gap has appeared on a BMC rather than an Elsevier deposit**. Carried on title because skeletal development is an endpoint the PM literature has barely touched, and a human-embryonic-stem-cell osteogenic model would be a new mechanistic line rather than another airway or endothelial assay. No finding is asserted: neither particle source, dose, duration nor the direction of the reported disruption is verifiable from metadata. On the retry list.

doi: 10.1186/s12989-026-00701-9

Wu et al. 2026 Ecotoxicol Environ Saf*β-carotene alleviates PM_{2.5}-induced pulmonary fibrosis via suppression of the Frizzled5-mediated Wnt5a/Ca²⁺ pathway*

C57BL/6 mice given **intratracheal PM_{2.5}**, paired with BEAS-2B human bronchial epithelial cells carrying Fzd5 overexpression. Instillation produced fibrosis, oxidative stress and calcium dyshomeostasis — more mitochondria-associated ER membranes, altered IP₃R, STIM1, ORAI1 and SERCA2 — with activation of a **Wnt5a/Fzd5/Ca²⁺** axis. Dietary β-carotene reversed these changes, and Fzd5 overexpression abolished the protection, identifying Fzd5 as the mediating receptor. **No instilled dose, particle source, treatment concentration, effect size or CI** is reported for any endpoint, so potency cannot be placed against ambient exposure. Intratracheal instillation also bypasses the deposition physics that a sensor-derived exposure metric is meant to approximate.

doi: 10.1016/j.ecoenv.2026.120816

e-waste children mucosal immunity 2026 Front Immunol*iNOS/Arg-1 balance and sIgA levels are associated with intestinal mucosal immune defense in children chronically exposed to air pollutants and lead.*

238 children aged 3-6 from the Guiyu e-waste area and the Haojiang reference area. Guiyu children carried higher blood Pb and higher estimated daily doses of all five pollutants, and showed lower iNOS, higher Arg-1 and IL-4, and a lower iNOS/Arg-1 ratio - an M2-skewed macrophage profile. BKMR linked the mixture inversely to the iNOS/Arg-1 ratio, but serum sIgA did not differ between areas and was not associated with the mixture; only a non-linear iNOS-sIgA relation emerged. The headline endpoint is therefore null and the abstract leads with the secondary. Pb and combustion aerosol co-vary almost perfectly across these two sites, so the design cannot attribute anything to particles specifically; network toxicology adds no empirical weight.

doi: 10.3389/fimmu.2026.1903043

Occupational & indoor

12 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (12 OF 12 RECORDS IN THIS SUBTOPIC)

Assan et al. 2026 Health Sci Rep*Ambient particulate matter, urinary arsenic and resident health in two Ghanaian mining communities*

65 residents each from an artisanal and an incorporated mining community, all resident at least five years; PM by Modular Air Quality System, urinary arsenic by AAS, kidney function by albumin–creatinine ratio. More artisanal residents fell in the at-risk band for kidney disease ($p = 0.006$); the incorporated community showed poorer general health status. Descriptive only: $n = 130$, chi-square on dichotomised uACR with no confounder adjustment and no exposure–response, and **neither PM concentrations nor arsenic levels are numerically reported** despite being the two measured exposures. The 35 and 70 $\mu\text{g}/\text{m}^3$ figures quoted as WHO standards are the 2005-era interim targets, not the 2021 guideline values of 15 and 45 $\mu\text{g}/\text{m}^3$ for 24-hour $\text{PM}_{2.5}$ and PM_{10} — so the exceedance framing understates the gap.

doi: 10.1002/hsr2.73246

Chen et al. 2026 Indoor Air*Indoor air protection under extreme air pollution risk: evidence from a sandstorm in China*

Difference-in-differences on Baidu search intensity for air purifiers and indoor air-quality information around China's most intense sandstorm in a decade (early 2021). Affected regions show roughly **18%** higher search intensity than unaffected ones, rising to about **33%** in historically cleaner areas and about **200%** where baseline purifier sales were lowest. The gradient is the useful result: protective response scales inversely with prior exposure experience and prior equipment stock. Search volume proxies intent, not purchase, use, or any realised reduction in indoor $\text{PM}_{2.5}$ — no concentration data enter the study at all — and regional treatment assignment with no individual exposure makes the effect ecological throughout.

doi: 10.1155/ina/2274208

Environ Int 2026 Environ Int*Portable HEPA air filtration effects on particulate matter, radioactivity, and chemical constituents in occupied buildings.*

METADATA ONLY. Crossref deposit of 19 September, abstract not deposited and not recoverable from any queried index. Carried on title and venue. The title's three-endpoint framing - particulate mass, radioactivity and chemical constituents from the same intervention - is unusual and would speak to whether HEPA removal efficiency is constituent-dependent, which is the question a mass-only sensor cannot answer when validating a filtration intervention. No reduction fraction, building count or study design is asserted here. Held for pickup.

doi: 10.1016/j.envint.2026.110528

Bellacov et al. 2026 J Occup Environ Hyg*Diesel-related pollutant migration in fire stations measured with low-cost direct-reading monitors*

Seven consecutive days of minute-level monitoring at fixed engine-bay and adjacent occupied locations in three fire stations, aggregated to daily means and fitted by hierarchical mixed-effects models. Coarse PM, NO_2 , VOCs and CO_2 were significantly associated with engine activity, source distance and station identity, but the *direction* varied by pollutant — and **$\text{PM}_{2.5}$ and PM_1 were not significantly associated with any of them**. Measurable concentrations beyond the bays indicate migration or shared indoor sources; this design cannot separate them. Daily aggregation of minute data discards precisely the transient peaks that dominate diesel exposure, so the null on the fine fractions reads as a resolution artefact rather than an absence.

doi: 10.1080/15459624.2026.2722264

Fermoso et al. 2026 Atmosphere*Integrated hospital IAQ monitoring: continuous sensing with targeted chemical characterisation*

Thirty months of continuous low-cost sensor monitoring across hospital microenvironments (June 2023 – December 2025) yielding **1.79 million raw records, 1.42 million retained indoor measurements and 264,311 hourly aggregates**, paired with targeted VOC and aldehyde sampling. Distinct recurrent signatures emerged per area: CO_2 tracked occupancy and ventilation demand, while formaldehyde response was persistent and seasonal in a pattern consistent with indoor sources plus ventilation heterogeneity. The formaldehyde inference rests on a low-cost channel with known cross-sensitivity, and the campaign sampling is too sparse to validate it continuously. The **79% retained fraction** is the number to take away when sizing a long-horizon deployment.

doi: 10.3390/atmos17090895

Hu et al. 2026 Front Public Health*Analysis of carbon fiber dust exposure status and related occupational health risk factors among workers during the molding process of carbon fiber products.*

955 molding-line workers across 30 carbon-fibre firms in Shenzhen. Dust exposure was identified for 8.38% of employees and varied by post; ICMM tiering returned 3 tolerable, 36 potential, 37 high, 2 very high and 2 intolerable workstation risks. Elevated risk was associated with years of prior frontline work (OR 5.018, 95% CI 2.728–9.231) and grinding-shop employment (3.111, 1.123–8.624); protective were high prevention awareness (0.739, 0.595–0.919) and tertiary education (0.131, 0.024–0.699). The outcome is a risk score, not disease, so every OR describes exposure classification rather than health. The education coefficient spans an order of magnitude on 955 workers and should be treated as noise. Carbon fibre is a rising industrial aerosol with fibre morphology that mass-based instruments handle badly - worth tracking as a sensing gap.

doi: 10.3389/fpubh.2026.1926373

Ikeda et al. 2026 — metadata only Build Environ*Characterizing layout-dependent breathing-zone contaminant concentration in office spaces with mixing ventilation using an occupant-to-exhaust distance metric*

Same-day Elsevier deposit; nothing in Crossref, Europe PMC core, Europe PMC fullTextXML or the OpenAlex per-work record. The title indicates a modelling study deriving breathing-zone contaminant concentration in mixing-ventilated offices from an **occupant-to-exhaust distance metric**. In scope for sensor siting specifically: it is a claim that indoor concentration is a predictable function of geometry, which is the assumption every single-point indoor monitor makes implicitly and almost never tests. Carried forward for pickup.

doi: 10.1016/j.buildenv.2026.115282

Yang et al. 2026 Environ Sci Technol*Organic contaminants of emerging concern in indoor air: global distribution and hazard burden*

Seven OCEC classes spanning over 130 compounds synthesised across residences, offices, schools and commercial spaces. Microenvironments separate cleanly — phthalate alternatives dominate residential air to $2.08 \times 10^4 \text{ ng}/\text{m}^3$, PFAS schools to $230 \text{ ng}/\text{m}^3$, organophosphate esters offices to $159 \text{ ng}/\text{m}^3$ — and the US, Germany, Sweden and China rank highest on inhalation hazard, which the authors say plainly may reflect monitoring intensity rather than contamination. **Microplastics were excluded from the hazard index because units and toxicity metrics are irreconcilable**: a candid admission that the particulate fraction of indoor emerging contaminants cannot currently be placed on a common scale with the rest.

doi: 10.1021/acs.est.6c06332

Khan et al. 2026 Front Public Health

Cement industry emissions and respiratory health: global systematic review and meta-analysis, 2000–2025

Eighteen observational studies, PROSPERO CRD420251268913. Random-effects pooling gave standardised mean differences of **−0.71 for FEV₁, −0.56 for FVC and −0.57 for FEV₁/FVC** in exposed groups; chronic cough was significantly more common while other symptoms were elevated but not significant. Occupational exposure produced greater FEV₁ impairment than residential — the expected dose ordering and the most reassuring internal check available. Reporting is the weakness: **no confidence interval, no I² and no prediction interval accompanies any pooled estimate**, so neither precision nor heterogeneity can be assessed, and SMDs pooled across studies using different spirometry reference equations with no common dust metric cannot be read as a dose–response.

doi: 10.3389/fpubh.2026.1887654

Ma et al. 2026 Front Public Health

Association between combined occupational exposure to coal dust and noise and high-frequency hearing loss among workers: a cross-sectional study.

1,187 workers examined in Baiyin City in 2023: 276 noise-only, 911 noise plus coal dust. High-frequency hearing loss prevalence 35.5% vs 46.8% ($P < 0.001$), with the characteristic 4 kHz notch and little speech-frequency involvement; adjusted OR for combined versus noise-alone exposure 1.620 (95% CI 1.198–2.190, $P = 0.002$). The inference is weak in a specific way: exposure is a job label, not a measured dust concentration or a measured noise dose, so the combined group may simply work louder jobs and the OR then encodes noise intensity rather than a dust interaction. A cross-sectional design in a workforce also selects out the already-deafened. It is carried because dust-noise co-exposure is common and because it is a reminder that a sensor network characterising an industrial site is measuring only one of two correlated hazards.

doi: 10.3389/fpubh.2026.1929586

Tam et al. 2026 Curr Environ Health Rep

Toxic air contaminant emissions from residential and commercial fossil gas appliances: a systematic review

25 studies, the first systematic synthesis of emission factors across gas appliance types. Nonzero emissions documented for **at least 41 toxic air contaminants** via both leakage and combustion pathways. Only formaldehyde and benzene had enough data for quantitative characterisation, and their emission factors **spanned two orders of magnitude** across studies and operating conditions — a spread that makes any single-value inventory estimate indefensible. Coverage is badly skewed toward cooking stoves; heaters, water heaters and dryers are barely studied despite longer duty cycles. The operationally interesting claim, correctly stated as emerging rather than established, is that **methane and CO measurements may proxy for leakage- and combustion-related TAC emissions** — which is exactly the kind of cheap-channel inference target a low-cost indoor node can carry.

doi: 10.1007/s40572-026-00562-6

Yoshizaki et al. 2026 J Am Coll Surg

Differences in the effectiveness of smoke evacuator types for reducing surgical smoke: a randomized controlled trial

Randomised, double-blind trial (UMIN000047033), 64 laparotomy patients at the University of Tokyo Hospital, Feb 2022 to May 2023, allocated 1:1 to electrocautery-incorporated evacuation at 8 mm tip distance ($n = 22$), 18 mm ($n = 23$) or hose-type ($n = 21$). The 8 mm system cut particle counts **96–97% across 0.3–5.0 μm** against hose-type; **the 18 mm arm did not reach significance on particles**, so effectiveness collapses over a 10 mm change in capture distance — the practically important finding. Median acetaldehyde was 1.90, 2.70 and 7.90 $\mu\text{g}/\text{m}^3$ in the 8 mm, 18 mm and hose arms, both electrocautery arms below hose, with formaldehyde behaving similarly. Endpoints are area concentrations rather than personal breathing-zone exposure and no health outcome was measured; single centre, one procedure type. The distance sensitivity is the transferable lesson for any point-source capture or near-field sensing geometry.

doi: 10.1097/xcs.0000000000001943

Burden, policy & mitigation

13 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (13 OF 13 RECORDS IN THIS SUBTOPIC)

Atmospheric infrastructures Czech activism 2026 Social Health Illn

Cities as Atmospheric Infrastructures: An Analysis of Activist Responses to Respiratory Illness in the Czech Republic

Proposes ‘atmospheric infrastructures’ to hold together two things usually analysed apart: the city as a physical producer of the air residents breathe, and the city as a social infrastructure that makes collective response possible. Grounded in Czech air-pollution activism, where organisers frame respiratory health as a citizenship claim, fight for the health of the city rather than of individuals, and treat civic education as equal in weight to protest. The framing is directly useful to anyone deploying community sensor networks, because it names why data alone rarely mobilises and why the collective rather than individual framing carries. This is conceptual and interpretive work from a single national context; no exposure or health measurements are involved and no generalisability is claimed.

doi: 10.1111/1467-9566.70260

Bao et al. 2026 J Environ Manage

Distribution-sensitive assessment of air-quality and health benefits from clean-heating pilots in northern China

88 clean-heating pilot cities, 2013–23, evaluated with machine-learning weather normalisation and the augmented synthetic control method against city-specific no-policy counterfactuals. The distributional result is the point: the policy cut the heating-season **mean only 3.01 $\mu\text{g}/\text{m}^3$** but **P95 by 5.76, P99 by 5.96 and exceedance days by 7.92** per season — the tail moved roughly twice as far as the mean, which a mean-only evaluation misses entirely. Health impact assessment gave **12,454 fewer attributable all-cause deaths (95% CI 8,564–16,222)**, with 88.9% from the 2017–18 rollout cohorts and 56.2% from the top decile of cities; high-pollution days were 27.8% of improvement days but **48.8% of the mortality difference**. Benefit is therefore concentrated in both time and space. ASCM counterfactuals depend on donor-pool comparability during a decade of nationwide decline, and the mortality figure is a transfer through an exposure-response function, not an observed death count.

doi: 10.1016/j.jenvman.2026.130960

Campania review 2026 Monaldi Arch Chest Dis

Air pollution and chronic respiratory diseases: a narrative review including a case study in Campania (Italy).

Narrative review of mechanism (oxidative stress, chronic inflammation, epithelial damage, airway remodelling) attached to a Campania case study, one of the Italian regions most persistently above national average. Between 2015 and 2024 national PM₁₀ and PM_{2.5} fell slightly while Campania stayed high, with measurable respiratory impact and marked intra-regional disparity that the authors tie to socioeconomic gradient. Closing sections argue for machine learning and satellite-based forecasting to improve monitoring, and for environmental justice targeting. No systematic search, no pooled estimate, and the regional numbers are cited rather than re-analysed. Included for the inequality framing and because the monitoring-gap argument it makes is the case for dense low-cost networks in exactly such regions.

doi: 10.4081/monaldi.2026.4216

Liu et al. 2026 Environ Monit Assess*Spatiotemporal evolution of PM_{2.5} and its health and economic effects in Central China*

Henan, Hubei and Hunan analysed with global Moran's I, centroid migration and BenMAP, with Henan carried forward for valuation over 2017–2023. Spatial clustering is strong and stable (**Moran's I 0.524–0.738, all $p < 0.001$**), high-value clusters sit in Henan and around Wuhan and Changsha, and the concentration centroid moved only **2.86–24.37 km** per year. The authors explicitly *reject* a persistent southwestward migration because most displacements fall below their noise threshold — a welcome negative result in a literature prone to over-reading centroid drift. Attributable premature deaths declined with a late rebound, but a rising value of a statistical life meant monetised benefit grew more slowly than health benefit. BenMAP output inherits its CRF and baseline-rate assumptions entirely.

doi: 10.1007/s10661-026-15911-z

Covenant of Mayors co-benefit tool 2026 Open Res Eur*A tool to maximize air quality co-benefits while planning local climate actions: application to the Covenant of Mayors initiative.*

JRC tool translating residential-building and transport mitigation actions in city climate action plans into emission changes for five pollutants, applied to 531 plans from 494 European and neighbouring cities. Most greenhouse-gas actions are synergistic with air quality, but two trade-offs are quantified as risks: transport fuel switching can raise ammonia, and promoting solid biomass for residential heating raises primary PM_{2.5} – the failure mode that has repeatedly turned European heating decarbonisation into a local PM problem. The tool maps actions to emission factors and does not run dispersion, so nothing here is a concentration or an exposure; city-reported plans are also self-declared and not verified against realised emissions.

doi: 10.12688/openreseurope.22919.1

EOTBL cancer 2026 Adv Sci*Global Burden and Temporal Trends of Early-Onset Tracheal, Bronchial, and Lung Cancer: An Analysis of GLOBOCAN 2022 and GBD 2023 Data.*

Early-onset tracheal, bronchial and lung cancer caused 137,705 incident cases and 72,646 deaths globally in 2022, with the highest incidence in Eastern Asia and in High-HDI settings, but disproportionate mortality in Medium- and Low-HDI regions – the familiar detection-versus-survival split. The PM-relevant claim is that ambient particulate pollution emerges as a leading attributable risk in rapidly developing economies such as China, where active smoking dominates in countries like Russia. That attribution is GBD machinery, not new observation: it inherits GBD's exposure surfaces and its concentration-response assumptions wholesale, and no uncertainty interval accompanies the risk-factor ranking here. Registry coverage is also weakest in exactly the Low-HDI regions carrying the mortality burden.

doi: 10.1002/advsc.77778

Habit-locked household emissions 2026 Environ Sci Ecotechnol*Habit-locked-in carbon-air-pollutant emissions dominate China's household footprints.*

Consumption-based accounting decomposed into emissions locked in by behavioural inertia versus those driven by demand growth. Roughly 84% of China's household-driven CO₂, PM_{2.5}, NO_x and SO₂ is habit-locked, rising to about 97% for food and 84–86% for residential expenditure; per-capita habit-locked emissions peak in East and North China, with 32–53% originating in Nei Mongol and Shandong production systems. Under business-as-usual, habit-locked CO₂ peaks near 2030 while habit-locked air pollutants already decline. Pairing carbon-neutrality policy with attenuated consumption inertia yields a further 1285 Mt CO₂, 218 kt PM_{2.5}, 729 kt SO₂ and 126 kt NO_x of abatement. The locked-in share depends entirely on how inertia is parameterised, which the abstract does not expose.

doi: 10.1016/j.j.ese.2026.100757

Ju et al. 2026 Lancet Planet Health*Premature mortality and economic losses attributable to landscape fire-related PM_{2.5} in ten countries and territories*

1,366 communities, ten countries and territories, **47,844,851 deaths**, 2000–19, with community-specific associations estimated by instrumental-variable models and pooled by random-effects meta-analysis. Each 1 µg/m³ of 2-day mean landscape-fire PM_{2.5} raised mortality **0.60% (95% CI 0.19–1.00)** and mortality-related economic losses **0.59% (0.21–0.97)**, yielding **906,869 deaths and US\$994.5 billion** attributable (attributable fractions 1.90% and 1.64%). The disproportion is the headline: fire PM_{2.5} is **19.0% of ambient mass but 30.6% of PM_{2.5}-attributable deaths**. IV estimates exceeded conventional association estimates by a **median 56.0% (2.2–111.3)**, so classical designs are biased toward the null here. The weak point is that the instrument is not named in the abstract and exclusion-restriction validity — against co-emitted gases and weather — carries the causal claim entirely; value-of-statistical-life transfer across ten economies adds uncertainty that is not propagated into the intervals.

doi: 10.1016/j.j.lanplh.2026.101506

Mairghani et al. 2026 Rev Environ Health*An umbrella review of public health interventions to mitigate air pollution in high-income countries*

Eight systematic reviews spanning city and transport policy, household exposure reduction and emission-reduction modelling. Transport interventions gave consistent cardiovascular benefit and fewer traffic injuries; portable air cleaners cut indoor PM_{2.5} with modest blood-pressure improvement; modelling projected a **median ~1.5% mortality reduction** under large-scale mitigation. The ordering is the useful output — structural policy beat individual and household measures, which were complementary rather than substitutive. Two limits constrain its use. **Equity was rarely assessed and inconsistent where it was**, so the review cannot say who benefits; and the 1.5% figure is modelled rather than measured, mixing evidence tiers inside one summary. Eight reviews, all high-income, so none of it transfers to the settings carrying the largest burden.

doi: 10.1515/reveh-2026-0061

Massimi et al. 2026 Environ Sci Technol*Urban green walls under traffic stress: elemental composition of PM deposition and plant response*

A nine-species green wall at a high-traffic urban site, followed for a year. PCA on the insoluble fraction of Al, Cr, Cs, Fe, Mn, Mo, Rb, Sb, Sn, Ti and Zn identified *Photinia × fraseri* as the highest and most stable retainer of non-exhaust traffic particles and *Pteris vittata* as the lowest, with SEM-EDS showing **up to 100-fold higher retention** in the former. The methodological finding travels furthest: **only insoluble elements traced deposition reliably**, because soluble and whole-leaf fractions are dominated by endogenous pools of nutrients and of tracers such as Cs and Rb, so conventional whole-leaf digestion overestimates deposition. *Photinia* showed marked seasonal physiological and transcriptional modulation; *Pteris* did not. One wall at one site with no paired ambient PM mass measurement, so species are ranked against each other but no removal flux is quantified and no air-quality benefit can be inferred.

doi: 10.1021/acs.est.6c10086

Jiang et al. 2026 Med Gas Res*Ambient particulate matter pollution and stroke burden*

GBD 2021 stroke deaths and DALYs attributable to ambient particulate matter, decomposed by region, sex and SDI over 1990–2021 with EAPC trends and concentration and slope indices of inequality. Absolute deaths rose **107%** and DALYs **101%** while age-standardised rates fell — the familiar decomposition of ageing and population growth against declining risk. Middle-SDI locations carry **46% of the global burden** and exceed high-SDI on both age-standardised measures. The concentration index suggests improving relative inequality while the slope index worsened against 1990, and the paper does not reconcile the two. Attributable burden is a modelled quantity resting on the exposure surface and the chosen CRF; this is not independent epidemiological evidence.

doi: 10.4103/mgr.MEDGASRES-D-25-00188

US net-zero multi-model 2026 Environ Sci Technol*A Multi-Model Assessment of the Air Quality and Health Impacts of U.S. Energy Decarbonization.*

Three independent US energy system models (US-REGEN, GCAM-USA, REPEAT) run to 2050 agree on direction but not magnitude: PM_{2.5}-attributable mortality rates fall 28–73% against 2017 under net-zero, despite population growth and ageing. The spread is the result. It tracks how each model allocates CO₂ mitigation across sectors, with carbon dioxide removal deployment in 2050 varying threefold and setting how much residual fossil combustion - and therefore precursor emission - remains. A sensitivity analysis shows plausible increases in non-energy emissions, wildfire in particular, can offset the energy-sector benefit outright. Concentration-response functions are shared across the three models, so the reported spread understates total uncertainty: model structure varies, epidemiological input does not.

doi: 10.1021/acs.est.6c02914

Western Pacific stroke burden 2026 JACC Asia*Stroke Burden, Trends, and Inequalities in the Western Pacific Region, 1990–2021: A Population-Based Study.*

GBD 2021 across 31 Western Pacific countries and areas: 34.56 million prevalent stroke cases (95% UI 32.11–37.21) and 64.53 million DALYs in 2021. Age-standardised DALY rates roughly halved from 4,155.11 to 2,326.92 per 100,000 while absolute DALYs rose, driven by ageing (219.51%) and population growth (70.27%); lower-SDI countries carried the larger absolute inequality. Environmental exposures accounted for 43.78% of stroke DALYs with ambient particulate matter the single largest contributor at 22.22%. That fraction is a modelled comparative-risk-assessment output resting on exposure-response functions imported from cohorts largely outside the region, and GBD uncertainty intervals do not propagate the uncertainty in those functions, so 22.22% should be read as an order of magnitude rather than an estimate.

doi: 10.1016/j.jacasi.2026.07.034

Other clinical endpoints**8 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (8 OF 8 RECORDS IN THIS SUBTOPIC)

Aerosol Air Qual Res 2026 Aerosol Air Qual Res*Association Between Air Pollution and Psoriasis-Related Outcomes: A Systematic Review and Meta-analysis.*

METADATA ONLY. Springer deposit of 19 September without an abstract; not yet indexed by Semantic Scholar. Carried on title and venue only. No pooled estimate, study count, heterogeneity statistic or pollutant metric is asserted, because none was deposited. Noted as the fourth systematic review or meta-analysis in a single day's window - alongside the venous thrombosis, climate-burden and ischaemic stroke reviews screened out or admitted above - which is itself a signal worth watching in the weekly rollup. Held for pickup.

doi: 10.1007/s44408-026-00166-7

Alhalwani et al. 2025 Saudi J Ophthalmol*Correlations of CO, SO₂, PM₁₀, NO₂, and O₃ with dry eye disease in Saudi Arabia.*

492 dry eye disease diagnoses at King Abdulaziz Medical City, Jeddah, 2017–18, against three monitoring stations. Adjusted ORs reached significance only for SO₂ (1.182) and CO (1.162); PM₁₀, the particulate metric, was not associated. The reported SO₂ interval is 0.908–1.111 with P=0.021, which is internally inconsistent - an interval spanning unity cannot accompany that P value - and that error is in the published abstract, so the estimate should not be used. Three monitors for a city of four million gives essentially no exposure contrast within Jeddah, and a clinic case series with no non-DED comparison group cannot support an odds ratio at all as described. Carried at tier C as a record of the ocular-surface literature only; the finding that PM₁₀ was null here should not be read as evidence against a particulate effect.

doi: 10.4103/sjopt.sjopt_286_24

Huang et al. 2026 Medicine (Baltimore)*Secondhand smoke exposure in never-smokers: A cross-sectional and prospective cohort study of chronic kidney disease prevalence and all-cause mortality in affected patients.*

13,276 never-smoking adults, NHANES 1999–2020, with 1,630 CKD cases followed for mortality. Moderate SHS exposure was associated with CKD prevalence (OR 1.37, 95% CI 1.07–1.76). Among those with CKD, moderate and heavy exposure carried all-cause mortality HRs of 2.03 (1.23–3.36) and 1.80 (1.11–2.93); each doubling of cotinine gave HR 1.04 (1.01–1.08). The non-monotonic ordering - moderate exceeding heavy for mortality, and heavy not associated with prevalence - undercuts a dose-response reading and points to residual confounding by household socioeconomic position. Cotinine is a good internal dose marker for the nicotine fraction and a poor one for the particulate fraction, which is the exposure of interest here; the two dissociate across ventilation conditions in a way an indoor PM sensor would resolve and a serum assay cannot.

doi: 10.1097/md.00000000000050614

Kim et al. 2026 J Am Geriatr Soc*Extreme Temperatures, Elevated PM_{2.5}, and Acute Care Events Among Dual-Eligible Nursing Facility Residents.*

24,784,081 facility-days from 24,760 nursing facilities in 47 states plus DC, Medicare-Medicaid linked claims 2017–20. On elevated-PM_{2.5} days ED visits rose 0.020 (95% CI 0.013–0.027) and hospitalisations 0.017 (0.012–0.022) per 1,000 resident-days; extreme heat gave 0.022 (0.003–0.041) and 0.025 (0.011–0.039), extreme cold 0.014 (0.001–0.027) hospitalisations. Effect sizes are tiny in absolute terms and the intervals for heat and cold nearly touch zero; what carries the paper is 24.8 million facility-days and a fixed-effects design that removes time-invariant facility confounding. The exposure is the weak link - an outdoor county-level value dichotomised at 9 µg m⁻³, applied to a population that is almost entirely indoors, with no measurement of the HVAC infiltration the authors themselves identify as the mechanism. The policy asymmetry is the point: nursing facilities are regulated on indoor temperature and not at all on indoor PM.

doi: 10.1111/jgs.70720

Liu et al. 2026 Zhonghua Liu Xing Bing Xue Za Zhi

Association of exposure to fine particulate matter and its components with frailty in middle-aged and older adults in Beijing-Tianjin-Hebei region and surrounding areas.

1,875 adults across ten cities, mean age 64.95 (SD 13.42), 49.4% men, deficit-accumulation frailty index mean 0.15 (SD 0.09). An IQR rise in 90-day $PM_{2.5}$ was associated with a 1.30% higher frailty index (95% CI 0.29-2.31%), strongest at lag0-90. Components: OC 3.63% (1.17-6.09), Cu 1.50% (0.76-2.24), Ca 1.35% (0.51-2.20), EC 0.83% (0.45-1.21) per IQR. Two problems. Exposure is assigned from the nearest superstation, so within-city gradient is entirely unmodelled and the IQR is a between-city contrast wearing individual-level clothing - precisely the misassignment a dense network exists to fix. And the components are collinear combustion tracers; the OC-over-EC ordering is not evidence that organic carbon is the causal fraction. No subgroup modification was detected on any of six strata, which at $n=1,875$ is uninformative rather than reassuring.

doi: 10.3760/cma.j.cn112338-20251220-00911

Mtenga et al. 2026 Cancer Epidemiol

Ambient $PM_{2.5}$ exposure and cancer incidence in Nigerian adults

First prospective sub-Saharan African cohort on this question. 5,621 adults, median follow-up 4.7 years, 147 incident cancers confirmed histopathologically, mean $PM_{2.5}$ $50.4 \mu g m^{-3}$ — tenfold the WHO guideline. Each $10 \mu g m^{-3}$ gives HR 1.12 (95% CI 1.06–1.19) for all-cancer incidence and the top quartile 1.82 (1.29–2.57); mediation analysis attributes 41.3% to oxidative stress and inflammation. Site-specific signals for lung, breast and haematopoietic cancer rest on very small strata at 147 total events, and biomarkers measured once in a subsample cannot establish the temporal ordering mediation requires. The 1 km satellite–land-use regression exposure surface has no ground reference network to anchor it.

doi: 10.1016/j.canep.2026.103232

Odiko et al. 2026 J Expo Sci Environ Epidemiol

Environmental lead exposure and clinical infectious respiratory disease in a sample of urban and disadvantaged children.

13,221 children aged 14 and under, Temple University Health System, Philadelphia, 2010-20; 28% above the CDC 3.5 ug/dL action level. Streptococcal pharyngitis was associated with BLC ≥ 3.5 ug/dL (aOR 2.74, 95% CI 1.92-3.90) and, almost identically, with BLC 1-3.4 ug/dL (2.45, 1.75-3.44); frequent strep showed the same pattern. Croup was null. That the two lead strata give the same estimate is the problem - a genuine dose-response should separate them, and the flatness suggests both strata are marking the same underlying thing, most plausibly housing age and healthcare contact frequency, neither of which is adjusted for beyond insurance type. Outcome ascertainment is ICD-coded within one health system, so a child who is screened for lead is also a child who is seen more often. The 'no safe level' conclusion is not supported by a contrast this flat.

doi: 10.1038/s41370-026-00978-0

Su et al. 2026 Lancet Oncol

Wildfire-related $PM_{2.5}$ and overall survival of older lung cancer patients

414,016 SEER-Medicare patients aged 65+, 1,211,490 person-years, 2008–2019, time-varying Cox on 3-year moving averages of zip-code wildfire and non-wildfire $PM_{2.5}$. Per $1 \mu g m^{-3}$: HR 1.0782 (95% CI 1.0641–1.0924) for wildfire against 1.0171 (1.0149–1.0192) for non-wildfire, a 4.6-fold difference in per-unit potency. The exposure contrast is the weakness — median wildfire $PM_{2.5}$ was $0.28 \mu g m^{-3}$ (IQR 0.21–0.39) against 8.11 for non-wildfire, so the per-unit hazard is extrapolated far outside the observed range, and the per-SD estimates (1.0229 versus 1.0384) reverse the ordering. Zip-code assignment with no residential history. Species-resolved rather than mass-only monitoring is the direct implication.

doi: 10.1016/s1470-2045(26)00295-0

Pooled corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
45 studies, 29 pooled, 2010-2024	Environmental exposure and risk of death by suicide, attempted suicide and suicide ideation: a systematic review and meta-analysis.	<i>J Epidemiol Community Health</i>	Systematic review (PRISMA)	Ambient $PM_{2.5}$ and PM_{10} , short-term and long-term contrasts pooled together	Global / multi-region
CHARLS neurodegenerative composite	The association of long-term air pollution with a composite neurodegenerative risk score among Chinese elderly: evidence from the CHARLS.	<i>Environ Health</i>	Prospective cohort (CHARLS)	2012-2015 mean $PM_{2.5}$, PM_{10} , NO ₂ and O ₃ gridded to residence	China
Corrigendum	Corrigendum to 'Prenatal and childhood air pollution exposure, cellular immune biomarkers, and brain connectivity in early adolescence'.	<i>Brain Behav Immun Health</i>	Metadata only (no abstract)	Corrects 10.1016/j.bbimh.2024.100799; (metadata original exposure was modelled prenatal and childhood $PM_{2.5}$)	Not stated only)
Hubei Memory & Aging	Associations between compound heat exposure and cognitive impairment and the potential mediating role of plasma sphingolipid metabolism.	<i>Environ Pollut</i>	Repeated-measures cohort + metabolite mediation	ERA5-Land hourly temperature, CMDSC humidity; $PM_{2.5}$ and O ₃ from the Tracking Air Pollution dataset as covariates	China
Morrel et al.	Outdoor Air Pollution, Perivascular Space Morphology, and Cognition in Preadolescence.	<i>Biol Psychiatry Glob Open Sci</i>	Cross-sectional cohort imaging + weighted quantile sum regression	Annual-average $PM_{2.5}$, O ₃ , NO ₂ and 15 $PM_{2.5}$ components, spatiotemporal models at residence	USA

First author	Focus	Journal	Design	Metric	Region
Temporao et al.	Daily associations between outdoor air pollution and sleep quality.	<i>medRxiv (preprint)</i>	Panel study with ambulatory EEG	Daytime outdoor PM _{2.5} , NO ₂ and O ₃ aggregated from city monitoring stations; same-day and previous-day	Hungary
Zhang et al.	Associations between household solid fuel use, kitchen ventilation, and depressive symptoms among older adults in China: a national cross-sectional study.	<i>Front Public Health</i>	Cross-sectional (multistage)	Self-reported household fuel type and kitchen ventilation, no measured concentration	China
Cardiovascular & metabolic					
COPDB Beijing	Associations between air pollution and cardiorespiratory function and metabolic indicators: New insights into mixture exposures.	<i>J Hazard Mater</i>	Panel study	Continuously measured PM _{2.5} , UFP and accumulation-mode counts, SO ₂ /NO/NO ₂ /O ₃ , and PM _{2.5} carbon, metal and water-soluble ion components	China
Chiang et al.	Ambient Carbon Monoxide as a Delayed Trigger of Acute Aortic Dissection: A Time-Stratified Case-Crossover Study in a Tropical Industrial Metropolis.	<i>J Am Heart Assoc</i>	Case-crossover	PM _{2.5} , PM ₁₀ , CO, NO ₂ , O ₃ , SO ₂ from 24 fixed stations; lags 0-3 d, per 1 ppm and per IQR	Taiwan
Fang et al.	Venous Thrombosis and Air Pollution: A Systematic Review and Meta-Analysis of 1.4 Million Adult Patients.	<i>J Thromb Haemost</i>	Systematic review + random-effects meta-analysis	Long-term (>=1 y average) and short-term PM _{2.5} , PM ₁₀ and gaseous pollutants, per 10 µg/m ³	Global
Fleming & Fleming	Tire Wear Pollution and Cardiovascular Risk.	<i>Am J Med</i>	Metadata only (no abstract)	Tire-wear (non-exhaust) particles; no exposure metric retrievable	Global
Huang et al.	Association between air pollution exposure and metabolic dysfunction-associated steatotic liver disease in adults in China.	<i>Zhonghua Liu Xing Bing Xue Za Zhi</i>	Observational - cross-sectional	PM _{2.5} , PM ₁₀ , NO ₂ and warm-season O ₃ , per 1 SD; weighted environmental risk score for the mixture	China
Hyperuricemia low-concentration Korea	Long-term exposure to ambient air pollution and an increased risk of hyperuricemia.	<i>Clin Rheumatol</i>	Retrospective cross-sectional (health screening)	Annual mean PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ and CO from CMAQ at residential address	South Korea
Li et al.	Joint effects of short term exposure to heat and ozone on circulatory disease mortality in Beijing-Tianjin-Hebei region and surrounding areas.	<i>Zhonghua Liu Xing Bing Xue Za Zhi</i>	Time-series	Daily maximum 8-h O ₃ (mean 138.4 +/- 57.5 µg/m ³) and daily mean temperature, warm season	China
Pollutant mixtures atherosclerosis	A Perfect Storm of Pollutants: Environmental Mixtures in the Pathogenesis of Atherosclerosis	<i>Int J Mol Sci</i>	Narrative review	PM in mixtures with toxic metals, pesticides, PAHs and endocrine disruptors	Global / multi-region
Shahriary et al.	The Silent Intruders: Cardiovascular and Pregnancy-Related Health Effects of Ultrafine Particulate Matter.	<i>Toxicol Lett</i>	Mechanistic review	Ultrafine particulate matter, number-based rather than mass-based metrics	Global / multi-region
Yancheng, 2,312 admissions 2022-2024	Association between ambient fine particulate matter and hemorrhagic stroke in Yancheng, China.	<i>Front Public Health</i>	Time-stratified case-crossover	Station-average daily PM _{2.5} , lag 0 to lag 01	China

First author	Focus	Journal	Design	Metric	Region
Zhang et al.	Cardiopulmonary hospitalization risks from wildfire-specific and non-wildfire PM _{2.5} in 20 US states.	<i>Nat Commun</i>	Self-controlled design	Two-year average wildfire-specific vs non-wildfire PM _{2.5} , modelled and source-separated, per 1 µg/m ³	USA
Reproductive & developmental					
Ammons et al.	Epidemiologic research on environmental chemical exposures and ovarian and endometrial cancers: a systematic review.	<i>Environ Res</i>	Systematic review + random-effects meta-analysis	Long-term ambient PM _{2.5} , per 5 µg/m ³ , alongside other environmental chemical exposures	North America
Liaoning fertility clinics, n=1,349	Ambient air pollutant mixtures in relation to sonographic endometrial receptivity parameters during the peri-implantation window: a clinic-based observational study.	<i>Environ Res</i>	Observational exposure study	Monthly mean PM _{2.5} , PM ₁₀ , CO, NO ₂ , SO ₂ , O ₃ from a high-resolution reanalysis grid	China
Nelin et al.	Associations of short-term changes in air pollution with spontaneous preterm birth: A time-stratified case-crossover analysis.	<i>Ann Epidemiol</i>	Time-stratified case-crossover + DLNM	Daily high-resolution modelled PM _{2.5} , week before delivery, three lag parameterisations	United States
Prenatal PM childhood NHL Spain	Exposure to Particulate Matter (PM) During Pregnancy and Non-Hodgkin Lymphomas in Children	<i>Cancers</i>	Case-control (registry)	Prenatal PM _{2.5} and PM ₁₀ from a random-forest exposure model, by trimester	Spain
Shriyan et al.	Ambient and Indoor Air Pollution in Pregnancy and the risk of Low birth weight and Ensuing Effects in Infants (APPLE): A cohort study in Bangalore, South India.	<i>Wellcome Open Res</i>	Study protocol	Paired indoor and ambient PM ₁₀ , PM _{2.5} and CO during pregnancy, nested in the MAASTHI cohort	India
Respiratory & allergic					
Chen et al.	Residential Green and Blue Space, the Natural Environment, Genetic Susceptibility, and Incident COPD.	<i>Int J Chron Obstruct Pulmon Dis</i>	Prospective cohort	Residential green space, blue space and natural environment at 300 m and 1000 m buffers; no PM term reported	UK
Heat-COPD meta, 36 studies	Heat exposure and its impact on COPD morbidity and mortality: a systematic review and meta-analysis.	<i>Int J Environ Health Res</i>	Systematic review and three-level random-effects meta-analysis	Co-exposure only: air-pollution adjustment tested as a moderator	Global / multi-region
IPF acute PM _{2.5} hospitalisation	Short-term air pollution exposure and hospitalizations in idiopathic pulmonary fibrosis: data from the IPF-PRO Registry.	<i>Ann Am Thorac Soc</i>	Time-stratified case-crossover	Daily PM _{2.5} at the nearest regulatory monitor, lags 0-6	United States
Korean J Intern Med	Health effects of air pollution on chronic respiratory diseases.	<i>Korean J Intern Med</i>	Narrative review	Outdoor and indoor air pollutants, life-course framing; no single exposure metric	Not stated
Meng et al.	Exposure-response relationship between ambient PM _{2.5} and SO ₂ with COPD and small airway dysfunction prevalence among individuals at high risk for COPD in Yunnan Province, China: a threshold analysis.	<i>Front Public Health</i>	Cross-sectional (multistage)	Ambient annual PM _{2.5} and SO ₂ , residential assignment	China
Nanjing, 62,167 paediatric asthma visits	Extreme Cold Events and Pediatric Asthma Outpatient Visits: The Modifying Role of Ambient Carbon Monoxide and Sulfur Dioxide.	<i>J Asthma Allergy</i>	Time-series	Daily PM _{2.5} and PM ₁₀ as stratifying co-pollutants, not the exposure of interest	China
Narrative review, prenatal and early postnatal window	Early Life Exposure to Air Pollutants and Childhood Asthma.	<i>Curr Allergy Asthma Rep</i>	Narrative review	Prenatal and early postnatal ambient air pollutants, reviewed qualitatively	Global / multi-region

First author	Focus	Journal	Design	Metric	Region
San Juan-Reyes et al.	Short-term effects of ambient air pollution exposure on respiratory hospitalizations among children in the State of Mexico.	<i>Environ Sci Pollut Res Int</i>	Case-crossover	Daily PM _{2.5} , PM ₁₀ , CO, NO ₂ and O ₃ from fixed monitoring stations; primary window lag 1-7 cumulative	Mexico
South Africa, 59,017 IPD cases, 4,350 genomes	Pneumococcal population structure influences the effects of air pollution on invasive disease risk in South Africa.	<i>Nat Microbiol</i>	Bayesian spatiotemporal model	Ambient PM _{2.5} , lagged to 8 weeks, linked to serotype and lineage	Sub-Saharan Africa
Tang et al.	Analysis on epidemiology trends of chronic obstructive pulmonary disease in residents aged 40 years and above in Hubei Province, 2014-2024.	<i>Zhonghua Liu Xing Bing Xue Za Zhi</i>	Cross-sectional survey	Self-reported occupational dust and harmful-gas exposure; household coal heating fuel use	China
Thoracic tissue microplastics malignancy	Microplastics in Human Thoracic Tissues: Increased Burden in Patients with Lung Malignancy	<i>J Clin Med</i>	Prospective tissue biomonitoring series	Tissue microplastic particle counts by confocal micro-Raman spectroscopy	Greece
Vance et al.	Greenspace, air pollution and respiratory health outcomes: a systematic review of cohort studies.	<i>Scand J Public Health</i>	Systematic review	Residential greenspace (NDVI and similar) with co-assigned ambient air pollution	Europe
Xia et al.	Short-term associations between air pollution and nocturnal cough frequency.	<i>Commun Health</i>	Multi-city time-series + random-effects meta-analysis	City-day ambient PM _{2.5} , daily mean	Global / multi-region
Mechanistic toxicology					
Chen et al.	A Quasi-Michaelis-Menten Mechanism Explains the Nonlinear Relationship between Transition Metals and PM _{2.5} Oxidative Potential Measured by the Dithiothreitol (DTT) Assay.	<i>Environ Sci Technol</i>	Laboratory metrology	OP-DTT on laboratory single-metal and Cu-Mn mixture solutions and on 62 ambient PM _{2.5} filter samples	China
Dulong et al.	Effects of fine particulate matter on neuroblastoma cell line: metabolic and lipid deregulations and their consequences on the onset of neurological diseases.	<i>Biochimie</i>	Narrative review	PM _{2.5} , in vitro dosing; no harmonised mass concentration or duration reported	Global
Gao et al.	Integrative transcriptomic analysis identifies CCL22-associated immune signatures in air pollution-related atopic dermatitis.	<i>Front Public Health</i>	Integrative transcriptomics + murine validation	No measured PM; pollutant-associated gene sets taken from a toxicogenomic database and intersected with atopic-dermatitis transcriptomes	Not stated
Gulyaev et al.	Association between polyphenols and inhalation-induced oxidative stress in chronic lung diseases: A systematic review.	<i>Biomed Rep</i>	Systematic review	Inhalation-induced oxidative stress across in vitro, animal and clinical exposure models	Global / multi-region
Kou et al.	Pollutants regulate changes in pathological markers of neurodegenerative diseases: a new perspective in environmental toxicology.	<i>Front Toxicol</i>	Narrative review	Air pollutants treated as one class alongside metals and POPs; no exposure metric	Global
Ma et al.	Combined PM _{2.5} and microplastics exposure exacerbates allergic asthma via OPA1/SLC7A11-mediated airway epithelial ferroptosis.	<i>Ecotoxicol Environ Saf</i>	Animal + in vitro	PM _{2.5} and microplastic co-exposure, ovalbumin-sensitised murine model and MRE cells	China
PM diabetic kidney MR transcriptomics	Particulate Matter Exposure and Diabetic Kidney Dysfunction: Insights from Integrated Transcriptomic and Bioinformatics Analyses	<i>Int J Mol Sci</i>	Mendelian randomisation + TWAS	PM _{2.5} -10 coarse-fraction genetic instruments	Global / multi-region

First author	Focus	Journal	Design	Metric	Region
PM skin AhR-Nrf2 framework	Particulate Matter-Induced Skin Injury: A Dual-Pathway AhR-Nrf2 Framework for Epidermal Homeostasis and Therapeutic Targeting	<i>Int J Mol Sci</i>	Mechanistic review	PM-bound reactive oxygen species, transition metals and PAHs	Taiwan
Scrap tire valorization risks	Is recycling sustainable: The promise and risks of scrap tire valorization.	<i>iScience</i>	Narrative review, technical and toxicological	Process emissions: VOCs, PAHs, heavy metals, microplastics, 6PPD-quinone	Global / multi-region
Sprague Dawley whole-body inhalation, GD5-19	Gestational inhalation of nanoparticles disrupts placental zone structure and induces vascular placentation in rats.	<i>Reprod Toxicol</i>	Animal model	nano-TiO ₂ aerosol, 9.23 +/- 0.39 mg/m ³ whole-body, as an ultrafine PM proxy	Laboratory
UPM instillation reproduction null	Repeated Short-Term Intratracheal Exposure to Urban Particulate Matter Has No Apparent Adverse Effects on Female Reproductive Outcomes in Mice	<i>Cells</i>	Animal experiment	Intratracheal urban PM instillation, 40-200 mg/kg over one month	South Korea
Vera-Colon et al.	Exposure to airborne fine particulate matter disrupts human embryonic stem cell osteogenic differentiation.	<i>Part Fibre Toxicol</i>	Metadata only (no abstract)	PM composition, dose and collection route not recoverable from metadata	Not stated
Wu et al.	Beta-carotene alleviates PM _{2.5} -induced pulmonary fibrosis via suppression of the Frizzled5-mediated Wnt5a/Ca ²⁺ pathway.	<i>Ecotoxicol Environ Saf</i>	Animal exposure study + in vitro receptor overexpression	Intratracheal PM _{2.5} instillation; dose, source and treatment concentration not reported	China
e-waste children mucosal immunity	iNOS/Arg-1 balance and sIgA levels are associated with intestinal mucosal immune defense in children chronically exposed to air pollutants and lead.	<i>Front Immunol</i>	Cross-sectional + ML	Estimated average daily doses of PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ and CO plus blood Pb	China
Occupational & indoor					
Assan et al.	Evaluation of Ambient Air Particulate Matter, Urine Arsenic Levels and the Effect on Residents Health in an Artisanal Mining Community and an Incorporated Mining Community in Ghana.	<i>Health Sci Rep</i>	Cross-sectional (comparative community)	Ambient PM _{2.5} and PM ₁₀ by Modular Air Quality System; urinary arsenic by atomic absorption spectrophotometry	Ghana
China 2021 sandstorm, Baidu search DID	Indoor Air Protection Under Extreme Air Pollution Risk: Evidence From a Sandstorm in China.	<i>Indoor Air</i>	Ecological panel	Sandstorm-driven outdoor PM episode as an exogenous shock; no PM measured directly	China
Environ Int	Portable HEPA air filtration effects on particulate matter, radioactivity, and chemical constituents in occupied buildings.	<i>Environ Int</i>	Metadata only (no abstract)	Not recoverable from the deposited record	Not stated
Fire-station diesel, 3 stations	Characterization of diesel-related pollutant migration in fire stations using low-cost direct-reading monitors.	<i>J Occup Environ Hyg</i>	Repeated-measures monitoring, mixed-effects models	Minute-level PM size fractions, NO ₂ , VOCs and CO ₂ from direct-reading low-cost monitors, 7 days per station	United States
Hospital IAQ, 1.79M records 2023-2025	Toward Integrated Hospital IAQ Monitoring: Continuous Sensing and Targeted Chemical Characterization.	<i>Atmosphere</i>	Long-term continuous sensing plus targeted chemical sampling	CO ₂ , PM _{2.5} , PM ₁₀ , formaldehyde, T, RH, TVOC from low-cost sensors; June 2023 - December 2025	Europe
Hu et al.	Analysis of carbon fiber dust exposure status and related occupational health risk factors among workers during the molding process of carbon fiber products.	<i>Front Public Health</i>	Observational - cross-sectional	Workplace carbon fibre dust measurement at molding workstations, ICM quantitative risk tiering	China

First author	Focus	Journal	Design	Metric	Region
Ikeda et al.	Characterizing layout-dependent breathing-zone contaminant concentration in office spaces with mixing ventilation using an occupant-to-exhaust distance metric.	<i>Build Environ</i>	Metadata only (no abstract)	Breathing-zone contaminant concentration under mixing ventilation; no quantities retrievable	Japan
Indoor OCEC hazard synthesis	Organic Contaminants of Emerging Concern (OCECs) in Indoor Air: A Perspective on Global Distribution and Hazard Burden.	<i>Environ Sci Technol</i>	Systematic synthesis with inhalation hazard prioritisation	Airborne concentrations of 130+ compounds across seven OCEC classes	Global / multi-region
Khan et al.	Exposure to cement industry emissions and respiratory health outcomes: a global systematic review and meta-analysis from 2000 to 2025.	<i>Front Public Health</i>	Systematic review + random-effects meta-analysis	Cement dust and cement-plant emissions, occupational and residential; no harmonised gravimetric metric across studies	Global / multi-region
Ma et al.	Association between combined occupational exposure to coal dust and noise and high-frequency hearing loss among workers: a cross-sectional study.	<i>Front Public Health</i>	Cross-sectional survey	Job-category coal dust exposure, dichotomous, combined with occupational noise; no gravimetric metric	China
Tam et al.	Toxic Air Contaminant Emissions from Residential and Commercial Fossil Gas Appliances: A Systematic Review.	<i>Curr Environ Health Rep</i>	Systematic review (PRISMA)	Appliance emission factors for 41 toxic air contaminants, leakage and combustion pathways	Global / multi-region
Yoshizaki et al.	Differences in the Effectiveness of Smoke Evacuator Types for Reducing Surgical Smoke: A Randomized Controlled Trial.	<i>J Am Coll Surg</i>	Randomised controlled trial	Optical particle counts at 0.3, 0.5, 1.0 and 5.0 μm plus acetaldehyde and formaldehyde, 10-min sampling	Japan
Sensing, forecasting & instrumentation					
AAQR	A Sensor-Driven State-Space Framework for $\text{PM}_{2.5}$ -Informed Natural Ventilation Opportunities Under Coupled Pollution-Thermal-Humidity Constraints.	<i>Aerosol Air Qual Res</i>	Monitoring-record analysis	Trailing 24-h mean $\text{PM}_{2.5}$ from AirGradient low-cost nodes via OpenAQ, threshold 15 $\mu\text{g}/\text{m}^3$ (WHO 2021 24-h AQG)	Southeast Asia
Aeolus transatlantic smoke, Sep	Characterization of aerosol properties during a huge transatlantic smoke transport event using Aeolus observations in synergy with multi-platform data.	<i>Atmos Chem Phys</i>	Spaceborne HSRL case study with trajectory modelling	Aerosol extinction and backscatter from ALADIN; AOD	Global / multi-region
Bai et al.	Role of low-level jet evolution in vertical aerosol redistribution: a Doppler wind lidar study over East China.	<i>Atmos Meas Tech</i>	Remote sensing / lidar retrieval	Coherent Doppler wind lidar backscatter, spectral width and shear profiles against surface PM_{10}	China
Central-eastern China, 14-day lead time	14-Day $\text{PM}_{2.5}$ and O_3 Forecasting over Eastern China Using a Hybrid CNN-Transformer Architecture Driven by Multilevel Meteorological Fields.	<i>Environ Sci Technol</i>	Deep-learning model	Gridded surface $\text{PM}_{2.5}$ and O_3 forecast from multilevel meteorology plus emissions	China
Clean air advocacy data approaches	Examining Data-Driven Approaches for Clean Air Advocacy in a Changing Climate: a Scoping Review.	<i>Curr Environ Health Rep</i>	Scoping review	Low-cost sensor, satellite and modelled air quality data used for advocacy	Global / multi-region
Embassy monitor shutdown, 46 cities	Air pollution after the shutdown of US embassy monitoring.	<i>Proc Natl Acad Sci U S A</i>	Matched difference-in-differences, satellite AOD	Real-time satellite aerosol optical depth, calibrated to pre-shutdown embassy $\text{PM}_{2.5}$	Global / multi-region

First author	Focus	Journal	Design	Metric	Region
Feature engineering without co-location	Machine Learning-Based Calibration of Low-Cost PM _{2.5} Sensors Using Location-Specific Environmental Covariates and Feature Engineering Strategies.	<i>Chemosensors</i>	Ablation study, time-based validation	Low-cost PM _{2.5} sensor in an outdoor residential setting	Turkiye
Front Bioeng	Bionic breathing-type airway exposure system for inhalation exposure studies.	<i>Front Bioeng Biotechnol</i>	Laboratory instrument development	Branch-level particle deposition in a CT-derived airway replica under programmed breathing waveforms	Laboratory
Hu et al.	Advanced insights into biomass burning aerosols during the 2023 Canadian wildfires from dual-site Raman and fluorescence lidar observations.	<i>Atmos Chem Phys</i>	Remote sensing / lidar retrieval	Multi-wavelength Raman, depolarisation and fluorescence lidar profiles of transported smoke	Global / multi-region
Huang et al.	Interpretable Spatiotemporal Coupling Network for Robust Emission Prediction and Implicit Mechanism Mining in Sludge Gasification.	<i>Environ Sci Technol</i>	Deep-learning forecasting	Continuous industrial stack monitoring of PM, SO ₂ and NO _x from a sludge-gasification plant, with feature-level optimal time lags	Not stated
J Aerosol Sci	High-efficiency plasma-near-passage unipolar aerosol charger for sub-10 nm particles.	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Not recoverable from the deposited record	Not stated
Kalluri et al.	Long-term (2008-2024) analysis of black carbon aerosols over a semi-arid rain-shadow region in southern peninsular India: trends, wavelet analysis, machine learning approaches and source regions.	<i>Atmos Environ</i>	Metadata only (no abstract)	Black carbon, presumably aethalometer-derived; instrument, wavelength set and trend magnitudes not recoverable	India
Krajnc Pogacnik et al.	Selective real-time monitoring of respiratory droplets for indoor air safety in educational environments.	<i>Build Environ</i>	Metadata only (no abstract)	Respiratory-droplet detection; sensing principle, size range and selectivity mechanism not recoverable	Not stated
LLM-compiled adaptive sampling policies	Agentic LLM Compilation of Adaptive Sampling Policies for Low-Power IoT Sensing	<i>Sensors</i>	Algorithm development	Beijing PM _{2.5} series used as the benchmark signal for duty-cycled sampling	China
Li et al.	A multi-angle and polarization-based retrieval algorithm for aerosol layer height of smoke and dust.	<i>Atmos Meas Tech</i>	Satellite retrieval evaluation	Aerosol layer height from HARP2/PACE multi-angle polarimetry, validated against EarthCARE ATLID lidar	Global / multi-region
Ni et al.	An AI-based algorithm for retrieving aerosol optical depth and single scattering albedo using All-Sky Imager observations.	<i>Atmos Meas Tech</i>	Algorithm development	AOD and single scattering albedo retrieved from all-sky camera imagery, sun-photometer targets	Global / multi-region
Organic aerosol self-diffusion MD	Predicting the Self-Diffusion Coefficient of Atmospheric Organic Aerosol Components via Molecular Dynamics Simulations.	<i>ACS Omega</i>	Modelling / inventory	Particle-phase self-diffusion of atmospheric organic components	Not applicable
PMS5003 and Sniffer4D, concrete drilling	Non-Cumulative, Size-Specific Calibration of Low-Cost Particulate Matter Sensors Under Simulated Construction Drilling Events.	<i>Atmosphere (preprint of record)</i>	Chamber / laboratory	Non-cumulative PM ₁ , PM _{2.5} , PM ₁₀ fractions under simulated concrete-slab drilling	Not stated

First author	Focus	Journal	Design	Metric	Region
Partector 2 Pro vs SMPS and CPC, 4 filaments	Evaluation of the Partector 2 Pro for measuring ultrafine particles emitted from 3D printing.	<i>J Occup Environ Hyg</i>	Multi-instrument field evaluation	UFP number concentration and D10-D95 mobility diameter percentiles	Laboratory
Peng et al.	Inhibition of condensable particulate matter by co-firing biomass and high sulfur coal on a 50 kW self-sustaining combustion furnace.	<i>Environ Pollut</i>	Chamber emission characterisation	Condensable particulate matter mass and speciation, mg/MJ, dilution sampling	China
PurpleAir Raleigh and Delhi	Evaluating and Calibrating Low-Cost Air Quality Sensors in Contrasting Aerosol Regimes.	<i>Aerosol Air Qual Res</i>	Co-location calibration campaign	PurpleAir PA-II PM _{2.5} against Federal Equivalent Method reference	United States; India
Raju et al.	Investigations of black carbon aerosol sources and strengths over a rural-valley location in Haryana, India.	<i>J Environ Sci Health A</i>	Multi-year aethalometer monitoring + source apportionment	Equivalent black carbon, seven-wavelength AE33, 880 nm; annual and seasonal means in $\mu\text{g}/\text{m}^3$	India
Rural wintertime SIA metal catalysis	Important role of acidity-regulated metal catalysis in rural wintertime secondary inorganic aerosol formation in the North China Plain	<i>Atmos Res</i>	Metadata only (no abstract)	Rural wintertime secondary inorganic aerosol in PM _{2.5}	China
Vishakhapatnam long-term co-location	Exploring Environmental and Temporal Performances of Machine Learning Models for Calibration of a Low-Cost PM _{2.5} Sensor.	<i>Aerosol Air Qual Res</i>	Sensor co-location	Low-cost PM _{2.5} against reference, binned by temperature, RH, hour and month	India
Zhang et al.	Adaptive robotic sensing for exposure-oriented mapping of hazardous airborne particle dispersion in indoor environments.	<i>J Hazard Mater</i>	Adaptive mobile-robot sampling + CFD validation	PM _{2.5} mapped by a mobile optical sensor on a robot; Kernel DM+V field reconstruction	Laboratory
Zhao et al.	A novel deep learning prediction method for PM _{2.5} integrating improved MSTL decomposition and FATA optimization algorithm.	<i>PLoS One</i>	Deep-learning forecasting	PM _{2.5} time series with meteorological and pollutant covariates, station-based	China
sMCPC supersaturation model	Performance modeling of a small mixing-type condensation particle counter.	<i>Atmos Meas Tech</i>	Numerical / CFD instrument model	Particle number, critical activation diameter Dp,50	Not applicable (laboratory / numerical)
Exposure assessment & modelling					
118 CMAQ scenarios, 17 regions x 7 sectors	A LASSO-based reduced-form CMAQ model for predicting ozone and PM _{2.5} responses to emission changes in South Korea.	<i>PLoS One</i>	Chemical transport modelling	CMAQ-simulated surface PM _{2.5} and O ₃ response to sectoral emission perturbation	South Korea
13 Java coal plants, 41,119 daily obs 2015-2025	Predictive modelling of fine-mode combustion aerosol events near Indonesian coal-fired power plants using AE-filtered Himawari-8 satellite data.	<i>Environ Monit Assess</i>	Machine-learning model	Himawari-8 AOD filtered by Angstrom exponent > 1.2 as a fine-mode combustion proxy	Not stated
33 microwave radiometers, China, 2024-2025	Surface-Based Temperature Inversions Climatology Across China from Microwave Radiometer Network and Their Relationship to PM _{2.5} Pollution.	<i>Environ Pollut</i>	Field measurement	PM _{2.5} at hotspot stations against hourly MWR-derived inversion and mixing-layer metrics	China
8 Korean agricultural sites, 547 days	Relationships Between PM _{2.5} and Its Precursor Gases in Agricultural Areas of South Korea Using a Loading-Rate-Based Analysis.	<i>Atmosphere</i>	Field measurement	Hourly PM _{2.5} and NH ₃ , NO ₂ , SO ₂ , expressed as concentration and as wind-scaled loading rate	South Korea
Atmos Pollut Res	Canyons and Canopies: Effects of building configuration and green infrastructure on urban air quality.	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Not recoverable from the deposited record	Not stated

First author	Focus	Journal	Design	Metric	Region
China NOx isotopes	Driving factors of atmospheric nitrogen oxides emissions and pollution in China.	<i>Environ Pollut</i>	Source apportionment	delta15N of particulate and precipitation nitrate across eight Chinese regions, 2001-2022	China
Delhi N2O5	Enhanced nocturnal N2O5-CINO2 chemistry in Delhi.	<i>Atmos Chem Phys</i>	Field campaign	Nocturnal N2O5 and CINO2 mixing ratios; particulate chloride 4.7 µg/m ³ nighttime mean	India
Dou et al.	Volatile organic compounds in Chengdu: Functional zone disparities and four-year dynamics in a basin city of Southwest China.	<i>Environ Pollut</i>	Metadata only (no abstract)	Ambient VOCs by functional zone, four-year record; no quantities retrievable	China
Environ Pollut	Underlying Surface Heterogeneity Modulates Meteorological Drivers of PM _{2.5} and O3 Pollution in the Sichuan Basin.	<i>Environ Pollut</i>	Metadata only (no abstract)	Not recoverable from the deposited record	China
Fireman et al.	Biomonitoring of urban ultrafine particle exposure and associated health effects.	<i>Int J Public Health</i>	Biomonitoring + in vitro	Ultrafine particle counts in exhaled breath condensate, industrial vs control urban setting	Israel
Firozabad size-segregated lung deposition	Seasonal variation of particulate matter deposition in human lungs through different exposure pathways in an industrial city.	<i>Environ Monit Assess</i>	Deposition sampling	Size-segregated PM _{1.0} , PM _{2.5} and PM ₁₀ with metal speciation	India
Frueh et al.	Source apportionment of volatile organic compounds in a Philadelphia fenceline community: results from THRIVEair.	<i>J Expo Sci Environ Epidemiol</i>	Community passive-sampler campaign + PMF source apportionment	38 VOCs, one-week integrated passive thermal-desorption tubes, 20 sites, 12 months	USA
Godri Pollitt	Beyond Inhalation: Air as an Integrative Matrix for Exposome Research.	<i>Environ Sci Technol</i>	Metadata only (no abstract)	No measured PM; air framed as an exposome sampling matrix	Global / multi-region
Han et al.	Environmental Mixture-Health Associations: Current Analytical Practice and Reporting Recommendations.	<i>China CDC Wkly</i>	Review / synthesis	Not applicable; the unit of analysis is the mixture-modelling study	Global
Kim et al.	Residential indoor air monitoring to assess long-term exposure to air pollution in susceptible populations for epidemiology: overview and early findings.	<i>Environ Health</i>	Long-term indoor monitoring campaign	Indoor and matched outdoor PM _{2.5} plus 36 chemical components, RTI MicroPEM, 5-day runs in each of four seasons, hourly resolution	South Korea
Konon et al.	Seasonal Variation in Volatile Organic Compound Emissions and Secondary Organic Aerosol Formation from Unconventional Oil and Gas Development.	<i>Atmos Environ</i>	Metadata only (no abstract)	VOC emissions and SOA formation potential; no quantities retrievable	USA
Lahore, 30 filters, Mar-Apr	Carbonaceous Composition and Multi-Wavelength Optical Properties of Particulate Matter in Lahore, Pakistan.	<i>Atmosphere</i>	Measurement campaign	Gravimetric PM _{2.5} , OC/EC, five-wavelength absorption (375-850 nm) by MWAA	Not stated
Li et al.	A Theoretical Model for Predicting Visibility During Dust Events.	<i>Atmosphere</i>	Theoretical / physical model	Horizontal visibility derived from saltation-controlled dust emission and near-surface concentration	Global / multi-region
Melbourne, 3 acellular OP assays, 2 sites	Source Apportionment of PM _{2.5} Oxidative Potential in metropolitan Melbourne: integrating PMF with multiple linear regression across three acellular assays.	<i>Environ Pollut</i>	Source apportionment	PM _{2.5} oxidative potential by DTT, AA and DCFH, regressed on PMF source contributions	Australia

First author	Focus	Journal	Design	Metric	Region
Melssen Pfahl et al.	Renoxification via Reduction of Inorganic Nitrate by Photoacids in Aerosols.	<i>Environ Sci Technol</i>	Chamber / laboratory	Nebulised gallic acid plus nitrate aerosol, UV-visible irradiation, NO _y and HONO yields across pH	Laboratory
Monte Carlo wind erosion PM, Sardinia	Refinement and Quantitative Evaluation of a Monte Carlo Model for Wind-Driven PM Emissions from Industrial Granular Materials.	<i>Atmosphere</i>	Physical emission model, numerical sensitivity analysis	PM emission mass from saltation-driven sandblasting of granular surfaces	Europe
Montreal smoke advisory mobility	Urban mobility responses to wildfire smoke-related air quality advisories in Montreal, Canada.	<i>Health Place</i>	Retrospective time series	Wildfire-smoke air quality advisory days, 2022-2025	Canada
Multi-Tree	Multi-Tree Model for Precision Environmental Health with Longitudinally Assessed Mixture Exposure.	<i>J R Stat Soc Ser C Appl Stat</i>	Method / algorithm	Weekly-average gestational PM _{2.5} and maximum daily temperature, distributed-lag structure	United States
ORACLES 4STAR	Subseasonal and spatial variability of biomass burning aerosol radiative properties observed over the Southeast Atlantic during ORACLES.	<i>Atmos Chem Phys</i>	Aircraft measurement campaign	4STAR sky-scan retrievals of single scattering albedo and absorption Angstrom exponent for biomass burning aerosol above cloud	Sub-Saharan Africa
Oradea Metropolitan Area, 4 reference stations	Spatiotemporal assessment of urban PM ₁₀ and population exposure using integrated satellite, meteorological, and socio-spatial data.	<i>Front Public Health</i>	Field campaign / remote sensing	MODIS-MAIAC AOD regressed onto station PM ₁₀ , plus a socio-spatial exposure index	Europe
PRD limona ketone	Ground-gridded observations of high concentrations of ambient limona ketone suggest overlooked limonene oxidation in subtropical megacities.	<i>Environ Pollut</i>	Kinetic model + field validation	Limona ketone as an observational constraint on limonene oxidation; secondary organic aerosol precursor chemistry	China
Paygude et al.	Heavy metal exposure and source profiling based on leaf-deposited particulate matter in Pune city.	<i>Environ Monit Assess</i>	Biomonitoring + source apportionment	Leaf-deposited dust, ten elements, enrichment factors and PCA source profiles	India
Portland segregation typologies, 342 tracts	Beyond redlining: Gentrification, displacement, disadvantages, and exclusivity predict urban environmental and health inequities.	<i>Proc Natl Acad Sci U S A</i>	Cross-sectional typology construction, PCA and rank correlation	Tract-level pollution exposure among 104 environmental, hazard and health variables	United States
Savannah and boreal fire vapour ageing	Organic vapors from Savannah and European Boreal fire emissions: insights from photochemical and dark aging.	<i>Atmos Chem Phys</i>	Controlled burn plus 29 m ³ chamber ageing, PTR-MS	Organic vapour emission factors, SOA precursors	Global / multi-region
Sen et al.	Genomic and Phenotypic Classification of Staphylococcus Species Isolated From Delhi Air.	<i>Environ Microbiol Rep</i>	Bioaerosol monitoring campaign	Culturable bioaerosol sampling (Merck MAS Eco-100), six sites, peak pollution season	India
Shenzhen, 2014-2024 filter series	Unraveling decadal changes in PM _{2.5} sources and secondary aerosol formation drivers using integrated receptor model and machine learning.	<i>J Environ Manage</i>	Source apportionment	Speciated PM _{2.5} filter samples, PMF factors plus ML driver attribution	China
Sichuan Basin	Topographic Modulation of High-Temperature Ozone Suppression: Core-Periphery Threshold Structure in the Sichuan Basin.	<i>Environ Pollut</i>	Observational + ML	Hourly O ₃ from 22 city monitoring stations, 2020-2024, with ERA5 meteorology	China

First author	Focus	Journal	Design	Metric	Region
Southern Hemisphere fire trend 2003-2022	Characteristics and trend of the wild-fire activity in the Southern Hemisphere (2003-2022).	<i>Sci Total Environ</i>	Satellite time-series analysis	MODIS Active Fire detections as a proxy for biomass-burning PM source strength	Global / multi-region
Sun et al.	Nonlinear Nitrate Formation Governed by Aerosol Water and VOC-Sensitive Oxidation in a Humid Urban Basin.	<i>Environ Sci Technol</i>	Field observations + box model + thermodynamic equilibrium	Wintertime PM _{2.5} composition, NO ₂ , O ₃ and aerosol liquid water content	China
Sunyer-Caldu	Beyond Blood and Urine: Decoding the Human Chemical Exposome across Matrices and Time.	<i>Environ Sci Technol</i>	Perspective / framework proposal	No measured PM; air positioned as one external matrix within a multi-matrix exposome design	Global / multi-region
Yu et al.	Continental pollutants modulate organic nitrogen and light absorption of marine organic aerosols over East Asian marginal seas.	<i>Atmos Chem Phys</i>	Field campaign / shipborne	Shipboard aerosol organic nitrogen, water-soluble and water-insoluble fractions, ON/OC ratio and light absorption over three seasons	China
Zero-filling DLM bias	Gestational misalignment with fixed exposure windows: the potential dangers of zero-filling in distributed lag models	<i>Int J Epidemiol</i>	Method / algorithm	Weekly gestational PM _{2.5} exposure histories of unequal length	United States
Zhao et al.	Unveiling the Importance of Organic Hydroxy Acids-Related Autoxidation Pathway in Isoprene Secondary Organic Aerosol Formation.	<i>Environ Sci Technol</i>	Year-long regional tracer measurements + box model	PM _{2.5} -phase isoprene SOA tracers, six iHOM species, ng/m ³ , nine sites	China
Burden, policy & mitigation					
Atmospheric infrastructures Czech activism	Cities as Atmospheric Infrastructures: An Analysis of Activist Responses to Respiratory Illness in the Czech Republic	<i>Sociol Health Illn</i>	Qualitative / ethnographic study	Urban air pollution as the object of civic mobilisation	Czech Republic
Bao et al.	Distribution-sensitive assessment of air-quality and health benefits from clean-heating pilots in northern China.	<i>J Environ Manage</i>	Augmented synthetic control + weather normalisation	City-daily PM _{2.5} from 1-km gridded product, ML weather-normalised, heating-season mean/P95/P99	China
Campania review	Air pollution and chronic respiratory diseases: a narrative review including a case study in Campania (Italy).	<i>Monaldi Arch Chest Dis</i>	Narrative review	Regional PM _{2.5} , PM ₁₀ , NO ₂ and O ₃ concentrations 2015-2024 against national averages	Italy
Central China BenMAP 2017-2024	Assessment of the spatiotemporal evolution of PM _{2.5} and its health and economic effects in Central China based on spatial statistics and BenMAP.	<i>Environ Monit Assess</i>	Spatial statistics plus BenMAP health impact assessment	Ground monitors and the Tracking Air Pollution in China (TAP) reanalysis, 2017-2024	China
Covenant of Mayors co-benefit tool	A tool to maximize air quality co-benefits while planning local climate actions: application to the Covenant of Mayors initiative.	<i>Open Res Eur</i>	Tool / software	PM _{2.5} , NO _x , SO ₂ , NH ₃ and VOC emission changes from city climate action plans	Europe
EOTBL cancer	Global Burden and Temporal Trends of Early-Onset Tracheal, Bronchial, and Lung Cancer: An Analysis of GLOBOCAN 2022 and GBD 2023 Data.	<i>Adv Sci</i>	Ecological (GLOBOCAN secondary analysis)	Ambient particulate matter pollution as a GBD comparative-risk-assessment attributable risk factor	Global / multi-region
Habit-locked household emissions	Habit-locked-in carbon-air-pollutant emissions dominate China's household footprints.	<i>Environ Sci Ecotechnol</i>	Emission inventory / decomposition	Household-driven PM _{2.5} , NO _x , SO ₂ and CO ₂ from consumption-based accounting	China

First author	Focus	Journal	Design	Metric	Region
Ju et al.	Premature mortality and economic losses attributable to landscape fire-related PM _{2.5} in ten countries and territories: a causal analysis.	<i>Lancet Planet Health</i>	Instrumental-variable time-series + random-effects meta-analysis	Population-weighted daily landscape-fire PM _{2.5} vs non-fire PM _{2.5} , 2-day moving average, modelled	Global / multi-region
Mairghani et al.	An umbrella review of public health interventions to mitigate air pollution and improve health outcomes in high-income countries.	<i>Rev Environ Health</i>	Umbrella review / meta-analysis	Intervention-level exposure change (traffic policy, portable air cleaners, net-zero pathways)	Global / multi-region
Massimi et al.	Urban Green Walls under Traffic Stress: Linking the Elemental Composition of Particulate Matter Deposition to Plant Physiological and Gene Expression Responses.	<i>Environ Sci Technol</i>	Biomonitoring + source apportionment	Multi-elemental fractionation of PM deposited on green-wall foliage - insoluble, soluble and whole-leaf fractions - with SEM-EDS, over one year	Italy
PM stroke burden GBD 1990-2021	Ambient particulate matter pollution and stroke burden.	<i>Med Gas Res</i>	Descriptive burden analysis, GBD 2021	GBD ambient particulate matter pollution risk-factor exposure	Global / multi-region
US net-zero multi-model	A Multi-Model Assessment of the Air Quality and Health Impacts of U.S. Energy Decarbonization.	<i>Environ Sci Technol</i>	Model intercomparison	Modelled ambient PM _{2.5} from energy-sector precursor emissions under net-zero scenarios to 2050	United States
Western Pacific stroke burden	Stroke Burden, Trends, and Inequalities in the Western Pacific Region, 1990-2021: A Population-Based Study.	<i>JACC Asia</i>	GBD secondary analysis	Ambient particulate matter attributable fraction of stroke DALYs	Global / multi-region
Other clinical endpoints					
Aerosol Air Qual Res	Association Between Air Pollution and Psoriasis-Related Outcomes: A Systematic Review and Meta-analysis.	<i>Aerosol Air Qual Res</i>	Metadata only (no abstract)	Not recoverable from the deposited record	Not stated
Alhalwani et al.	Correlations of CO, SO ₂ , PM ₁₀ , NO ₂ , and O ₃ with dry eye disease in Saudi Arabia.	<i>Saudi J Ophthalmol</i>	Observational exposure study	Daily PM ₁₀ , SO ₂ , NO ₂ , O ₃ and CO from three fixed monitoring stations in Jeddah	Saudi Arabia
Huang et al.	Secondhand smoke exposure in never-smokers: A cross-sectional and prospective cohort study of chronic kidney disease prevalence and all-cause mortality in affected patients.	<i>Medicine (Baltimore)</i>	Retrospective cohort	Serum cotinine as a biomarker of secondhand smoke particulate exposure, graded moderate/heavy	USA
Kim et al.	Extreme Temperatures, Elevated PM _{2.5} , and Acute Care Events Among Dual-Eligible Nursing Facility Residents.	<i>J Am Geriatr Soc</i>	Panel study	Daily outdoor PM _{2.5} dichotomised at >9 µg/m ³ ; extreme heat and extreme cold day indicators	USA
Liu et al.	Association of exposure to fine particulate matter and its components with frailty in middle-aged and older adults in Beijing-Tianjin-Hebei region and surrounding areas.	<i>Zhonghua Liu Xing Bing Xue Za Zhi</i>	Observational - cross-sectional	Superstation PM _{2.5} and speciated components, lag0-90 to lag0-180 day moving averages; 90-day mean 53.92 ± 10.51 µg/m ³	China
Nigeria 3-state cohort, n=5,621	Ambient PM _{2.5} exposure and cancer incidence in Nigerian adults: A prospective cohort study.	<i>Cancer Epidemiol</i>	Prospective cohort	Hybrid satellite + land-use regression, 1 km annual PM _{2.5} , PM ₁₀ , NO ₂	Sub-Saharan Africa

First author	Focus	Journal	Design	Metric	Region
Odiko et al.	Environmental lead exposure and clinical infectious respiratory disease in a sample of urban and disadvantaged children.	<i>J Expo Sci Environ Epidemiol</i>	Retrospective cohort	Blood lead concentration; lead is a PM-borne metal but no airborne measurement is made	USA
SEER-Medicare wildfire PM _{2.5} , 414,016 lung cancers	Long-term exposure to wildfire-related PM _{2.5} and overall survival of older patients with lung cancer: a retrospective, registry-based cohort study.	<i>Lancet Oncol</i>	Retrospective cohort	3-year moving average wildfire vs non-wildfire PM _{2.5} , zip-code, model-derived	United States

Provenance and method

Period **13–19 September 2026**, Sunday to Saturday, **seven daily issues, 148 records, 148 unique DOIs**, no record without a DOI. This rollup is **aggregation only**: it re-reads `state/corpus/*.json`, `state/metrics.csv` and `state/trials.json` over the period and **queries no literature API**, so every record and every number here was gated by `check_dois.py` at daily-issue time and nothing has been re-screened or re-assessed. The only live query made for this issue was the ClinicalTrials.gov reconciliation sweep reported above, which is state maintenance and contributes no record to the corpus. Tier mix across the week: **35 A, 71 B, 42 C**. The 19 September issue was a recovery build for a scheduled run that did not fire; it covers its full day and the week therefore has no gap. Effect estimates are transcribed verbatim from the daily issues and use heterogeneous exposure contrasts — see the forest-plot caption — and are **not** meta-analysed. Abstracts remain the copyright of their respective publishers.